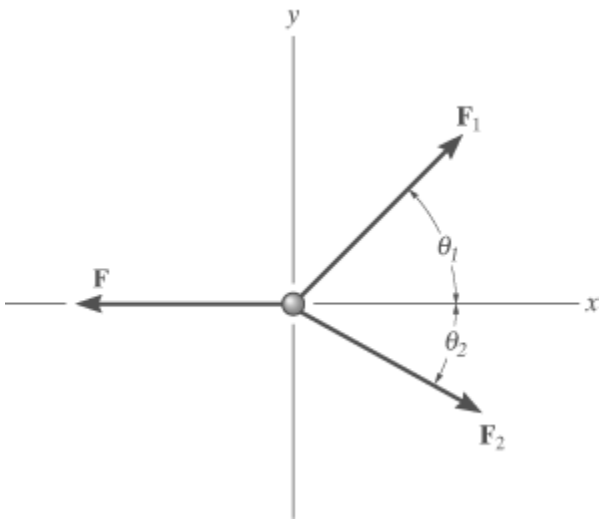
Problem 3-1

Determine the magnitudes of $\mathbf{F_1}$ and $\mathbf{F_2}$ so that the particle is in equilibrium.



Given:

$F = 500$ N

$\theta_1 = 45$ deg

$\theta_2 = 30$deg

Solution:

Initial Guesses

$F_1 = 1$N $\quad F_2 = 1$N

Given

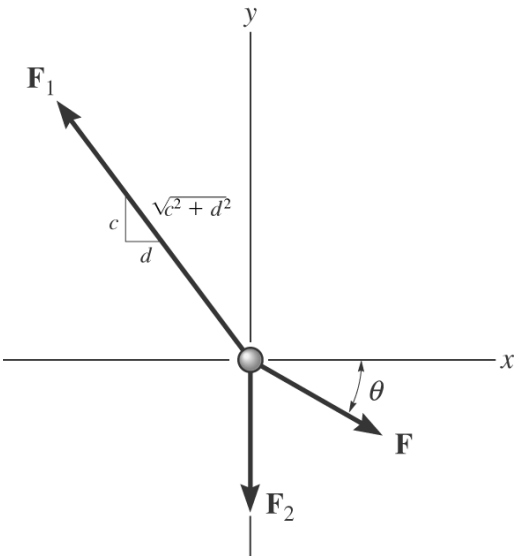
$\xrightarrow{+} \Sigma F_x = 0; \quad F_1 \cos(\theta_1) + F_2 \cos(\theta_2) - F = 0$

$+\uparrow \Sigma F_y = 0; \quad F_1 \sin(\theta_1) - F_2 \sin(\theta_2) = 0$

$\begin{pmatrix} F_1 \\ F_2 \end{pmatrix} = \text{Find}(F_1, F_2)$ $\quad \begin{pmatrix} F_1 \\ F_2 \end{pmatrix} = \begin{pmatrix} 259 \\ 366 \end{pmatrix}$ N

Problem 3-2

Determine the magnitude and direction θ of $\mathbf{F}$ so that the particle is in equilibrium.



Units Used:

kN $= 10^3$ N

Given:

$F_1 = 7$ kN

$F_2 = 3$ kN

$c = 4$

$d = 3$

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Solution:

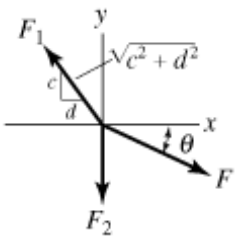
The initial guesses: $F = 1\text{kN}$ $\theta = 30\text{deg}$

Given

Equations of equilibrium:

$\xrightarrow{+} \Sigma F_x = 0;$ $\left(\frac{-d}{\sqrt{c^2 + d^2}}\right) F_1 + F\cos(\theta) = 0$

$+\uparrow \Sigma F_y = 0;$ $\left(\frac{c}{\sqrt{c^2 + d^2}}\right) F_1 - F_2 - F\sin(\theta) = 0$



$\begin{pmatrix} F \\ \theta \end{pmatrix} = \text{Find}(F, \theta)$ $F = 4.94\,\text{kN}$ $\theta = 31.8\,\text{deg}$

Problem 3-3

Determine the magnitude of **F** and the orientation θ of the force $\mathbf{F}_3$ so that the particle is in equilibrium.

Given:

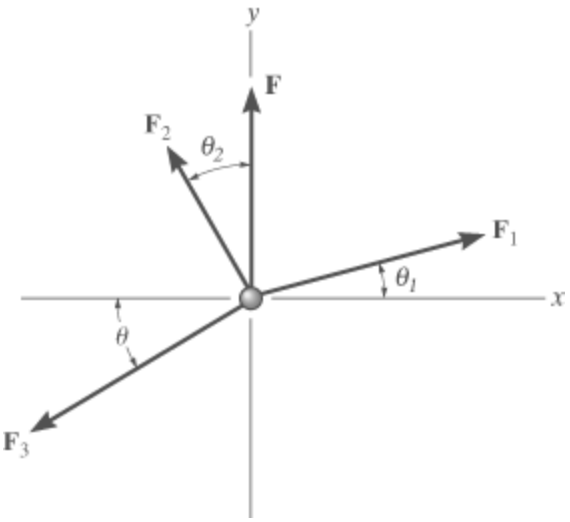
$F_1 = 700\ \text{N}$

$F_2 = 450\ \text{N}$

$F_3 = 750\ \text{N}$

$\theta_1 = 15\ \text{deg}$

$\theta_2 = 30\ \text{deg}$



Solution:

Initial Guesses: $F = 1\text{N}$ $\theta = 10\text{deg}$

Given

$\xrightarrow{+} \Sigma F_x = 0;$ $F_1\cos(\theta_1) - F_2\sin(\theta_2) - F_3\cos(\theta) = 0$

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$+\uparrow \Sigma F_y = 0; \qquad F + F_2\cos(\theta_2) + F_1\sin(\theta_1) - F_3\sin(\theta) = 0$

$\begin{pmatrix} F \\ \theta \end{pmatrix} = \text{Find}(F, \theta) \qquad F = 28.25\ \text{N} \qquad \theta = 53.02\ \text{deg}$

Problem 3-4

Determine the magnitude and angle θ of **F** so that the particle is in equilibrium.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

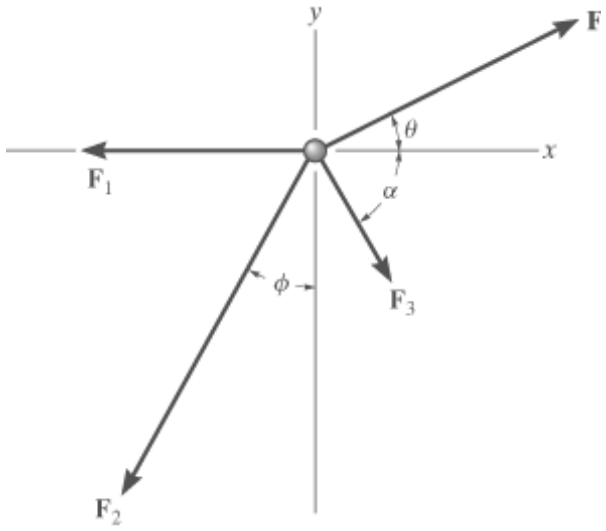
$F_1 = 4.5\ \text{kN}$

$F_2 = 7.5\ \text{kN}$

$F_3 = 2.25\ \text{kN}$

$\alpha = 60\ \text{deg}$

$\phi = 30\ \text{deg}$



Solution:

Guesses:

$F = 1\ \text{kN} \qquad \theta = 1$

Given

Equations of Equilibrium:

$\xrightarrow{+} \Sigma F_x = 0; \qquad F\cos(\theta) - F_2\sin(\phi) - F_1 + F_3\cos(\alpha) = 0$

$+\uparrow \Sigma F_y = 0; \qquad F\sin(\theta) - F_2\cos(\phi) - F_3\sin(\alpha) = 0$

$\begin{pmatrix} F \\ \theta \end{pmatrix} = \text{Find}(F, \theta) \qquad F = 11.05\,\text{kN} \qquad \theta = 49.84\,\text{deg}$

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Problem 3-5

The members of a truss are connected to the gusset plate. If the forces are concurrent at point O, determine the magnitudes of **F** and **T** for equilibrium.

Units Used:

$\text{kN} = 10^3\ \text{N}$

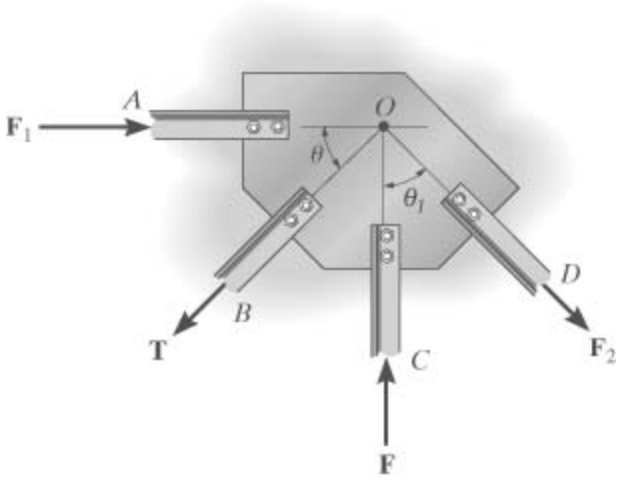
Given:

$F_1 = 8\ \text{kN}$

$F_2 = 5\ \text{kN}$

$\theta_1 = 45\ \text{deg}$

$\theta = 30\ \text{deg}$



Solution:

$\xrightarrow{+} \Sigma F_x = 0; \quad -T\cos(\theta) + F_1 + F_2\sin(\theta_1) = 0$

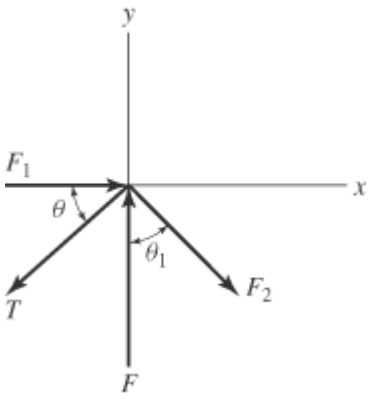
$$T = \frac{F_1 + F_2\sin(\theta_1)}{\cos(\theta)}$$

$T = 13.3\ \text{kN}$

$+\uparrow \Sigma F_y = 0; \quad F - T\sin(\theta) - F_2\cos(\theta_1) = 0$

$F = T\sin(\theta) + F_2\cos(\theta_1)$

$F = 10.2\ \text{kN}$



Problem 3-6

The gusset plate is subjected to the forces of four members. Determine the force in member B and its proper orientation θ for equilibrium. The forces are concurrent at point O.

Units Used:

$\text{kN} = 10^3\ \text{N}$

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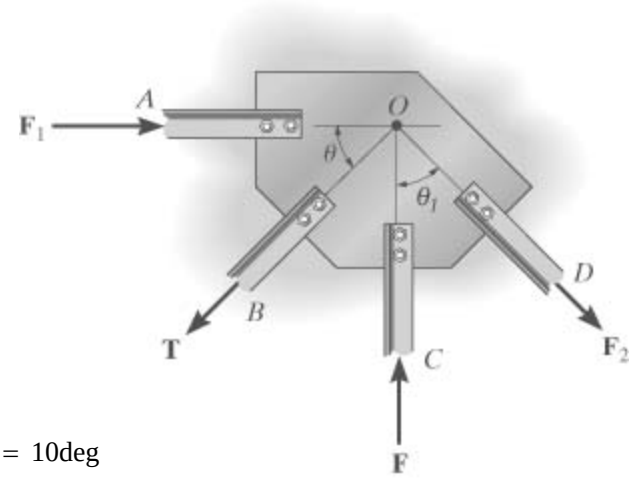
Given:

$F = 12$ kN

$F_1 = 8$ kN

$F_2 = 5$ kN

$\theta_1 = 45$ deg



Solution:

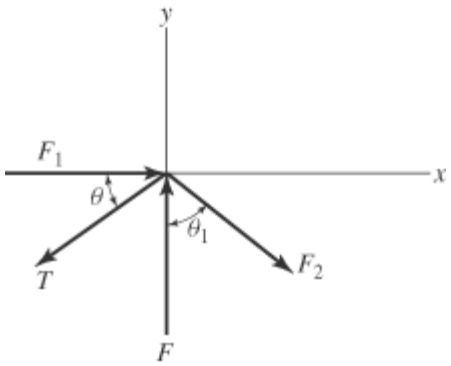
Initial Guesses $T = 1$kN $\theta = 10$deg

Given

$\xrightarrow{+} \Sigma F_x = 0; \quad F_1 - T\cos(\theta) + F_2\sin(\theta_1) = 0$

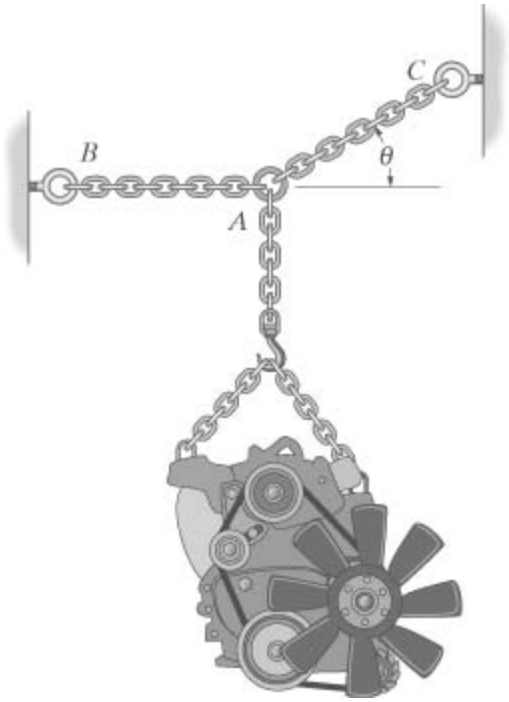
$+\uparrow \Sigma F_y = 0; \quad -T\sin(\theta) - F_2\cos(\theta_1) + F = 0$

$\begin{pmatrix} T \\ \theta \end{pmatrix} = \text{Find}(T, \theta)$ $T = 14.31$ kN $\theta = 36.27$ deg



Problem 3-7

Determine the maximum weight of the engine that can be supported without exceeding a tension of $\mathbf{T_1}$ in chain *AB* and $\mathbf{T_2}$ in chain *AC*.



Given:

$\theta = 30$ deg

$T_1 = 450$ lb

$T_2 = 480$ lb

Solution:

Initial Guesses

$F_{AB} = T_1$

$F_{AC} = T_2$

$W = 1$lb

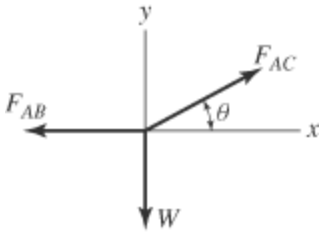
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Given

Assuming cable *AB* reaches the maximum tension $F_{AB} = T_1$.

$\xrightarrow{+} \Sigma F_x = 0; \qquad F_{AC}\cos(\theta) - F_{AB} = 0$

$+\uparrow \Sigma F_y = 0; \qquad F_{AC}\sin(\theta) - W = 0$



$\begin{pmatrix} F_{AC1} \\ W_1 \end{pmatrix} = \text{Find}(F_{AC}, W) \qquad W_1 = 259.81\,\text{lb}$

Given Assuming cable *AC* reaches the maximum tension $F_{AC} = T_2$.

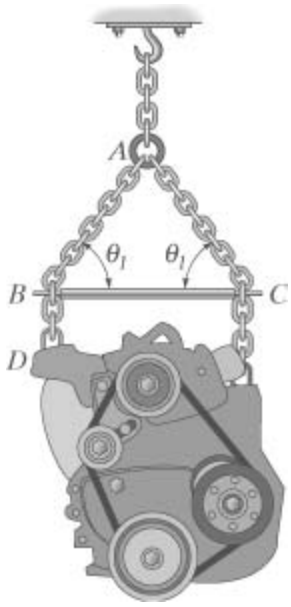
$\xrightarrow{+} \Sigma F_x = 0; \qquad F_{AC}\cos(\theta) - F_{AB} = 0$

$+\uparrow \Sigma F_y = 0; \qquad F_{AC}\sin(\theta) - W = 0$

$\begin{pmatrix} F_{AB2} \\ W_2 \end{pmatrix} = \text{Find}(F_{AB}, W) \qquad W_2 = 240.00\,\text{lb}$

$W = \min(W_1, W_2) \qquad W = 240.00\,\text{lb}$

Problem 3-8



The engine of mass *M* is suspended from a vertical chain at *A*. A second chain is wrapped around the engine and held in position by the spreader bar *BC*. Determine the compressive force acting along the axis of the bar and the tension forces in segments *BA* and *CA* of the chain. *Hint:* Analyze equilibrium first at *A*, then at *B*.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$M = 200\ \text{kg}$

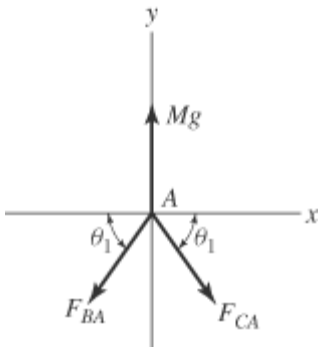
$\theta_1 = 55\ \text{deg}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$

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Solution:

Initial guesses: $F_{BA} = 1$ kN $F_{CA} = 2$ kN

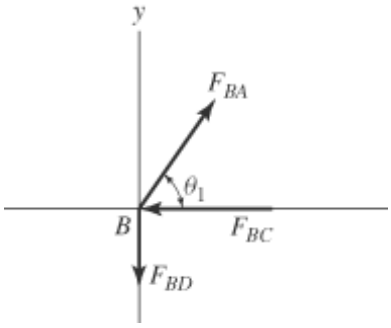


Given Point *A*

$\overset{+}{\rightarrow} \Sigma F_x = 0; \quad F_{CA}\cos(\theta_1) - F_{BA}\cos(\theta_1) = 0$

$+\uparrow \Sigma F_y = 0; \quad M(g) - F_{CA}\sin(\theta_1) - F_{BA}\sin(\theta_1) = 0$

$\begin{pmatrix} F_{BA} \\ F_{CA} \end{pmatrix} = \text{Find}(F_{BA}, F_{CA}) \qquad \begin{pmatrix} F_{BA} \\ F_{CA} \end{pmatrix} = \begin{pmatrix} 1.20 \\ 1.20 \end{pmatrix}$ kN



At point *B:*

$\overset{+}{\rightarrow} \Sigma F_x = 0; \quad F_{BA}\cos(\theta_1) - F_{BC} = 0$

$F_{BC} = F_{BA}\cos(\theta_1) \qquad F_{BC} = 687$ N

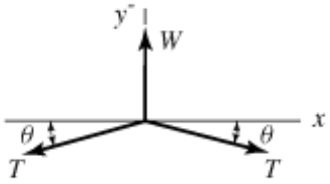
Problem 3-9

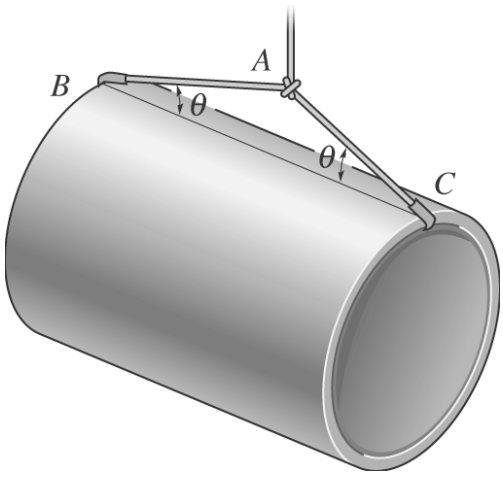
Cords *AB* and *AC* can each sustain a maximum tension *T*. If the drum has weight *W*, determine the smallest angle θ at which they can be attached to the drum.

Given:

$T = 800$ lb

$W = 900$ lb





Solution:

$+\uparrow \Sigma F_y = 0; \quad W - 2T\sin(\theta) = 0$

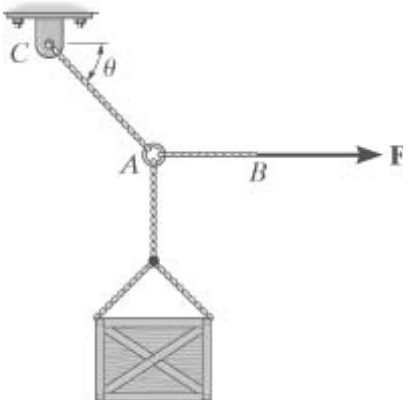
$\theta = \text{asin}\left(\frac{W}{2T}\right)$

$\theta = 34.2$ deg

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Problem 3-10

The crate of weight *W* is hoisted using the ropes *AB* and *AC*. Each rope can withstand a maximum tension *T* before it breaks. If *AB* always remains horizontal, determine the smallest angle θ to which the crate can be hoisted.



Given:

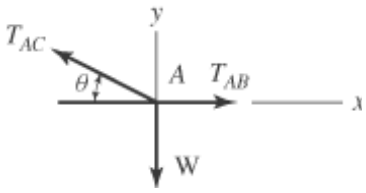
$W = 500$ lb

$T = 2500$ lb

Solution:

Case 1: Assume $T_{AB} = T$

The initial guess $\theta = 30$ deg $T_{AC} = 2000$ lb



Given

$\xrightarrow{+} \Sigma F_x = 0; \quad T_{AB} - T_{AC}\cos(\theta) = 0$

$+\uparrow \Sigma F_y = 0; \quad T_{AC}\sin(\theta) - W = 0$

$\begin{pmatrix} T_{AC1} \\ \theta_1 \end{pmatrix} = \text{Find}(T_{AC}, \theta) \qquad \theta_1 = 11.31\,\text{deg} \qquad T_{AC1} = 2550\,\text{lb}$

Case 1: Assume $T_{AC} = T$

The initial guess $\theta = 30$ deg $T_{AB} = 2000$ lb

Given

$\xrightarrow{+} \Sigma F_x = 0; \quad T_{AB} - T_{AC}\cos(\theta) = 0$

$+\uparrow \Sigma F_y = 0; \quad T_{AC}\sin(\theta) - W = 0$

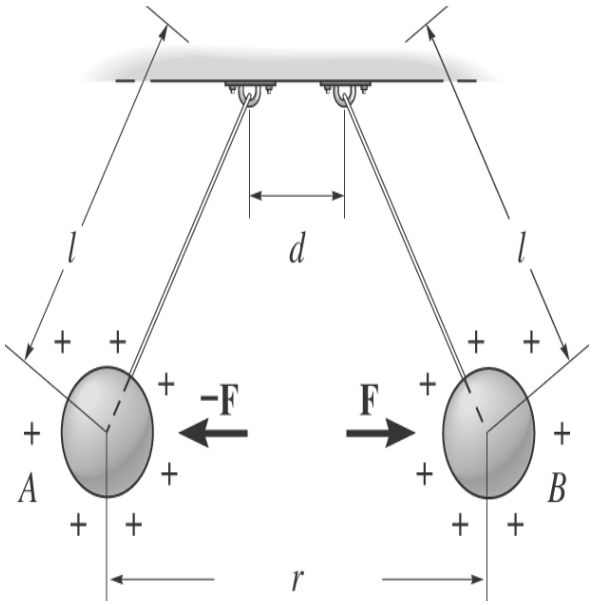
$\begin{pmatrix} T_{AB2} \\ \theta_2 \end{pmatrix} = \text{Find}(T_{AB}, \theta) \qquad \theta_2 = 11.54\,\text{deg} \qquad T_{AB2} = 2449\,\text{lb}$

$\theta = \max(\theta_1, \theta_2) \qquad \theta = 11.54\,\text{deg}$

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Problem 3-11

Two electrically charged pith balls, each having mass M, are suspended from light threads of equal length. Determine the resultant horizontal force of repulsion, F, acting on each ball if the measured distance between them is r.



Given:

$M = 0.2$ gm

$r = 200$ mm

$l = 150$ mm

$d = 50$ mm

Solution:

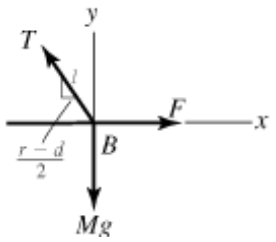
The initial guesses:

$T = 200$ N $\quad F = 200$ N

Given

$$+\uparrow \Sigma F_x = 0; \quad F - T\left(\frac{r-d}{2l}\right) = 0$$

$$\xrightarrow{+} \Sigma F_y = 0; \quad T\left[\frac{\sqrt{l^2 - \left(\frac{r-d}{2}\right)^2}}{l}\right] - Mg = 0$$



$$\begin{pmatrix} T \\ F \end{pmatrix} = \text{Find}(T, F) \qquad F = 1.13 \times 10^{-3}\ \text{N} \qquad T = 0.00\ \text{N}$$

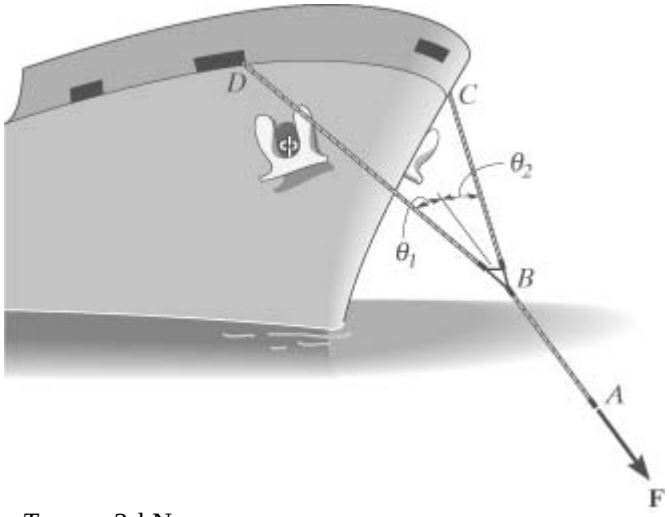
Problem 3-12

The towing pendant AB is subjected to the force **F** which is developed from a tugboat. Determine the force that is in each of the bridles, BC and BD, if the ship is moving forward with constant velocity.

Units Used:

$\text{kN} = 10^3$ N

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Given:

$F = 50$ kN

$\theta_1 = 20$ deg

$\theta_2 = 30$ deg

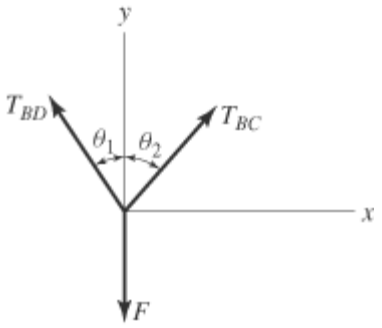
Solution:

Initial guesses: $T_{BC} = 1$ kN $T_{BD} = 2$ kN

Given

$\xrightarrow{+} \Sigma F_x = 0; \quad T_{BC}\sin(\theta_2) - T_{BD}\sin(\theta_1) = 0$

$+\uparrow \Sigma F_y = 0; \quad T_{BC}\cos(\theta_2) + T_{BD}\cos(\theta_1) - F = 0$

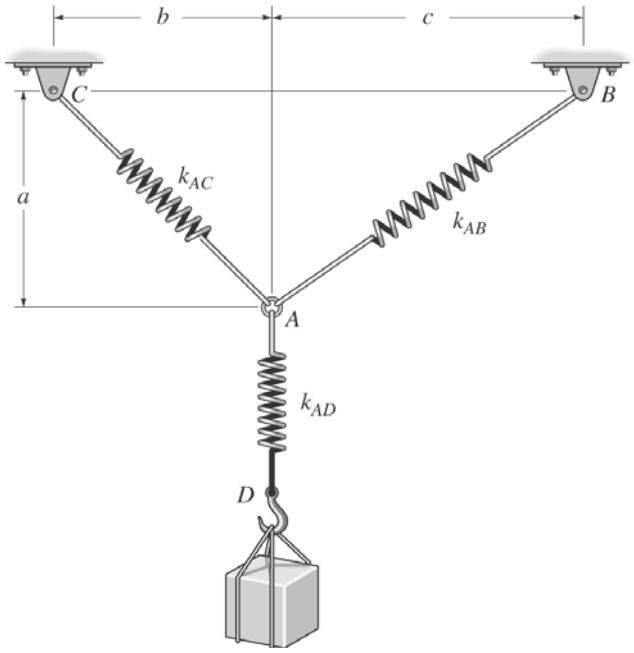


$$\begin{pmatrix} T_{BC} \\ T_{BD} \end{pmatrix} = \text{Find}(T_{BC}, T_{BD}) \qquad \begin{pmatrix} T_{BC} \\ T_{BD} \end{pmatrix} = \begin{pmatrix} 22.32 \\ 32.64 \end{pmatrix} \text{kN}$$

Problem 3-13

Determine the stretch in each spring for equilibrium of the block of mass *M*. The springs are shown in the equilibrium position.

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Given:

$M = 2$ kg

$a = 3$ m

$b = 3$ m

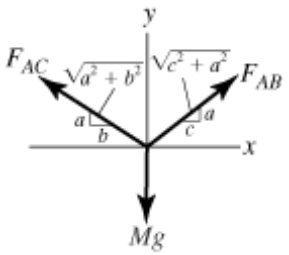
$c = 4$ m

$k_{AB} = 30 \dfrac{\text{N}}{\text{m}}$

$k_{AC} = 20 \dfrac{\text{N}}{\text{m}}$

$k_{AD} = 40 \dfrac{\text{N}}{\text{m}}$

$g = 9.81 \dfrac{\text{m}}{\text{s}^2}$



Solution:

The initial guesses:

$F_{AB} = 1$ N $\quad F_{AC} = 1$ N

Given

$$\xrightarrow{+} \Sigma F_x = 0; \qquad F_{AB}\left(\frac{c}{\sqrt{a^2+c^2}}\right) - F_{AC}\left(\frac{b}{\sqrt{a^2+b^2}}\right) = 0$$

$$+\uparrow \Sigma F_y = 0; \qquad F_{AC}\left(\frac{a}{\sqrt{a^2+b^2}}\right) + F_{AB}\left(\frac{a}{\sqrt{a^2+c^2}}\right) - Mg = 0$$

$$\begin{pmatrix} F_{AC} \\ F_{AB} \end{pmatrix} = \text{Find}\left(F_{AC}, F_{AB}\right) \qquad \begin{pmatrix} F_{AC} \\ F_{AB} \end{pmatrix} = \begin{pmatrix} 15.86 \\ 14.01 \end{pmatrix} \text{N}$$

$$x_{AC} = \frac{F_{AC}}{k_{AC}} \qquad x_{AC} = 0.79 \text{ m}$$

$$x_{AB} = \frac{F_{AB}}{k_{AB}} \qquad x_{AB} = 0.47 \text{ m}$$

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Problem 3-14

The unstretched length of spring *AB* is δ. If the block is held in the equilibrium position shown, determine the mass of the block at *D*.

Given:

$\delta = 2\ \text{m}$

$a = 3\ \text{m}$

$b = 3\ \text{m}$

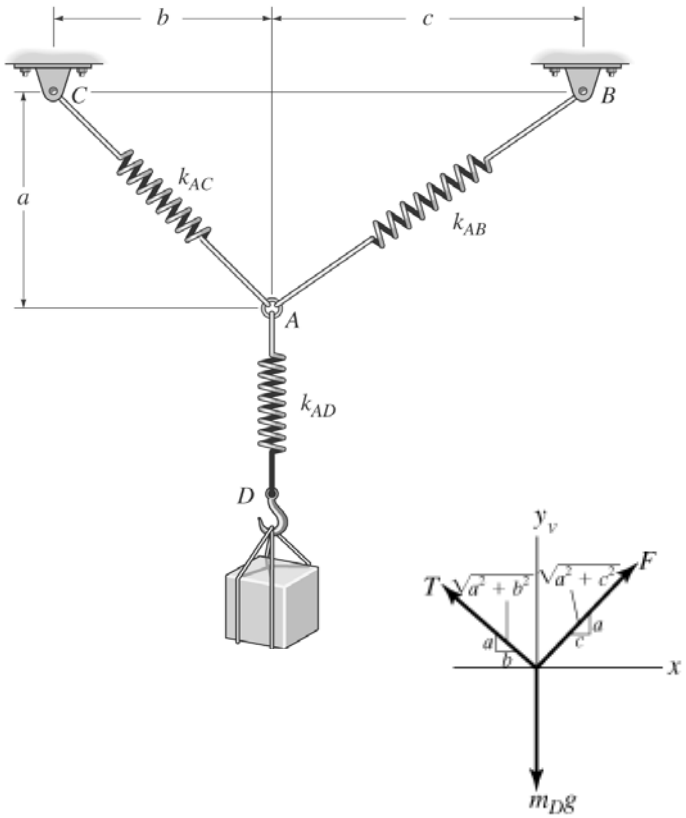
$c = 4\ \text{m}$

$k_{AB} = 30\ \frac{\text{N}}{\text{m}}$

$k_{AC} = 20\ \frac{\text{N}}{\text{m}}$

$k_{AD} = 40\ \frac{\text{N}}{\text{m}}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$



Solution:

$$F_{AB} = k_{AB}\left(\sqrt{a^2 + c^2} - \delta\right)$$

The initial guesses:

$m_D = 1\ \text{kg}$ $F_{AC} = 1\ \text{N}$

Given

$$\xrightarrow{+} \Sigma F_x = 0; \qquad F_{AB}\left(\frac{c}{\sqrt{a^2 + c^2}}\right) - F_{AC}\left(\frac{b}{\sqrt{a^2 + b^2}}\right) = 0$$

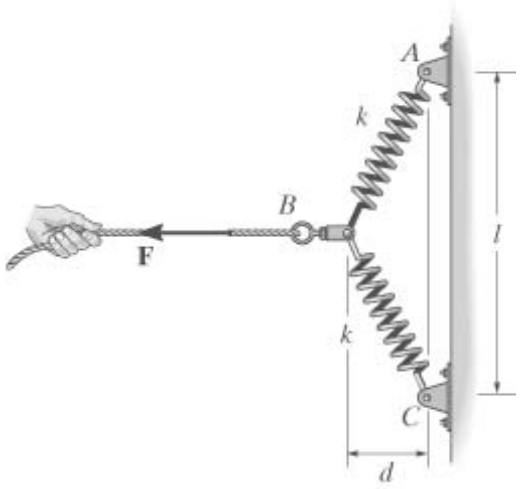
$$+\uparrow \Sigma F_y = 0; \qquad F_{AC}\left(\frac{a}{\sqrt{a^2 + b^2}}\right) + F_{AB}\left(\frac{a}{\sqrt{a^2 + c^2}}\right) - m_D g = 0$$

$$\begin{pmatrix} F_{AC} \\ m_D \end{pmatrix} = \text{Find}\left(F_{AC}, m_D\right) \qquad F_{AC} = 101.8\ \text{N} \qquad m_D = 12.8\ \text{kg}$$

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Problem 3-15

The springs *AB* and *BC* have stiffness *k* and unstretched lengths *l*/2. Determine the horizontal force **F** applied to the cord which is attached to the *small* pulley *B* so that the displacement of the pulley from the wall is *d*.



Given:

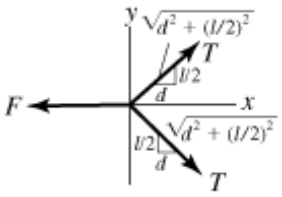
$l = 6\ \text{m}$

$k = 500\ \frac{\text{N}}{\text{m}}$

$d = 1.5\ \text{m}$

Solution:

$$T = k\left[\sqrt{\left(\frac{l}{2}\right)^2 + d^2} - \frac{l}{2}\right] \qquad T = 177.05\ \text{N}$$



$$\xrightarrow{+} \Sigma F_x = 0; \qquad \frac{d}{\sqrt{d^2 + \left(\frac{l}{2}\right)^2}}(2T) - F = 0$$

$$F = \frac{d}{\sqrt{\left(\frac{l}{2}\right)^2 + d^2}}(2T) \qquad F = 158.36\ \text{N}$$

Problem 3-16

The springs *AB* and *BC* have stiffness *k* and an unstretched length of *l*. Determine the displacement *d* of the cord from the wall when a force *F* is applied to the cord.

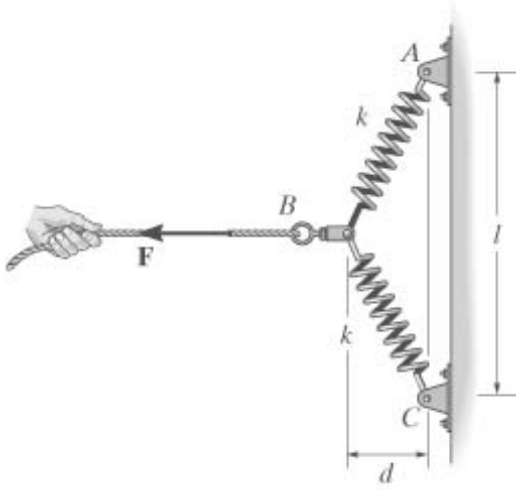
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Given:

$l = 6$ m

$k = 500 \dfrac{\text{N}}{\text{m}}$

$F = 175$ N



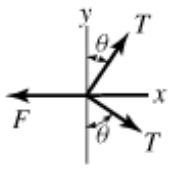
Solution:

The initial guesses:

$d = 1$ m $\quad T = 1$ N

Given

$$\xrightarrow{+} \Sigma F_x = 0; \quad -F + (2T)\frac{d}{\sqrt{d^2 + \left(\frac{l}{2}\right)^2}} = 0$$



Spring $\quad T = k\left[\sqrt{d^2 + \left(\frac{l}{2}\right)^2} - \frac{l}{2}\right]$

$$\begin{pmatrix} T \\ d \end{pmatrix} = \text{Find}(T, d) \qquad T = 189.96 \text{ N} \qquad d = 1.56 \text{ m}$$

Problem 3-17

Determine the force in each cable and the force **F** needed to hold the lamp of mass M in the position shown. *Hint:* First analyze the equilibrium at B; then, using the result for the force in BC, analyze the equilibrium at C.

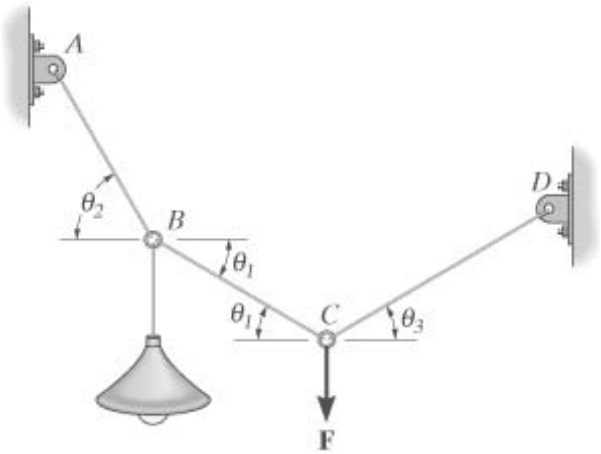
Given:

$M = 4$ kg

$\theta_1 = 30$ deg

$\theta_2 = 60$ deg

$\theta_3 = 30$ deg

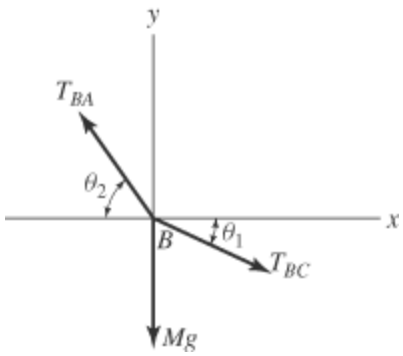


Solution:

Initial guesses:

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$T_{BC} = 1$ N $\quad T_{BA} = 2$ N



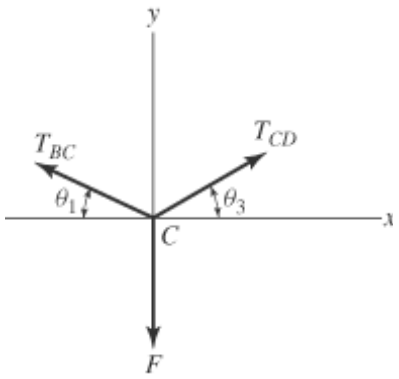
Given

At *B*:

$$\xrightarrow{+} \Sigma F_x = 0; \quad T_{BC}\cos(\theta_1) - T_{BA}\cos(\theta_2) = 0$$

$$+\uparrow \Sigma F_y = 0; \quad T_{BA}\sin(\theta_2) - T_{BC}\sin(\theta_1) - Mg = 0$$

$$\begin{pmatrix} T_{BC} \\ T_{BA} \end{pmatrix} = \text{Find}(T_{BC}, T_{BA}) \qquad \begin{pmatrix} T_{BC} \\ T_{BA} \end{pmatrix} = \begin{pmatrix} 39.24 \\ 67.97 \end{pmatrix} \text{N}$$



At *C*: $\quad T_{CD} = 1$ N $\quad F = 2$ N

Given

$$\xrightarrow{+} \Sigma F_x = 0; \quad -T_{BC}\cos(\theta_1) + T_{CD}\cos(\theta_3) = 0$$

$$+\uparrow \Sigma F_y = 0; \quad T_{BC}\sin(\theta_1) + T_{CD}\sin(\theta_3) - F = 0$$

$$\begin{pmatrix} T_{CD} \\ F \end{pmatrix} = \text{Find}(T_{CD}, F) \qquad \begin{pmatrix} T_{CD} \\ F \end{pmatrix} = \begin{pmatrix} 39.24 \\ 39.24 \end{pmatrix} \text{N}$$

Problem 3-18

The motor at *B* winds up the cord attached to the crate of weight *W* with a constant speed. Determine the force in cord *CD* supporting the pulley and the angle θ for equilibrium. Neglect the

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size of the pulley at *C*.

Given:

$W = 65$ lb $\quad c = 12 \quad d = 5$

Solution:

The initial guesses: $\theta = 100$ deg $\quad F_{CD} = 200$ lb

Given

Equations of Equilibrium:

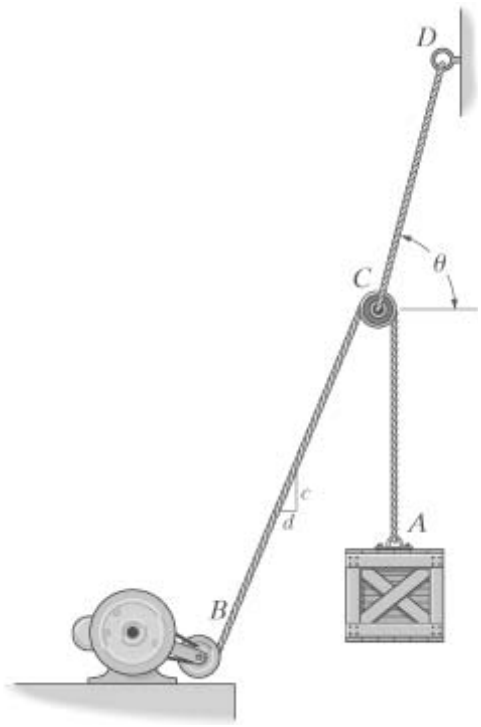
$$\overset{+}{\rightarrow} \Sigma F_x = 0; \quad F_{CD}\cos(\theta) - W\left(\frac{d}{\sqrt{c^2 + d^2}}\right) = 0$$

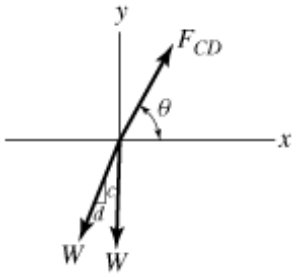
$$+\uparrow \Sigma F_y = 0; \quad F_{CD}\sin(\theta) - W\left(\frac{c}{\sqrt{c^2 + d^2}}\right) - W = 0$$

$$\begin{pmatrix} \theta \\ F_{CD} \end{pmatrix} = \text{Find}(\theta, F_{CD})$$

$\theta = 78.69$ deg

$F_{CD} = 127.5$ lb





Problem 3-19

The cords *BCA* and *CD* can each support a maximum load *T*. Determine the maximum weight of the crate that can be hoisted at constant velocity, and the angle θ for equilibrium.

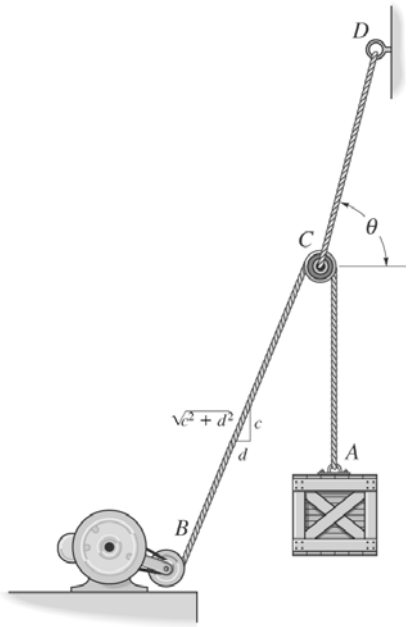
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Given:

$T = 100$ lb

$c = 12$

$d = 5$



The maximum will occur in *CD* rather than in *BCA*.

Solution :

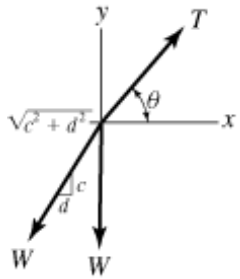
The initial guesses: $\theta = 100$ deg $W = 200$ lb

Given

Equations of Equilibrium:

$$\xrightarrow{+} \Sigma F_x = 0; \quad T\cos(\theta) - W\left(\frac{d}{\sqrt{c^2 + d^2}}\right) = 0$$

$$+\uparrow \Sigma F_y = 0; \quad T\sin(\theta) - W\left(\frac{c}{\sqrt{c^2 + d^2}}\right) - W = 0$$



$$\begin{pmatrix} \theta \\ W \end{pmatrix} = \text{Find}(\theta, W)$$

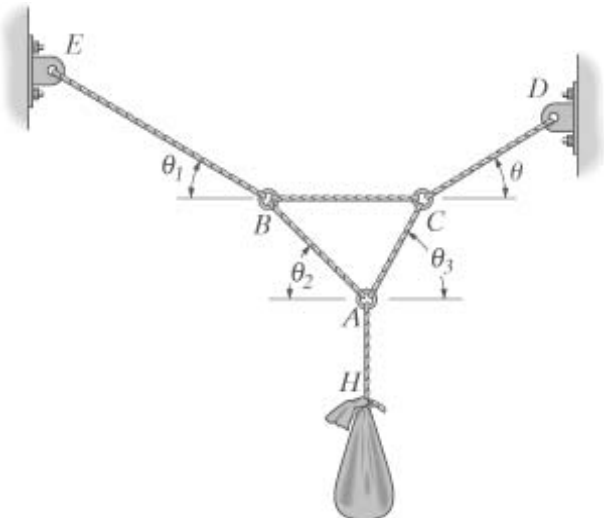
$\theta = 78.69$ deg

$W = 51.0$ lb

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Problem 3-20

The sack has weight W and is supported by the six cords tied together as shown. Determine the tension in each cord and the angle θ for equilibrium. Cord BC is horizontal.



Given:

$W = 15$ lb

$\theta_1 = 30$ deg

$\theta_2 = 45$ deg

$\theta_3 = 60$ deg

Solution:

Guesses $T_{BE} = 1$ lb $T_{AB} = 1$ lb

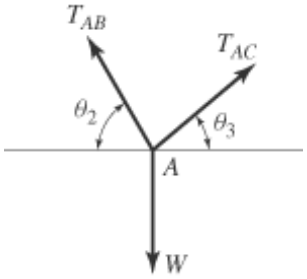
$\theta = 20$deg $T_{BC} = 1$ lb $T_{AC} = 1$ lb

$T_{CD} = 1$ lb $T_{AH} = 1$ lb

Given

At H:

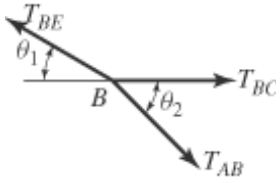
$+\uparrow \Sigma F_y = 0; \quad T_{AH} - W = 0$



At A:

$\rightarrow + \Sigma F_x = 0; \quad -T_{AB}\cos(\theta_2) + T_{AC}\cos(\theta_3) = 0$

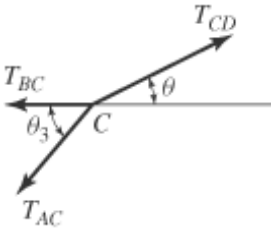
$+\uparrow \Sigma F_y = 0; \quad T_{AB}\sin(\theta_2) + T_{AC}\sin(\theta_3) - W = 0$



At B:

$\rightarrow + \Sigma F_x = 0; \quad T_{BC} - T_{BE}\cos(\theta_1) + T_{AB}\cos(\theta_2) = 0$

$+\uparrow \Sigma F_y = 0; \quad T_{BE}\sin(\theta_1) - T_{AB}\sin(\theta_2) = 0$



At C:

$\rightarrow + \Sigma F_x = 0; \quad T_{CD}\cos(\theta) - T_{BC} - T_{BE}\cos(\theta_3) = 0$

$+\uparrow \Sigma F_y = 0; \quad T_{CD}\sin(\theta) - T_{AC}\sin(\theta_3) = 0$

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$$\begin{pmatrix} T_{BE} \\ T_{AB} \\ T_{BC} \\ T_{AC} \\ T_{CD} \\ T_{AH} \\ \theta \end{pmatrix} = \text{Find}\left(T_{BE}, T_{AB}, T_{BC}, T_{AC}, T_{CD}, T_{AH}, \theta\right) \qquad \begin{pmatrix} T_{BE} \\ T_{AB} \\ T_{BC} \\ T_{AC} \\ T_{CD} \\ T_{AH} \end{pmatrix} = \begin{pmatrix} 10.98 \\ 7.76 \\ 4.02 \\ 10.98 \\ 13.45 \\ 15.00 \end{pmatrix} \text{lb}$$

$$\theta = 45.00 \text{ deg}$$

Problem 3-21

Each cord can sustain a maximum tension T. Determine the largest weight of the sack that can be supported. Also, determine θ of cord DC for equilibrium.

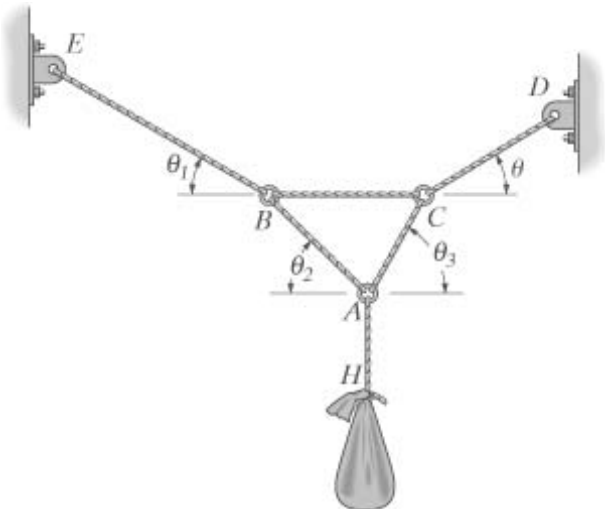
Given:

$T = 200$ lb

$\theta_1 = 30$ deg

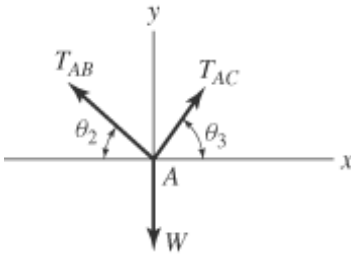
$\theta_2 = 45$ deg

$\theta_3 = 60$ deg



Solution:

Solve for $W = 1$ and then scale the answer at the end.



Guesses $T_{BE} = 1$ $T_{AB} = 1$

$T_{BC} = 1$ $T_{AC} = 1$

$T_{CD} = 1$ $T_{AH} = 1$

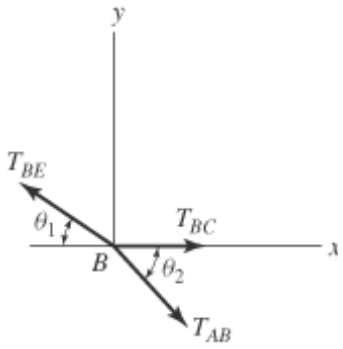
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$\theta = 20 \text{ deg}$

Given

At *H*:

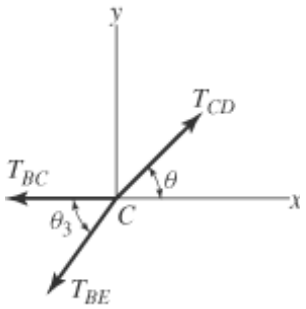
$+\uparrow \Sigma F_y = 0; \quad T_{AH} - W = 0$



At *A*:

$\overset{+}{\rightarrow} \Sigma F_x = 0; \quad -T_{AB}\cos(\theta_2) + T_{AC}\cos(\theta_3) = 0$

$+\uparrow \Sigma F_y = 0; \quad T_{AB}\sin(\theta_2) + T_{AC}\sin(\theta_3) - W = 0$



At *B*:

$\overset{+}{\rightarrow} \Sigma F_x = 0; \quad T_{BC} - T_{BE}\cos(\theta_1) + T_{AB}\cos(\theta_2) = 0$

$+\uparrow \Sigma F_y = 0; \quad T_{BE}\sin(\theta_1) - T_{AB}\sin(\theta_2) = 0$

At *C*:

$\overset{+}{\rightarrow} \Sigma F_x = 0; \quad T_{CD}\cos(\theta) - T_{BC} - T_{BE}\cos(\theta_3) = 0$

$+\uparrow \Sigma F_y = 0; \quad T_{CD}\sin(\theta) - T_{AC}\sin(\theta_3) = 0$

$$\begin{pmatrix} T_{BE} \\ T_{AB} \\ T_{BC} \\ T_{AC} \\ T_{CD} \\ T_{AH} \\ \theta \end{pmatrix} = \text{Find}(T_{BE}, T_{AB}, T_{BC}, T_{AC}, T_{CD}, T_{AH}, \theta) \qquad \begin{pmatrix} T_{BE} \\ T_{AB} \\ T_{BC} \\ T_{AC} \\ T_{CD} \\ T_{AH} \end{pmatrix} = \begin{pmatrix} 0.73 \\ 0.52 \\ 0.27 \\ 0.73 \\ 0.90 \\ 1.00 \end{pmatrix}$$

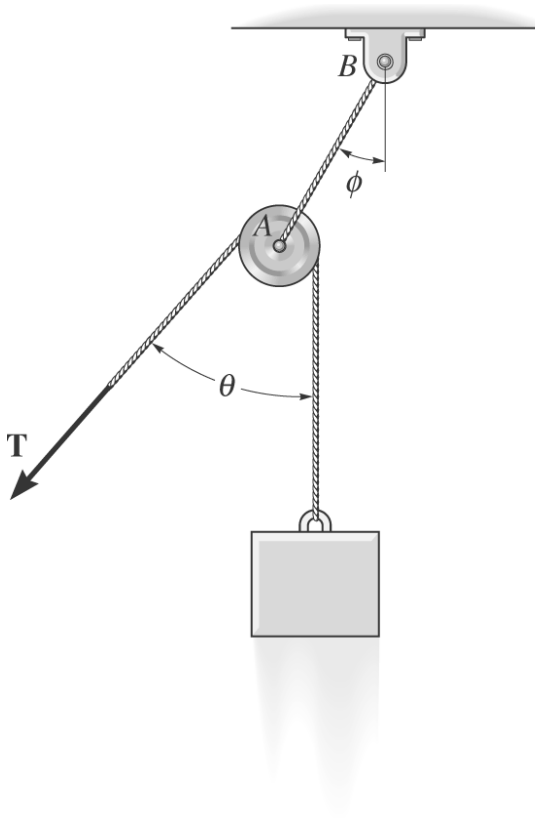
$$W = \frac{T}{\max(T_{BE}, T_{AB}, T_{BC}, T_{AC}, T_{CD}, T_{AH})} \qquad W = 200.00 \text{ lb}$$

$\theta = 45.00 \text{ deg}$

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Problem 3-22

The block has weight W and is being hoisted at uniform velocity. Determine the angle θ for equilibrium and the required force in each cord.



Given:

$W = 20$ lb

$\phi = 30$ deg

Solution:

The initial guesses:

$\theta = 10$ deg $\quad T_{AB} = 50$ lb

Given

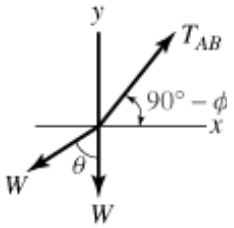
$T_{AB}\sin(\phi) - W\sin(\theta) = 0$

$T_{AB}\cos(\phi) - W - W\cos(\theta) = 0$

$$\begin{pmatrix} \theta \\ T_{AB} \end{pmatrix} = \text{Find}(\theta, T_{AB})$$

$\theta = 60.00$ deg

$T_{AB} = 34.6$ lb



Problem 3-23

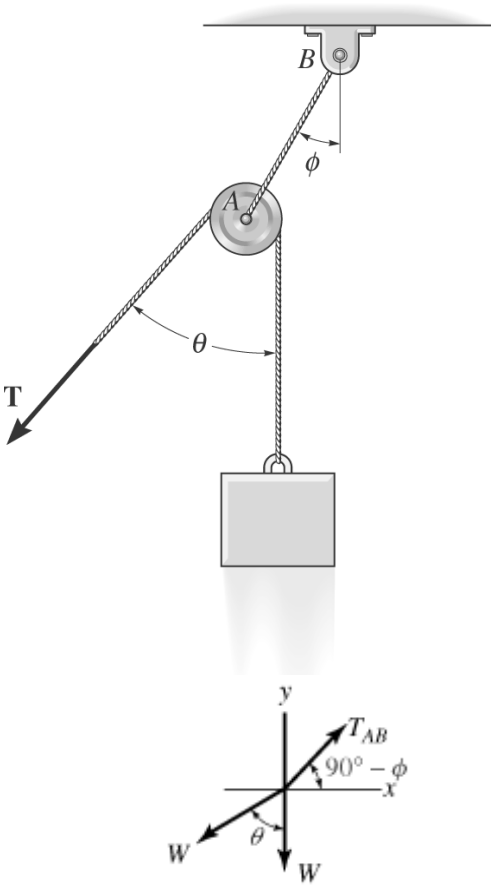
Determine the maximum weight W of the block that can be suspended in the position shown if each cord can support a maximum tension T. Also, what is the angle θ for equilibrium?

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Given:

$T = 80\ \text{lb}$

$\phi = 30\ \text{deg}$



The maximum load will occur in cord *AB*.

Solution:

$T_{AB} = T$

The initial guesses:

$\theta = 100\text{deg} \quad W = 200\text{lb}$

Given

$+\uparrow \Sigma F_y = 0; \qquad T_{AB}\cos(\phi) - W - W\cos(\theta) = 0$

$\xrightarrow{+} \Sigma F_x = 0; \qquad T_{AB}\sin(\phi) - W\sin(\theta) = 0$

$\begin{pmatrix} \theta \\ W \end{pmatrix} = \text{Find}(\theta, W)$

$W = 46.19\,\text{lb} \qquad \theta = 60.00\,\text{deg}$

Problem 3-24

Two spheres *A* and *B* have an equal mass *M* and are electrostatically charged such that the repulsive force acting between them has magnitude *F* and is directed along line *AB*. Determine the angle θ, the tension in cords *AC* and *BC*, and the mass *M* of each sphere.

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Unit used:

$\text{mN} = 10^{-3}\ \text{N}$

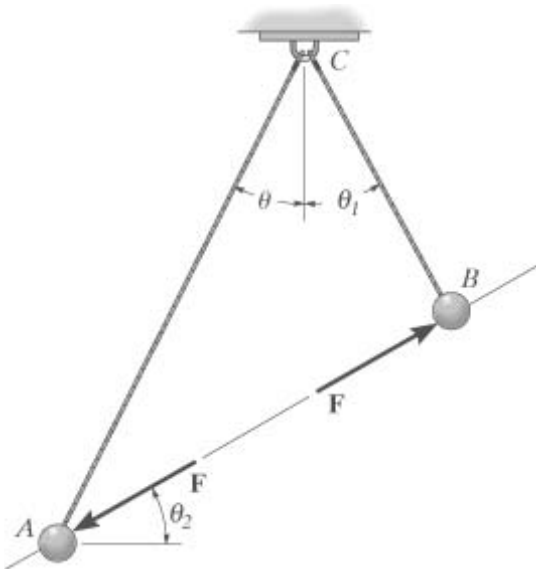
Given:

$F = 20\ \text{mN}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$

$\theta_1 = 30\ \text{deg}$

$\theta_2 = 30\ \text{deg}$



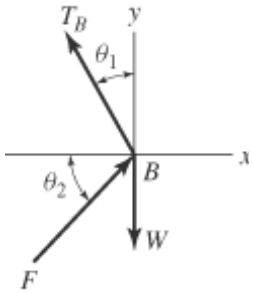
Solution:

Guesses $T_B = 1\ \text{mN}$ $M = 1\ \text{gm}$

$T_A = 1\ \text{mN}$ $\theta = 30\ \text{deg}$

Given

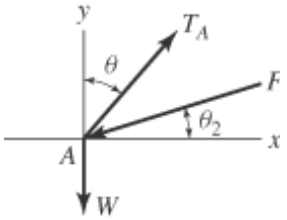
For *B:*



$\xrightarrow{+} \Sigma F_x = 0;\quad F\cos(\theta_2) - T_B\sin(\theta_1) = 0$

$+\uparrow \Sigma F_y = 0;\quad F\sin(\theta_2) + T_B\cos(\theta_1) - Mg = 0$

For *A*:



$\xrightarrow{+} \Sigma F_x = 0;\quad T_A\sin(\theta) - F\cos(\theta_2) = 0$

$+\uparrow \Sigma F_y = 0;\quad T_A\cos(\theta) - F\sin(\theta_2) - Mg = 0$

$\begin{pmatrix} T_A \\ T_B \\ \theta \\ M \end{pmatrix} = \text{Find}(T_A, T_B, \theta, M)$ $\begin{pmatrix} T_A \\ T_B \end{pmatrix} = \begin{pmatrix} 52.92 \\ 34.64 \end{pmatrix}\text{mN}$ $\theta = 19.11\ \text{deg}$ $M = 4.08\ \text{gm}$

Problem 3-25

Blocks *D* and *F* weigh W_1 each and block *E* weighs W_2. Determine the sag *s* for equilibrium. Neglect the size of the pulleys.

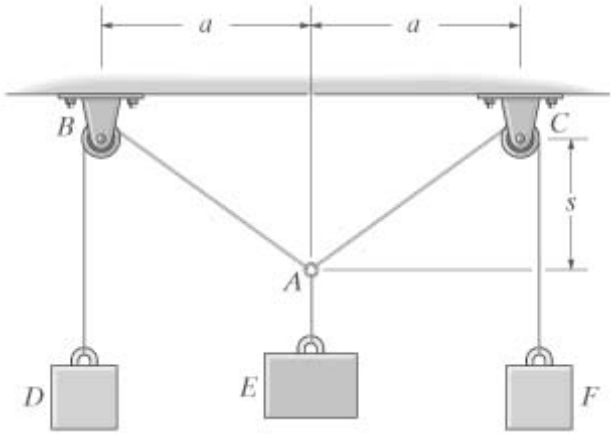
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Given:

$W_1 = 5$ lb

$W_2 = 8$ lb

$a = 4$ ft

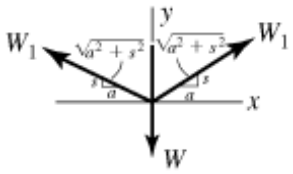


Solution:

Sum forces in the y direction

Guess $s = 1$ ft

Given $2\left(\frac{s}{\sqrt{s^2 + a^2}}\right) W_1 - W_2 = 0$ $s = \text{Find}(s)$ $s = 5.33\,\text{ft}$



Problem 3-26

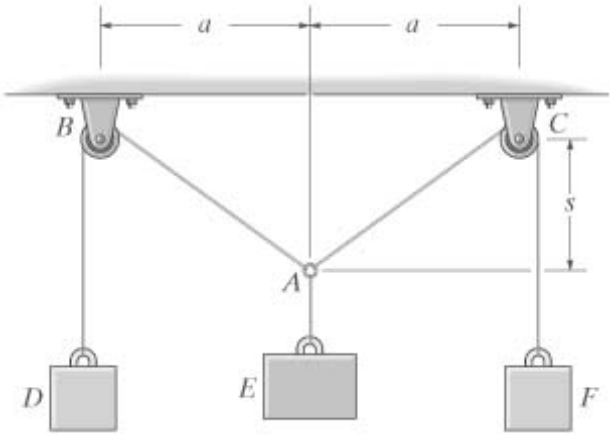
If blocks D and F each have weight W_1, determine the weight of block E if the sag is s. Neglect the size of the pulleys.

Given:

$W_1 = 5$ lb

$s = 3$ ft

$a = 4$ ft



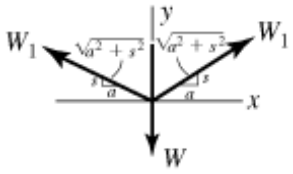
Solution:

Sum forces in the y direction

$$2\left(\frac{s}{\sqrt{s^2 + a^2}}\right) W_1 - W = 0$$

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$$W = 2\left(\frac{s}{\sqrt{s^2 + a^2}}\right)W_1 \qquad W = 6.00\,\text{lb}$$



Problem 3-27

The block of mass M is supported by two springs having the stiffness shown. Determine the unstretched length of each spring.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$M = 30\ \text{kg}$

$l_1 = 0.6\ \text{m}$

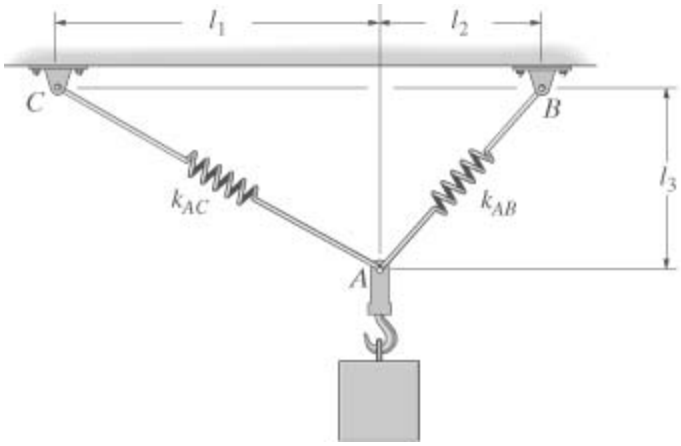
$l_2 = 0.4\ \text{m}$

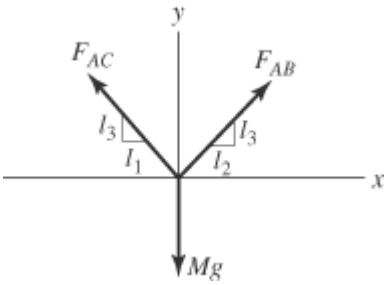
$l_3 = 0.5\ \text{m}$

$k_{AC} = 1.5\ \frac{\text{kN}}{\text{m}}$

$k_{AB} = 1.2\ \frac{\text{kN}}{\text{m}}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$





Solution:

Initial guesses: $F_{AC} = 20\ \text{N}$ $F_{AB} = 30\ \text{N}$

Given

$$\xrightarrow{+} \Sigma F_x = 0; \qquad \frac{l_2 F_{AB}}{\sqrt{l_2^2 + l_3^2}} - \frac{l_1 F_{AC}}{\sqrt{l_1^2 + l_3^2}} = 0$$

$$+\uparrow \Sigma F_y = 0; \qquad \frac{l_3 F_{AB}}{\sqrt{l_2^2 + l_3^2}} + \frac{l_3 F_{AC}}{\sqrt{l_1^2 + l_3^2}} - M\,g = 0$$

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$$\begin{pmatrix} F_{AC} \\ F_{AB} \end{pmatrix} = \text{Find}(F_{AC}, F_{AB}) \qquad \begin{pmatrix} F_{AC} \\ F_{AB} \end{pmatrix} = \begin{pmatrix} 183.88 \\ 226.13 \end{pmatrix} \text{N}$$

Then guess $L_{AB} = 0.1$ m $L_{AC} = 0.1$ m

Given $F_{AC} = k_{AC}\left(\sqrt{l_1^2 + l_3^2} - L_{AC}\right)$

$$F_{AB} = k_{AB}\left(\sqrt{l_2^2 + l_3^2} - L_{AB}\right)$$

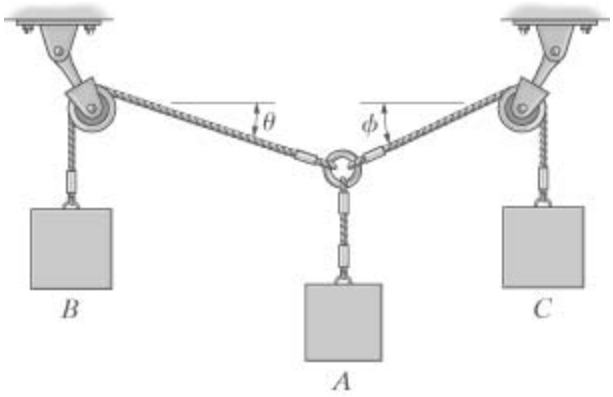
$$\begin{pmatrix} L_{AB} \\ L_{AC} \end{pmatrix} = \text{Find}(L_{AB}, L_{AC}) \qquad \begin{pmatrix} L_{AB} \\ L_{AC} \end{pmatrix} = \begin{pmatrix} 0.452 \\ 0.658 \end{pmatrix} \text{m}$$

Problem 3-28

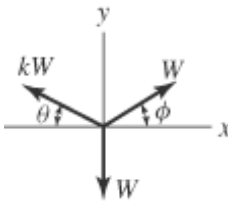
Three blocks are supported using the cords and two pulleys. If they have weights of $W_A = W_C = W$, $W_B = kW$, determine the angle θ for equilibrium.

Given:

$k = 0.25$



Solution:



$\overset{+}{\rightarrow} \Sigma F_x = 0;$ $\quad W\cos(\phi) - kW\cos(\theta) = 0$

$+\uparrow \Sigma F_y = 0;$ $\quad W\sin(\phi) + kW\sin(\theta) - W = 0$

$$\cos(\phi) = k\cos(\theta)$$

$$\sin(\phi) = 1 - k\sin(\theta)$$

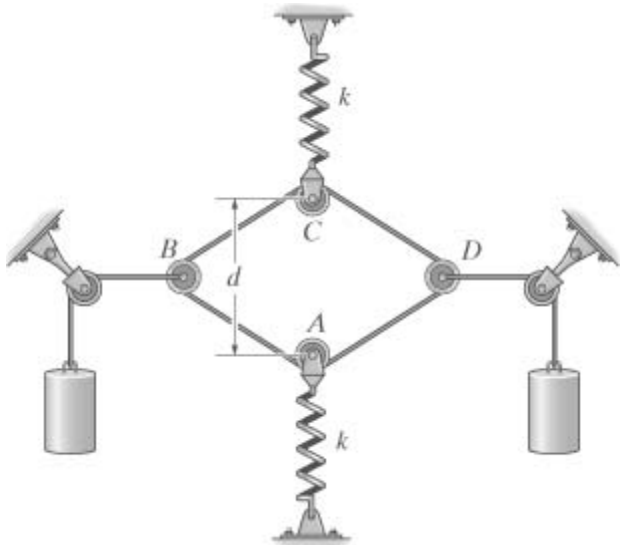
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$$1 = k^2 \cos(\theta)^2 + (1 - k\sin(\theta))^2 = 1 + k^2 - 2k\sin(\theta)$$

$$\theta = \operatorname{asin}\left(\frac{k}{2}\right) \qquad \theta = 7.18\,\text{deg}$$

Problem 3-29

A continuous cable of total length l is wrapped around the *small* pulleys at *A*, *B*, *C*, and *D*. If each spring is stretched a distance *b*, determine the mass *M* of each block. Neglect the weight of the pulleys and cords. The springs are unstretched when *d = l/2*.



Given:

$l = 4\ \text{m}$

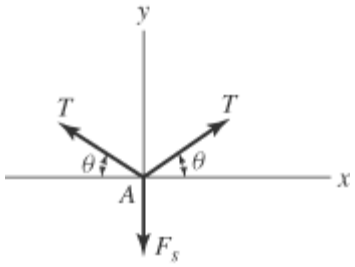
$k = 500\ \frac{\text{N}}{\text{m}}$

$b = 300\ \text{mm}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$

Solution:

$F_S = k\,b \qquad F_S = 150.00\ \text{N}$



Guesses

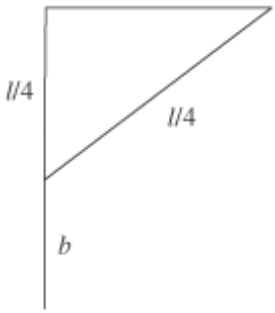
$T = 1\text{N} \quad \theta = 10\ \text{deg} \quad M = 1\ \text{kg}$

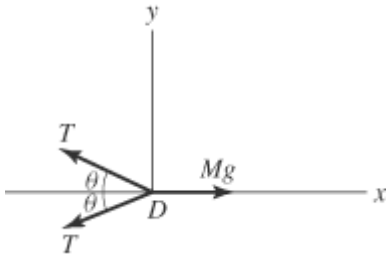
Given

$2T\sin(\theta) - F_S = 0$

$-2T\cos(\theta) + M\,g = 0$

$b + \frac{l}{4}\sin(\theta) = \frac{l}{4}$





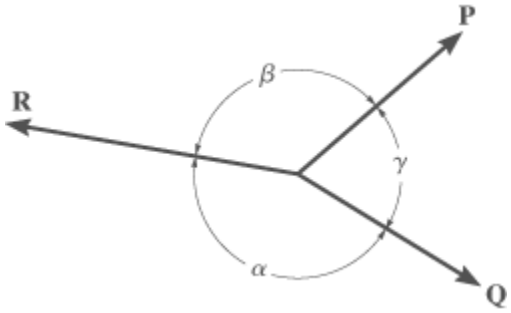
$$\begin{pmatrix} T \\ \theta \\ M \end{pmatrix} = \text{Find}(T, \theta, M)$$

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$T = 107.14\ \text{N}$ $\theta = 44.43\,\text{deg}$ $M = 15.60\ \text{kg}$

Problem 3-30

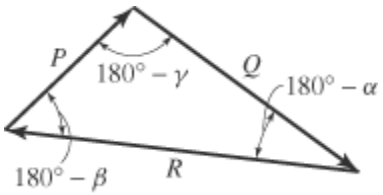
Prove *Lami's theorem*, which states that if three concurrent forces are in equilibrium, each is proportional to the sine of the angle of the other two; that is, $P/\sin\alpha = Q/\sin\beta = R/\sin\gamma$.



Solution:

Sine law:

$$\frac{R}{\sin(180\text{deg} - \gamma)} = \frac{Q}{\sin(180\text{deg} - \beta)} = \frac{P}{\sin(180\text{deg} - \alpha)}$$

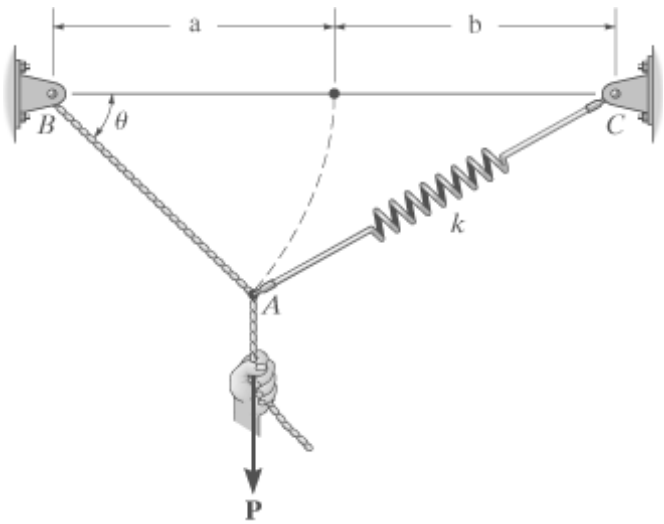


However, in general $\sin(180\text{deg} - \phi) = \sin(\phi)$, hence

$$\frac{R}{\sin(\gamma)} = \frac{Q}{\sin(\beta)} = \frac{P}{\sin(\alpha)}$$ Q.E.D.

Problem 3-31

A vertical force **P** is applied to the ends of cord *AB* of length *a* and spring *AC*. If the spring has an unstretched length δ, determine the angle θ for equilibrium.



Given:

$P = 10\ \text{lb}$

$\delta = 2\ \text{ft}$

$k = 15\ \dfrac{\text{lb}}{\text{ft}}$

$a = 2\ \text{ft}$

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$b = 2$ ft

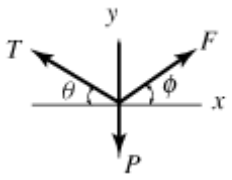
Guesses

$\theta = 10$ deg $\quad \phi = 10$ deg

$T = 1$ lb $\quad F = 1$ lb

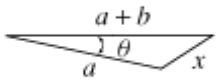
$x = 1$ft

Given



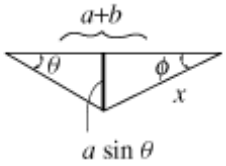
$-T\cos(\theta) + F\cos(\phi) = 0$

$T\sin(\theta) + F\sin(\phi) - P = 0$



$F = k(x - \delta)$

$a\sin(\theta) = x\sin(\phi)$



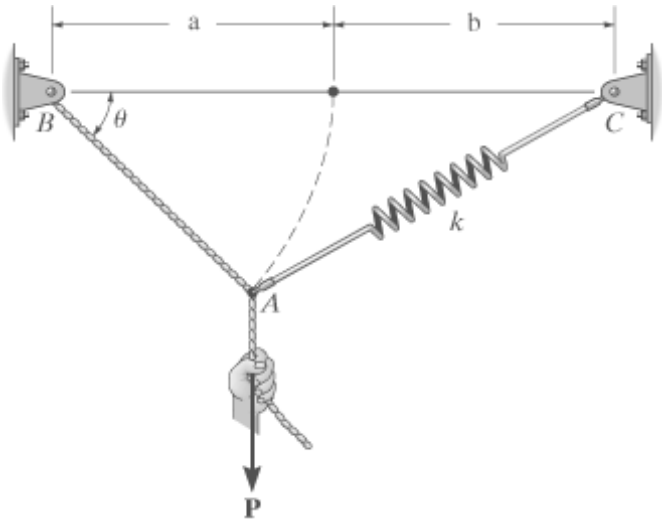
$a\cos(\theta) + x\cos(\phi) = a + b$

$$\begin{pmatrix} \theta \\ \phi \\ T \\ F \\ x \end{pmatrix} = \text{Find}(\theta, \phi, T, F, x) \qquad \begin{pmatrix} T \\ F \end{pmatrix} = \begin{pmatrix} 10.30 \\ 9.38 \end{pmatrix} \text{lb} \qquad \theta = 35\,\text{deg}$$

Problem 3-32

Determine the unstretched length δ of spring *AC* if a force **P** causes the angle θ for equilibrium. Cord *AB* has length *a*.

Given:



$P = 80$ lb

$\theta = 60$ deg

$k = 50\ \dfrac{\text{lb}}{\text{ft}}$

$a = 2$ ft

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$b = 2$ ft

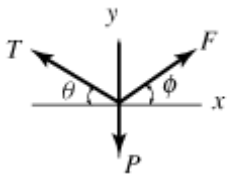
Guesses

$\delta = 1$ft $\quad \phi = 10$ deg

$T = 1$ lb $\quad F = 1$ lb

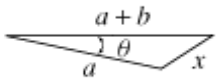
$x = 1$ft

Given



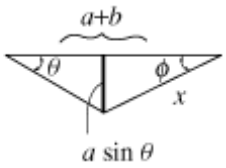
$-T\cos(\theta) + F\cos(\phi) = 0$

$T\sin(\theta) + F\sin(\phi) - P = 0$



$F = k(x - \delta)$

$a\sin(\theta) = x\sin(\phi)$

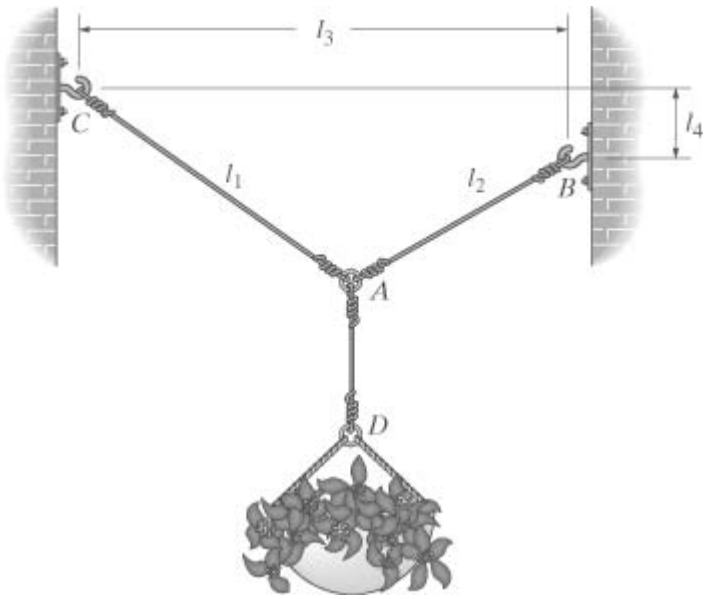


$a\cos(\theta) + x\cos(\phi) = a + b$

$$\begin{pmatrix} \delta \\ \phi \\ T \\ F \\ x \end{pmatrix} = \text{Find}(\delta, \phi, T, F, x) \qquad \begin{pmatrix} T \\ F \end{pmatrix} = \begin{pmatrix} 69.28 \\ 40.00 \end{pmatrix} \text{lb} \qquad \delta = 2.66\,\text{ft}$$

Problem 3-33

The flowerpot of mass M is suspended from three wires and supported by the hooks at B and C. Determine the tension in AB and AC for equilibrium.



Given:

$M = 20$ kg

$l_1 = 3.5$ m

$l_2 = 2$ m

$l_3 = 4$ m

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$l_4 = 0.5\ \text{m}$

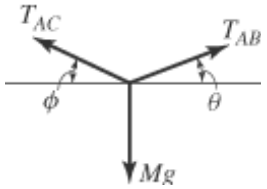
$g = 9.81\,\frac{\text{m}}{\text{s}^2}$

Solution:

Initial guesses:

$T_{AB} = 1\ \text{N}$ $T_{AC} = 1\ \text{N}$

$\theta = 10\ \text{deg}$ $\phi = 10\ \text{deg}$



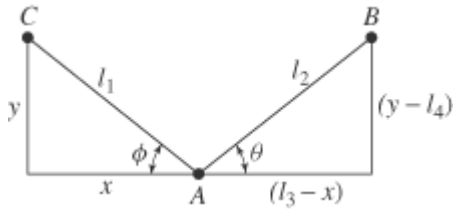
Given

$-T_{AC}\cos(\phi) + T_{AB}\cos(\theta) = 0$

$T_{AC}\sin(\phi) + T_{AB}\sin(\theta) - Mg = 0$

$l_1\cos(\phi) + l_2\cos(\theta) = l_3$

$l_1\sin(\phi) = l_2\sin(\theta) + l_4$



$$\begin{pmatrix} T_{AB} \\ T_{AC} \\ \theta \\ \phi \end{pmatrix} = \text{Find}(T_{AB}, T_{AC}, \theta, \phi) \qquad \begin{pmatrix} \theta \\ \phi \end{pmatrix} = \begin{pmatrix} 53.13 \\ 36.87 \end{pmatrix}\text{deg} \qquad \begin{pmatrix} T_{AB} \\ T_{AC} \end{pmatrix} = \begin{pmatrix} 156.96 \\ 117.72 \end{pmatrix}\text{N}$$

Problem 3-34

A car is to be towed using the rope arrangement shown. The towing force required is P. Determine the minimum length l of rope AB so that the tension in either rope AB or AC does not exceed T. *Hint:* Use the equilibrium condition at point A to determine the required angle θ for attachment, then determine l using trigonometry applied to triangle ABC.

Given:

$P = 600\ \text{lb}$

$T = 750\ \text{lb}$

$\phi = 30\ \text{deg}$

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$d = 4$ ft

Solution:

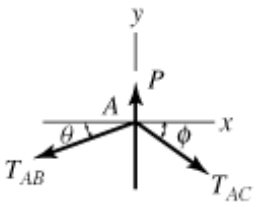
The initial guesses

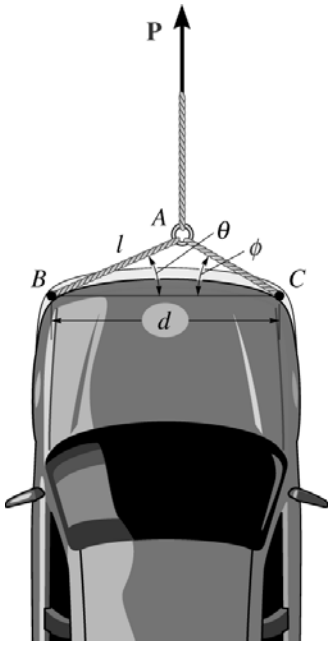
$T_{AB} = T$

$T_{AC} = T$

$\theta = 30$ deg

$l = 2$ ft





Case 1: Assume $T_{AC} = T$

Given

$\xrightarrow{+} \Sigma F_x = 0; \quad T_{AC}\cos(\phi) - T_{AB}\cos(\theta) = 0$

$+\uparrow \Sigma F_y = 0; \quad P - T_{AC}\sin(\phi) - T_{AB}\sin(\theta) = 0$

$$\frac{l}{\sin(\phi)} = \frac{d}{\sin(180\text{deg} - \theta - \phi)}$$

$$\begin{pmatrix} T_{AB} \\ \theta \\ l_1 \end{pmatrix} = \text{Find}(T_{AB}, \theta, l) \qquad T_{AB} = 687.39\,\text{lb} \qquad \theta = 19.11\,\text{deg} \qquad l_1 = 2.65\,\text{ft}$$

Case 2: Assume $T_{AB} = T$

Given

$\xrightarrow{+} \Sigma F_x = 0; \quad T_{AC}\cos(\phi) - T_{AB}\cos(\theta) = 0$

$+\uparrow \Sigma F_y = 0; \quad P - T_{AC}\sin(\phi) - T_{AB}\sin(\theta) = 0$

$$\frac{l}{\sin(\phi)} = \frac{d}{\sin(180\text{deg} - \theta - \phi)}$$

$$\begin{pmatrix} T_{AC} \\ \theta \\ l_2 \end{pmatrix} = \text{Find}(T_{AC}, \theta, l) \qquad T_{AC} = 840.83\,\text{lb} \qquad \theta = 13.85\,\text{deg} \qquad l_2 = 2.89\,\text{ft}$$

$l = \min(l_1, l_2) \qquad l = 2.65\,\text{ft}$

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Problem 3-35

Determine the mass of each of the two cylinders if they cause a sag of distance d when suspended from the rings at A and B. Note that $s = 0$ when the cylinders are removed.

Given:

$d = 0.5$ m

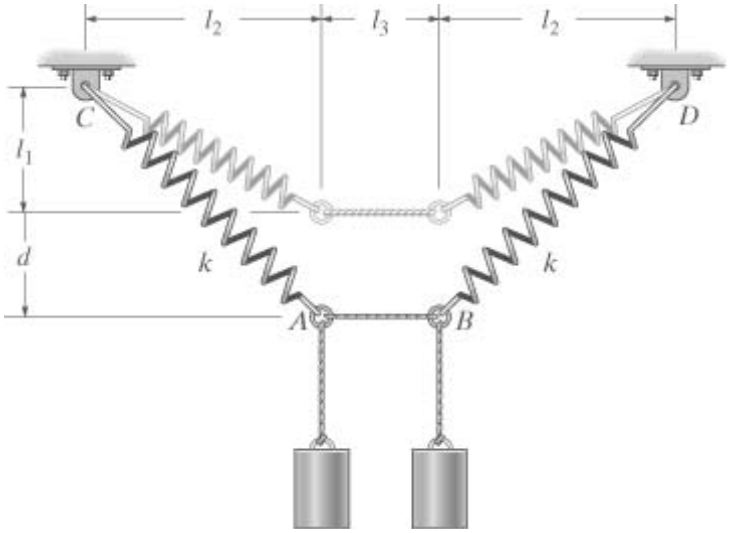
$l_1 = 1.5$ m

$l_2 = 2$ m

$l_3 = 1$ m

$k = 100 \dfrac{\text{N}}{\text{m}}$

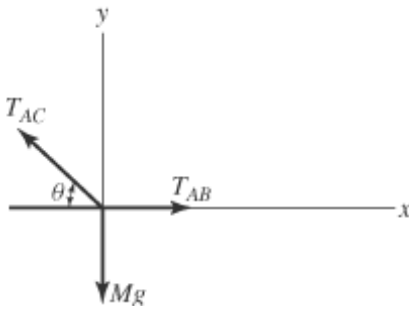
$g = 9.81 \dfrac{\text{m}}{\text{s}^2}$



Solution:

$$T_{AC} = k\left[\sqrt{\left(l_1 + d\right)^2 + {l_2}^2} - \sqrt{{l_1}^2 + {l_2}^2}\right]$$

$T_{AC} = 32.84$ N



$$\theta = \operatorname{atan}\left(\frac{l_1 + d}{l_2}\right)$$

$\theta = 45\,\text{deg}$

$$M = \frac{T_{AC}\sin(\theta)}{g}$$

$M = 2.37$ kg

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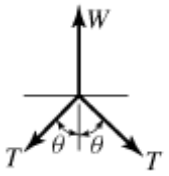
Problem 3-36

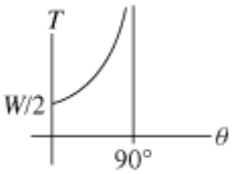
The sling *BAC* is used to lift the load *W* with constant velocity. Determine the force in the sling and plot its value *T* (ordinate) as a function of its orientation θ, *where* $0 \le \theta \le 90°$.

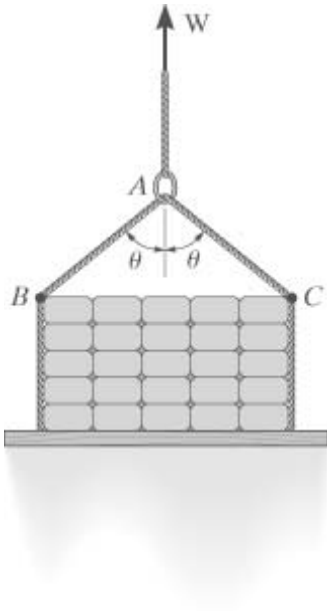
Solution:

$$W - 2T\cos(\theta) = 0$$

$$T = \frac{1}{2}\left(\frac{W}{\cos(\theta)}\right)$$







Problem 3-37

The lamp fixture has weight *W* and is suspended from two springs, each having unstretched length *L* and stiffness *k*. Determine the angle θ for equilibrium.

Units Used:

$$\text{kN} = 10^3\ \text{N}$$

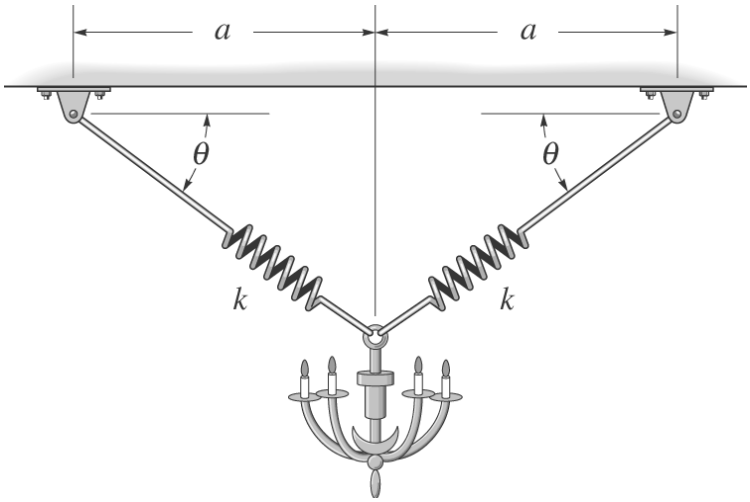
Given:

$$W = 10\ \text{lb}$$

$$L = 4\ \text{ft}$$

$$k = 5\ \frac{\text{lb}}{\text{ft}}$$

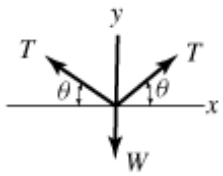
$$a = 4\ \text{ft}$$



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Solution:

The initial guesses: $T = 200\ \text{lb} \quad \theta = 10\ \text{deg}$



Given

Spring $T = k\left(\frac{a}{\cos(\theta)} - L\right)$

$+\uparrow \Sigma F_y = 0; \qquad 2T\sin(\theta) - W = 0$

$\begin{pmatrix} T \\ \theta \end{pmatrix} = \text{Find}(T, \theta) \qquad T = 7.34\,\text{lb} \qquad \theta = 42.97\,\text{deg}$

Problem 3-38

The uniform tank of weight W is suspended by means of a cable, of length l, which is attached to the sides of the tank and passes over the small pulley located at O. If the cable can be attached at either points A and B, or C and D, determine which attachment produces the least amount of tension in the cable.What is this tension?

Given:

$W = 200$ lb

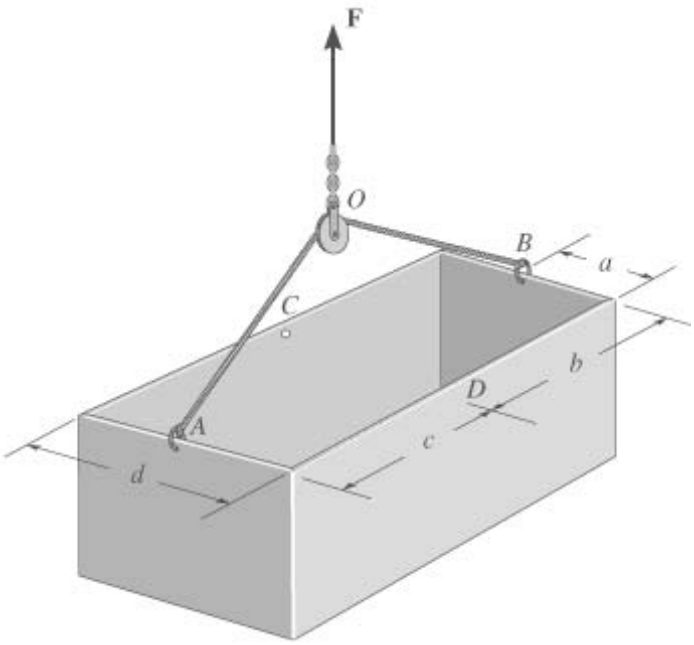
$l = 6$ ft

$a = 1$ ft

$b = 2$ ft

$c = b$

$d = 2a$

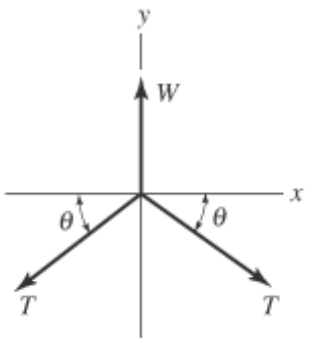


Solution:

Free Body Diagram: By observation, the force **F** has to support the entire weight of the tank. Thus, $F = W$. The tension in

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cable is the same throughout the cable.



Equations of Equilibrium

$\Sigma F_y = 0; \qquad W - 2T\sin(\theta) = 0$

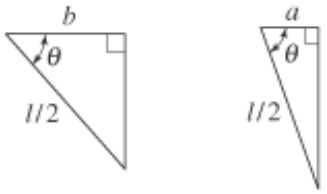
Attached to CD $\qquad \theta_1 = \text{acos}\left(\frac{2\,a}{l}\right) \qquad \theta_1 = 70.53\,\text{deg}$

Attached to AB $\qquad \theta_2 = \text{acos}\left(\frac{2\,b}{l}\right) \qquad \theta_2 = 48.19\,\text{deg}$

We choose the largest angle (which will produce the smallest force)

$\theta = \max(\theta_1, \theta_2) \qquad \theta = 70.53\,\text{deg}$

$T = \frac{1}{2}\left(\frac{W}{\sin(\theta)}\right) \qquad T = 106\,\text{lb}$



Problem 3-39

A sphere of mass m_s rests on the smooth parabolic surface. Determine the normal force it exerts on the surface and the mass m_B of block B needed to hold it in the equilibrium position shown.

Given:

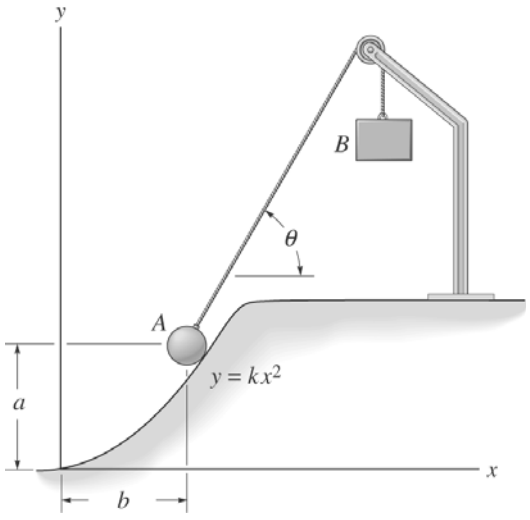
$m_S = 4\ \text{kg}$

$a = 0.4\ \text{m}$

$b = 0.4\ \text{m}$

$\theta = 60\ \text{deg}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$



Solution:

$k = \frac{a}{b^2}$

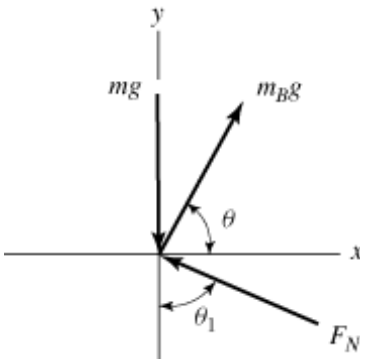
Geometry: The angle θ_1 which the surface make with the horizontal is to be determined first.

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$\tan(\theta_1) = \frac{dy}{dx} = 2\,k\,x$ evaluated at $x = a$ $\theta_1 = \text{atan}(2\,k\,a)$ $\theta_1 = 63.43\,\text{deg}$

Free Body Diagram : The tension in the cord is the same throughout the cord and is equal to the weight of block *B*, $m_B g$.

The initial guesses: $m_B = 200\ \text{kg}$ $F_N = 200\ \text{N}$



Given

$\xrightarrow{+} \Sigma F_x = 0; \quad m_B g \cos(\theta) - F_N \sin(\theta_1) = 0$

$+\uparrow \Sigma F_y = 0; \quad m_B g \sin(\theta) + F_N \cos(\theta_1) - m_s g = 0$

$\begin{pmatrix} m_B \\ F_N \end{pmatrix} = \text{Find}(m_B, F_N)$ $F_N = 19.66\ \text{N}$

$m_B = 3.58\ \text{kg}$

Problem 3-40

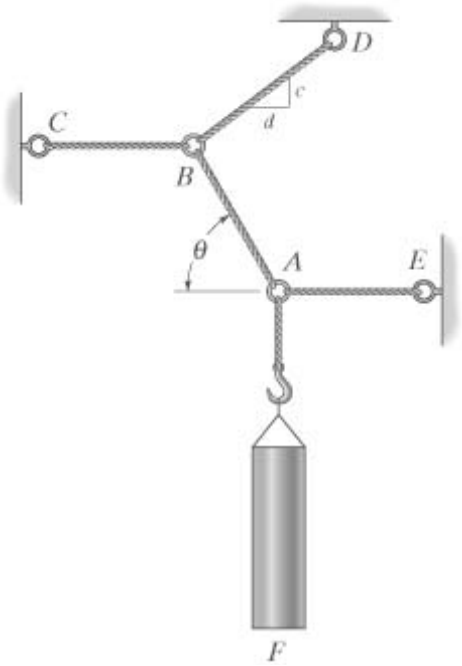
The pipe of mass *M* is supported at *A* by a system of five cords. Determine the force in each cord for equilibrium.

Given:

$M = 30\ \text{kg}$ $c = 3$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$ $d = 4$

$\theta = 60\ \text{deg}$



Solution:

Initial guesses:

$T_{AB} = 1\ \text{N}$ $T_{AE} = 1\ \text{N}$

$T_{BC} = 1\ \text{N}$ $T_{BD} = 1\ \text{N}$

Given

$T_{AB} \sin(\theta) - M g = 0$

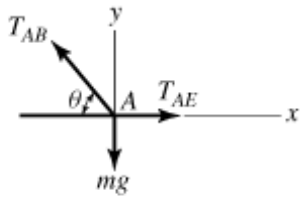
$T_{AE} - T_{AB} \cos(\theta) = 0$

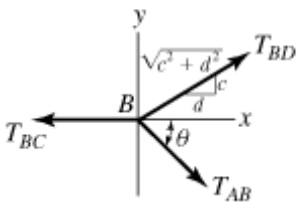
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$$T_{BD}\left(\frac{c}{\sqrt{c^2+d^2}}\right) - T_{AB}\sin(\theta) = 0$$

$$T_{BD}\left(\frac{d}{\sqrt{c^2+d^2}}\right) + T_{AB}\cos(\theta) - T_{BC} = 0$$

$$\begin{pmatrix} T_{AB} \\ T_{AE} \\ T_{BC} \\ T_{BD} \end{pmatrix} = \text{Find}\left(T_{AB}, T_{AE}, T_{BC}, T_{BD}\right)$$





$$\begin{pmatrix} T_{AB} \\ T_{AE} \\ T_{BC} \\ T_{BD} \end{pmatrix} = \begin{pmatrix} 339.8 \\ 169.9 \\ 562.3 \\ 490.5 \end{pmatrix} \text{ N}$$

Problem 3-41

The joint of a space frame is subjected to four forces. Strut *OA* lies in the *x-y* plane and strut *OB* lies in the *y-z* plane. Determine the forces acting in each of the three struts required for equilibrium.

Units Used:

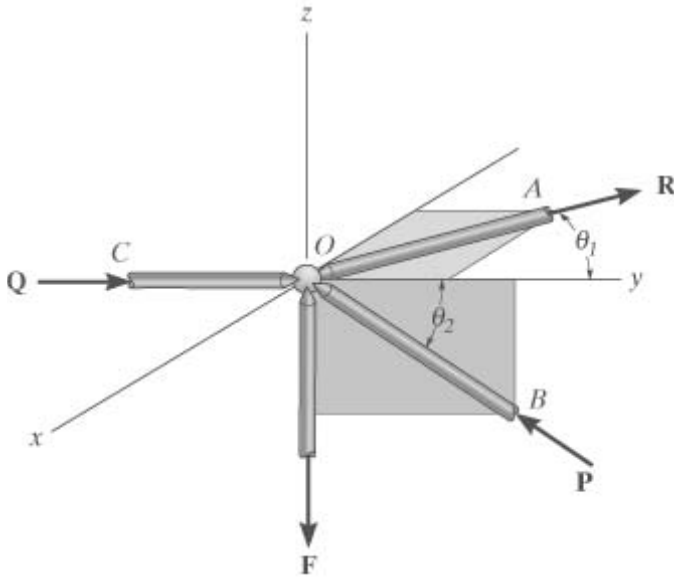
$$\text{kN} = 10^3 \text{ N}$$

Given:

$$F = 2 \text{ kN}$$

$$\theta_1 = 45 \text{ deg}$$

$$\theta_2 = 40 \text{ deg}$$



Solution:

$$\Sigma F_x = 0; \qquad -R\sin(\theta_1) = 0$$

$$R = 0$$

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$\Sigma F_z = 0; \qquad P \sin(\theta_2) - F = 0$

$$P = \frac{F}{\sin(\theta_2)}$$

$P = 3.11\,\text{kN}$

$\Sigma F_y = 0; \qquad Q - P \cos(\theta_2) = 0$

$$Q = P \cos(\theta_2)$$

$Q = 2.38\,\text{kN}$

Problem 3-42

Determine the magnitudes of $\mathbf{F_1}$, $\mathbf{F_2}$, and $\mathbf{F_3}$ for equilibrium of the particle.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F_4 = 800\ \text{N}$

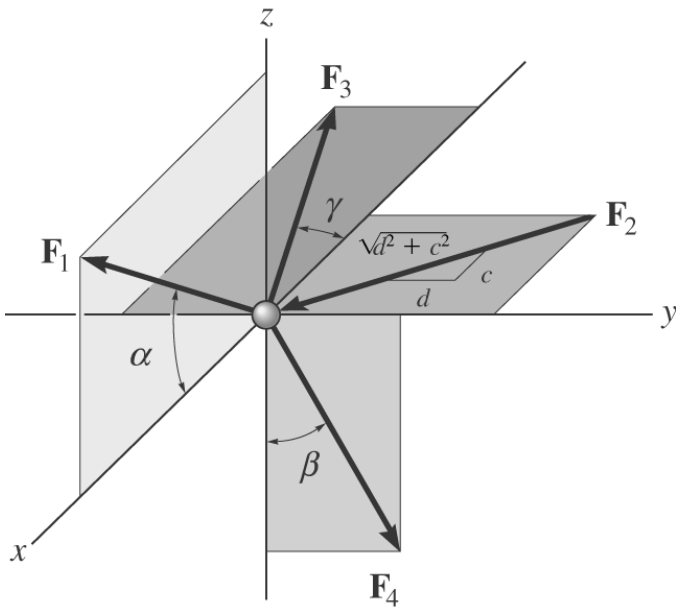
$\alpha = 60\ \text{deg}$

$\beta = 30\ \text{deg}$

$\gamma = 30\ \text{deg}$

$c = 3$

$d = 4$



Solution:

The initial guesses: $F_1 = 100\ \text{N} \quad F_2 = 100\ \text{N} \quad F_3 = 100\ \text{N}$

Given

$$F_1 \begin{pmatrix} \cos(\alpha) \\ 0 \\ \sin(\alpha) \end{pmatrix} + \frac{F_2}{\sqrt{c^2 + d^2}} \begin{pmatrix} c \\ -d \\ 0 \end{pmatrix} + F_3 \begin{pmatrix} -\cos(\gamma) \\ -\sin(\gamma) \\ 0 \end{pmatrix} + F_4 \begin{pmatrix} 0 \\ \sin(\beta) \\ -\cos(\beta) \end{pmatrix} = 0$$

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$$\begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = \text{Find}(F_1, F_2, F_3) \qquad \begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = \begin{pmatrix} 800 \\ 147 \\ 564 \end{pmatrix} \text{N}$$

Problem 3-43

Determine the magnitudes of $\mathbf{F_1}$, $\mathbf{F_2}$, and $\mathbf{F_3}$ for equilibrium of the particle.

Units Used:

kN = 1000 N

Given:

$F_4 = 8.5$ kN

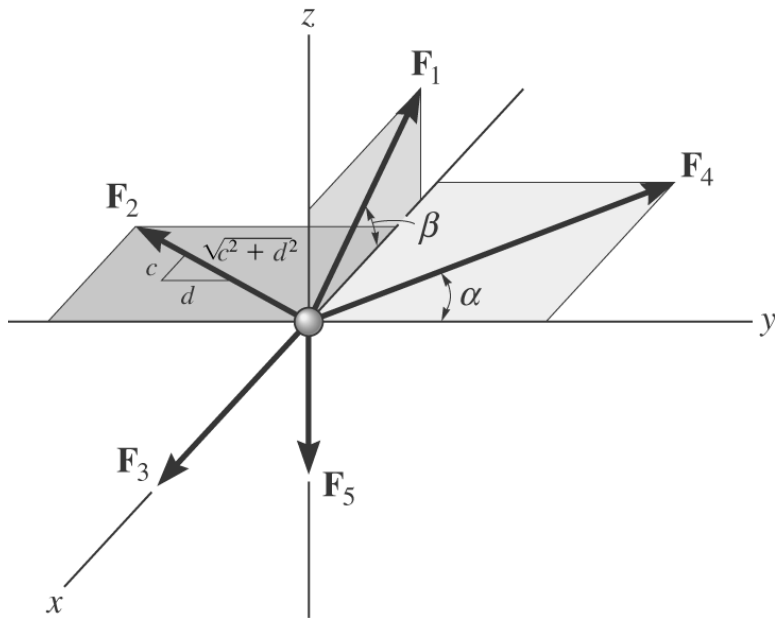
$F_5 = 2.8$ kN

$\alpha = 15$ deg

$\beta = 30$ deg

$c = 7$

$d = 24$



Solution:

Initial Guesses: $F_1 = 1$ kN $F_2 = 1$ kN $F_3 = 1$ kN

Given

$$F_1\begin{pmatrix} -\cos(\beta) \\ 0 \\ \sin(\beta) \end{pmatrix} + \frac{F_2}{\sqrt{c^2 + d^2}}\begin{pmatrix} -c \\ -d \\ 0 \end{pmatrix} + F_3\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + F_4\begin{pmatrix} -\sin(\alpha) \\ \cos(\alpha) \\ 0 \end{pmatrix} + F_5\begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} = 0$$

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$$\begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = \text{Find}(F_1, F_2, F_3) \qquad \begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = \begin{pmatrix} 5.60 \\ 8.55 \\ 9.44 \end{pmatrix} \text{kN}$$

Problem 3-44

Determine the magnitudes of $\mathbf{F_1}$, $\mathbf{F_2}$ and $\mathbf{F_3}$ for equilibrium of the particle $\mathbf{F} = \{-9\mathbf{i} - 8\mathbf{j} - 5\mathbf{k}\}$.

Units Used:

$\text{kN} = 10^3 \text{ N}$

Given:

$$F = \begin{pmatrix} -9 \\ -8 \\ -5 \end{pmatrix} \text{kN}$$

$a = 4 \text{ m}$

$b = 2 \text{ m}$

$c = 4 \text{ m}$

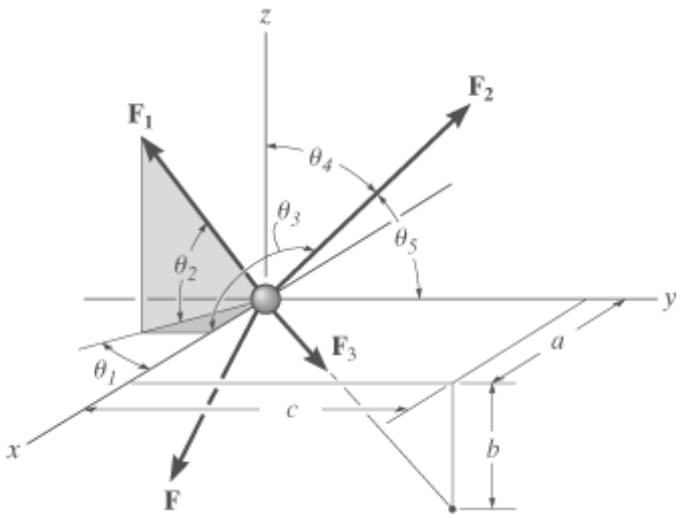
$\theta_1 = 30 \text{ deg}$

$\theta_2 = 60 \text{ deg}$

$\theta_3 = 135 \text{ deg}$

$\theta_4 = 60 \text{ deg}$

$\theta_5 = 60 \text{ deg}$



Solution:

Initial guesses: $F_1 = 8 \text{ kN}$ $F_2 = 3 \text{ kN}$ $F_3 = 12 \text{ kN}$

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Given

$$F_1\begin{pmatrix}\cos(\theta_2)\cos(\theta_1)\\ -\cos(\theta_2)\sin(\theta_1)\\ \sin(\theta_2)\end{pmatrix}+F_2\begin{pmatrix}\cos(\theta_3)\\ \cos(\theta_5)\\ \cos(\theta_4)\end{pmatrix}+\frac{F_3}{\sqrt{a^2+b^2+c^2}}\begin{pmatrix}a\\ c\\ -b\end{pmatrix}+F=0$$

$$\begin{pmatrix}F_1\\ F_2\\ F_3\end{pmatrix}=\text{Find}(F_1,F_2,F_3)\qquad \begin{pmatrix}F_1\\ F_2\\ F_3\end{pmatrix}=\begin{pmatrix}8.26\\ 3.84\\ 12.21\end{pmatrix}\text{kN}$$

Problem 3-45

The three cables are used to support the lamp of weight W. Determine the force developed in each cable for equilibrium.

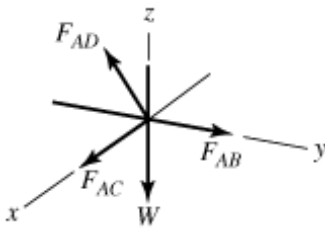
Units Used:

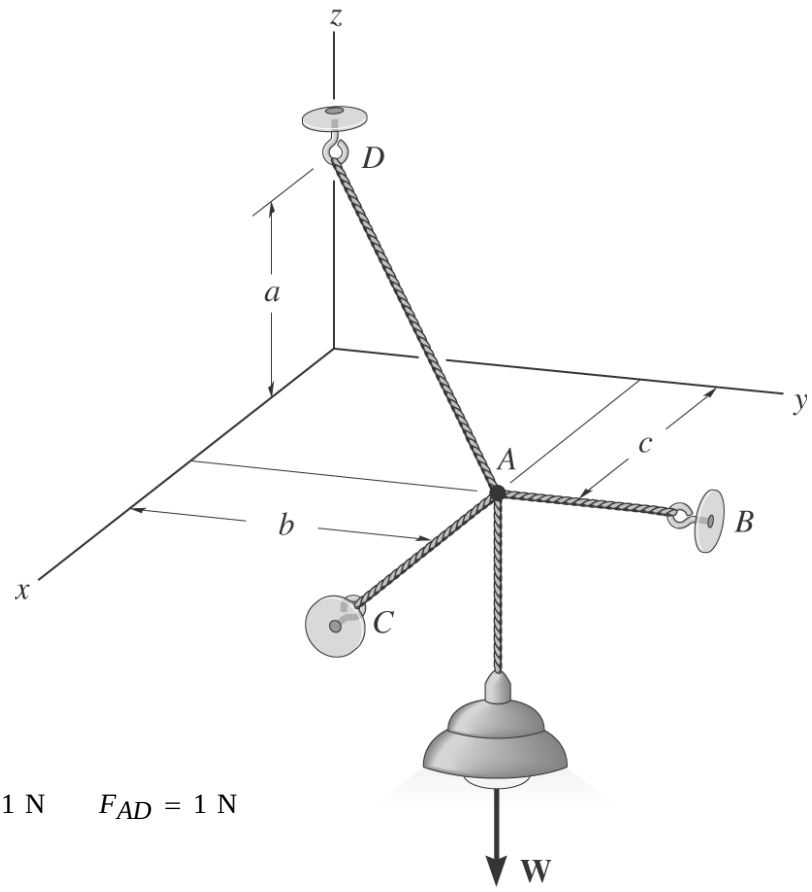
$\text{kN} = 10^3\ \text{N}$

Given:

$W = 800\ \text{N}$ $b = 4\ \text{m}$

$a = 4\ \text{m}$ $c = 2\ \text{m}$





Solution:

Initial Guesses:

$F_{AB} = 1\ \text{N}$ $F_{AC} = 1\ \text{N}$ $F_{AD} = 1\ \text{N}$

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Given

$$F_{AB}\begin{pmatrix}0\\1\\0\end{pmatrix}+F_{AC}\begin{pmatrix}1\\0\\0\end{pmatrix}+\frac{F_{AD}}{\sqrt{a^2+b^2+c^2}}\begin{pmatrix}-c\\-b\\a\end{pmatrix}+W\begin{pmatrix}0\\0\\-1\end{pmatrix}=0$$

$$\begin{pmatrix}F_{AB}\\F_{AC}\\F_{AD}\end{pmatrix}=\text{Find}\left(F_{AB},F_{AC},F_{AD}\right)\qquad\begin{pmatrix}F_{AB}\\F_{AC}\\F_{AD}\end{pmatrix}=\begin{pmatrix}800\\400\\1200\end{pmatrix}\text{ N}$$

Problem 3-46

Determine the force in each cable needed to support the load W.

Given:

$a = 8$ ft

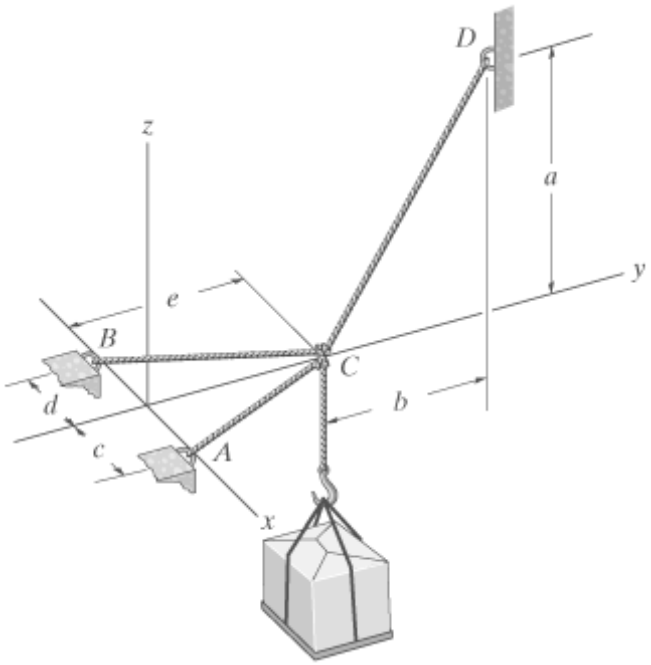
$b = 6$ ft

$c = 2$ ft

$d = 2$ ft

$e = 6$ ft

$W = 500$ lb



Solution:

Initial guesses:

$F_{CD} = 600$ lb $\quad F_{CA} = 195$ lb $\quad F_{CB} = 195$ lb

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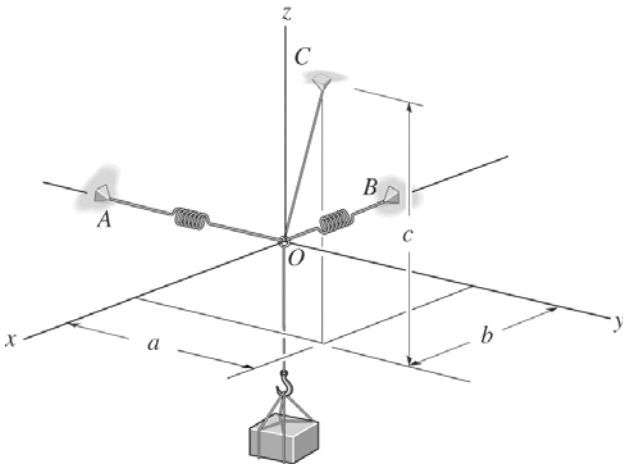
Given

$$\frac{F_{CA}}{\sqrt{c^2+e^2}}\begin{pmatrix} c \\ -e \\ 0 \end{pmatrix} + \frac{F_{CB}}{\sqrt{d^2+e^2}}\begin{pmatrix} -d \\ -e \\ 0 \end{pmatrix} + \frac{F_{CD}}{\sqrt{a^2+b^2}}\begin{pmatrix} 0 \\ b \\ a \end{pmatrix} + W\begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} = 0$$

$$\begin{pmatrix} F_{CD} \\ F_{CA} \\ F_{CB} \end{pmatrix} = \text{Find}\left(F_{CD}, F_{CA}, F_{CB}\right) \qquad \begin{pmatrix} F_{CD} \\ F_{CA} \\ F_{CB} \end{pmatrix} = \begin{pmatrix} 625 \\ 198 \\ 198 \end{pmatrix} \text{lb}$$

Problem 3-47

Determine the stretch in each of the two springs required to hold the crate of mass m_c in the equilibrium position shown. Each spring has an unstretched length δ and a stiffness k.



Given:

$m_c = 20$ kg

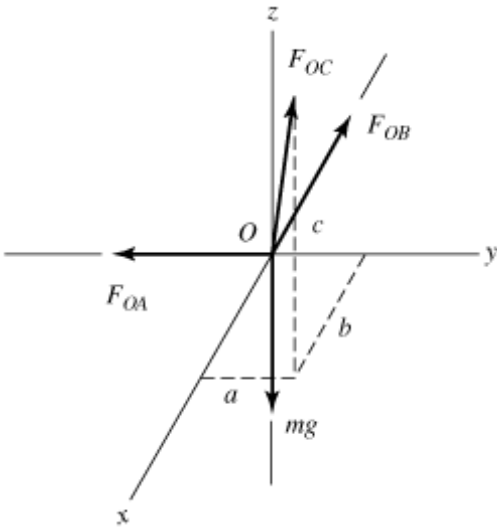
$\delta = 2$ m

$k = 300 \dfrac{\text{N}}{\text{m}}$

$a = 4$ m

$b = 6$ m

$c = 12$ m



Solution:

Initial Guesses

$F_{OA} = 1$ N

$F_{OB} = 1$ N

$F_{OC} = 1$ N

Given

$$F_{OA}\begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} + F_{OB}\begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} + \frac{F_{OC}}{\sqrt{a^2+b^2+c^2}}\begin{pmatrix} b \\ a \\ c \end{pmatrix} + m_c g\begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} = 0$$

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$$\begin{pmatrix} F_{OA} \\ F_{OB} \\ F_{OC} \end{pmatrix} = \text{Find}\left(F_{OA}, F_{OB}, F_{OC}\right) \qquad \begin{pmatrix} F_{OA} \\ F_{OB} \\ F_{OC} \end{pmatrix} = \begin{pmatrix} 65.40 \\ 98.10 \\ 228.90 \end{pmatrix} \text{N}$$

$$\delta_{OA} = \frac{F_{OA}}{k} \qquad \delta_{OA} = 218\,\text{mm}$$

$$\delta_{OB} = \frac{F_{OB}}{k} \qquad \delta_{OB} = 327\,\text{mm}$$

Problem 3-48

If the bucket and its contents have total weight W, determine the force in the supporting cables *DA*, *DB*, and *DC*.

Given:

$W = 20$ lb

$a = 3$ ft

$b = 4.5$ ft

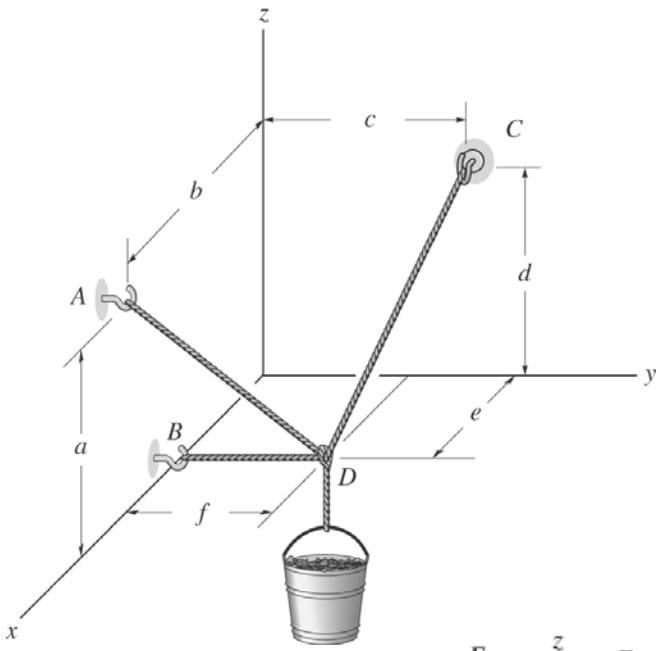
$c = 2.5$ ft

$d = 3$ ft

$e = 1.5$ ft

$f = 1.5$ ft

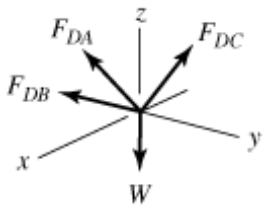
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Solution:

The initial guesses:

$F_{DA} = 40$ lb $F_{DB} = 20$ lb $F_{DC} = 30$ lb



Given

$$\Sigma F_x = 0; \quad \left[\frac{b-e}{\sqrt{(b-e)^2+f^2+a^2}}\right]F_{DA} - \left[\frac{e}{\sqrt{e^2+d^2+(c-f)^2}}\right]F_{DC} = 0$$

$$\Sigma F_y = 0; \quad \left[\frac{-f}{\sqrt{(b-e)^2+f^2+a^2}}\right]F_{DA} + \left[\frac{c-f}{\sqrt{e^2+d^2+(c-f)^2}}\right]F_{DC} - F_{DB} = 0$$

$$\Sigma F_z = 0; \quad \left[\frac{a}{\sqrt{(b-e)^2+f^2+a^2}}\right]F_{DA} + \left[\frac{d}{\sqrt{e^2+d^2+(c-f)^2}}\right]F_{DC} - W = 0$$

$$\begin{pmatrix} F_{DA} \\ F_{DB} \\ F_{DC} \end{pmatrix} = \text{Find}\left(F_{DA}, F_{DB}, F_{DC}\right) \qquad \begin{pmatrix} F_{DA} \\ F_{DB} \\ F_{DC} \end{pmatrix} = \begin{pmatrix} 10.00 \\ 1.11 \\ 15.56 \end{pmatrix} \text{lb}$$

Problem 3-49

The crate which of weight *F* is to be hoisted with constant velocity from the hold of a ship using the cable arrangement shown. Determine the tension in each of the three cables for equilibrium.

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Units Used:

$kN = 10^3 N$

Given:

$F = 2.5$ kN

$a = 3$ m

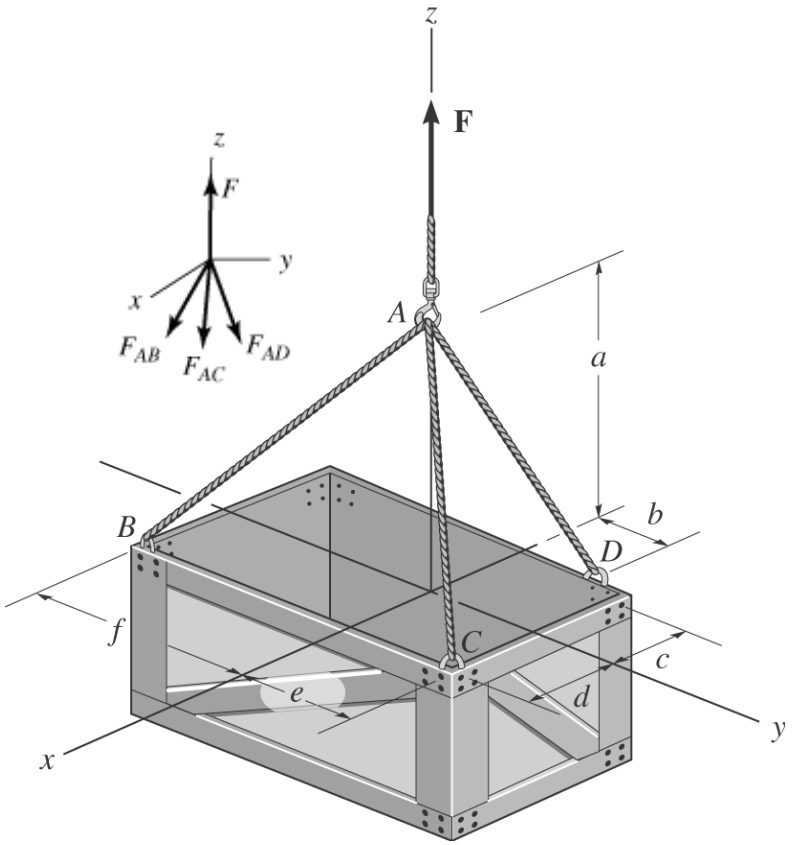
$b = 1$ m

$c = 0.75$ m

$d = 1$ m

$e = 1.5$ m

$f = 3$ m



Solution:

The initial guesses

$F_{AD} = 3$ kN $\quad F_{AC} = 3$ kN $\quad F_{AB} = 3$ kN

Given

$$\frac{-c}{\sqrt{c^2+b^2+a^2}}F_{AD}+\frac{d}{\sqrt{d^2+e^2+a^2}}F_{AC}+\frac{d}{\sqrt{d^2+f^2+a^2}}F_{AB}=0$$

$$\frac{b}{\sqrt{c^2+b^2+a^2}}F_{AD}+\frac{e}{\sqrt{d^2+e^2+a^2}}F_{AC}+\frac{-f}{\sqrt{d^2+f^2+a^2}}F_{AB}=0$$

$$\frac{-a}{\sqrt{c^2+b^2+a^2}}F_{AD}+\frac{-a}{\sqrt{d^2+e^2+a^2}}F_{AC}+\frac{-a}{\sqrt{d^2+f^2+a^2}}F_{AB}+F=0$$

$$\begin{pmatrix}F_{AD}\\F_{AC}\\F_{AB}\end{pmatrix}=\text{Find}\left(F_{AD},F_{AC},F_{AB}\right)$$

$$\begin{pmatrix}F_{AD}\\F_{AC}\\F_{AB}\end{pmatrix}=\begin{pmatrix}1.55\\0.46\\0.98\end{pmatrix}\text{kN}$$

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Problem 3-50

The lamp has mass m_l and is supported by pole *AO* and cables *AB* and *AC*. If the force in the pole acts along its axis, determine the forces in *AO, AB*, and *AC* for equilibrium.

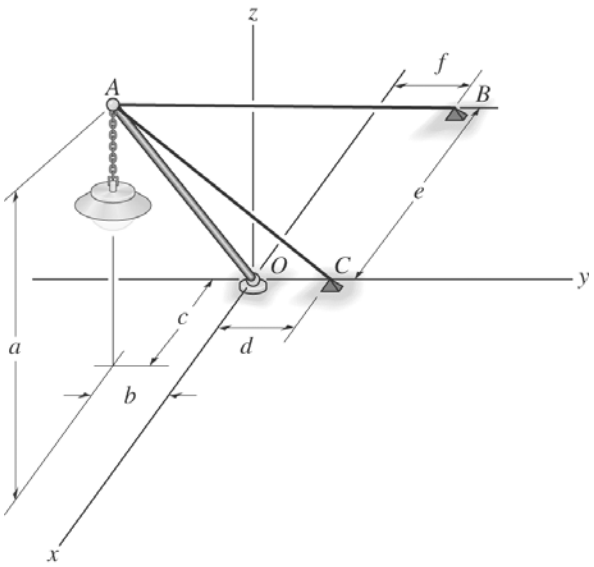
Given:

$m_l = 15\ \text{kg}$ $\quad d = 1.5\ \text{m}$

$a = 6\ \text{m}$ $\quad e = 4\ \text{m}$

$b = 1.5\ \text{m}$ $\quad f = 1.5\ \text{m}$

$c = 2\ \text{m}$ $\quad g = 9.81\ \frac{\text{m}}{\text{s}^2}$



Solution:

The initial guesses**:**

$F_{AO} = 100\ \text{N}$

$F_{AB} = 200\ \text{N}$

$F_{AC} = 300\ \text{N}$

Given



Equilibrium equations:

$$\frac{c}{\sqrt{c^2+b^2+a^2}}F_{AO}-\frac{c+e}{\sqrt{(c+e)^2+(b+f)^2+a^2}}F_{AB}-\frac{c}{\sqrt{c^2+(b+d)^2+a^2}}F_{AC}=0$$

$$-\frac{b}{\sqrt{c^2+b^2+a^2}}F_{AO}+\frac{b+f}{\sqrt{(c+e)^2+(b+f)^2+a^2}}F_{AB}+\frac{b+d}{\sqrt{c^2+(b+d)^2+a^2}}F_{AC}=0$$

$$\frac{a}{\sqrt{c^2+b^2+a^2}}F_{AO}-\frac{a}{\sqrt{(c+e)^2+(b+f)^2+a^2}}F_{AB}-\frac{a}{\sqrt{c^2+(b+d)^2+a^2}}F_{AC}-m_l g=0$$

$$\begin{pmatrix}F_{AO}\\F_{AB}\\F_{AC}\end{pmatrix}=\text{Find}\left(F_{AO},F_{AB},F_{AC}\right)\qquad\begin{pmatrix}F_{AO}\\F_{AB}\\F_{AC}\end{pmatrix}=\begin{pmatrix}318.82\\110.36\\85.84\end{pmatrix}\ \text{N}$$

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Problem 3-51

Cables *AB* and *AC* can sustain a maximum tension *Tmax*, and the pole can support a maximum compression *Pmax*. Determine the maximum weight of the lamp that can be supported in the position shown. The force in the pole acts along the axis of the pole.

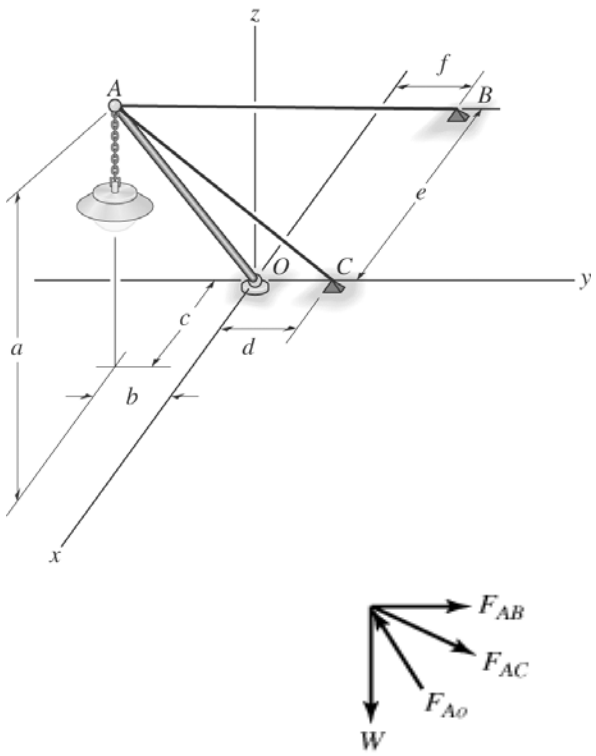
Given:

$Tmax = 500\ \text{N}$ $c = 2\ \text{m}$

$Pmax = 300\ \text{N}$ $d = 1.5\ \text{m}$

$a = 6\ \text{m}$ $e = 4\ \text{m}$

$b = 1.5\ \text{m}$ $f = 1.5\ \text{m}$



Solution:

Lengths

$$AO = \sqrt{a^2 + b^2 + c^2}$$

$$AB = \sqrt{a^2 + (c+e)^2 + (b+d)^2}$$

$$AC = \sqrt{a^2 + c^2 + (b+d)^2}$$

The initial guesses**:**

$F_{AO} = Pmax$ $F_{AB} = Tmax$ $F_{AC} = Tmax$ $W = 300\text{N}$

Case 1 Assume the pole reaches maximum compression

Given

$$\frac{F_{AO}}{AO}\begin{pmatrix} c \\ -b \\ a \end{pmatrix} + \frac{F_{AB}}{AB}\begin{pmatrix} -c-e \\ b+f \\ -a \end{pmatrix} + \frac{F_{AC}}{AC}\begin{pmatrix} -c \\ b+d \\ -a \end{pmatrix} + W\begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} = 0$$

$$\begin{pmatrix} W_1 \\ F_{AB1} \\ F_{AC1} \end{pmatrix} = \text{Find}\left(W, F_{AB}, F_{AC}\right) \qquad \begin{pmatrix} W_1 \\ F_{AB1} \\ F_{AC1} \end{pmatrix} = \begin{pmatrix} 138.46 \\ 103.85 \\ 80.77 \end{pmatrix}\ \text{N}$$

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Case 2 Assume that cable AB reaches maximum tension

Given

$$\frac{F_{AO}}{AO}\begin{pmatrix} c \\ -b \\ a \end{pmatrix} + \frac{F_{AB}}{AB}\begin{pmatrix} -c-e \\ b+f \\ -a \end{pmatrix} + \frac{F_{AC}}{AC}\begin{pmatrix} -c \\ b+d \\ -a \end{pmatrix} + W\begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} = 0$$

$$\begin{pmatrix} W_2 \\ F_{AO2} \\ F_{AC2} \end{pmatrix} = \text{Find}\left(W, F_{AO}, F_{AC}\right) \qquad \begin{pmatrix} W_2 \\ F_{AO2} \\ F_{AC2} \end{pmatrix} = \begin{pmatrix} 666.67 \\ 1444.44 \\ 388.89 \end{pmatrix} \text{ N}$$

Case 3 Assume that cable AC reaches maximum tension

Given

$$\frac{F_{AO}}{AO}\begin{pmatrix} c \\ -b \\ a \end{pmatrix} + \frac{F_{AB}}{AB}\begin{pmatrix} -c-e \\ b+f \\ -a \end{pmatrix} + \frac{F_{AC}}{AC}\begin{pmatrix} -c \\ b+d \\ -a \end{pmatrix} + W\begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} = 0$$

$$\begin{pmatrix} W_3 \\ F_{AO3} \\ F_{AB3} \end{pmatrix} = \text{Find}\left(W, F_{AO}, F_{AB}\right) \qquad \begin{pmatrix} W_3 \\ F_{AO3} \\ F_{AB3} \end{pmatrix} = \begin{pmatrix} 857.14 \\ 1857.14 \\ 642.86 \end{pmatrix} \text{ N}$$

Final Answer $W = \min\left(W_1, W_2, W_3\right)$ $W = 138.46$ N

Problem 3-52

Determine the tension in cables *AB, AC,* and *AD,* required to hold the crate of weight W in equilibrium.

Given:

$W = 60$ lb

$a = 6$ ft

$b = 12$ ft

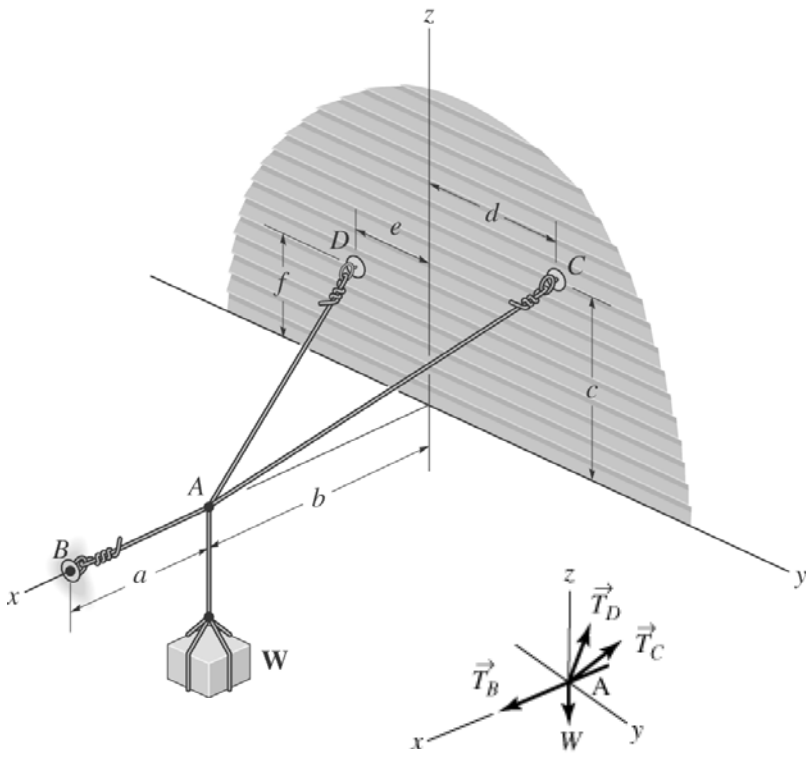
$c = 8$ ft

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$d = 9$ ft

$e = 4$ ft

$f = 6$ ft



Solution:

The initial guesses:

$T_B = 100$ lb

$T_C = 100$ lb

$T_D = 100$ lb

Given

$$\Sigma F_x = 0; \qquad T_B - \frac{b}{\sqrt{b^2 + c^2 + d^2}} T_C - \frac{b}{\sqrt{b^2 + e^2 + f^2}} T_D = 0$$

$$\Sigma F_y = 0; \qquad \frac{d}{\sqrt{b^2 + c^2 + d^2}} T_C - \frac{e}{\sqrt{b^2 + e^2 + f^2}} T_D = 0$$

$$\Sigma F_z = 0; \qquad -W + \frac{c}{\sqrt{b^2 + c^2 + d^2}} T_C + \frac{f}{\sqrt{b^2 + e^2 + f^2}} T_D = 0$$

Solving

$$\begin{pmatrix} T_B \\ T_C \\ T_D \end{pmatrix} = \text{Find}\left(T_B, T_C, T_D\right) \qquad \begin{pmatrix} T_B \\ T_C \\ T_D \end{pmatrix} = \begin{pmatrix} 108.84 \\ 47.44 \\ 87.91 \end{pmatrix} \text{lb}$$

Problem 3-53

The bucket has weight W. Determine the tension developed in each cord for equilibrium.

Given:

$W = 20$ lb

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$a = 2$ ft

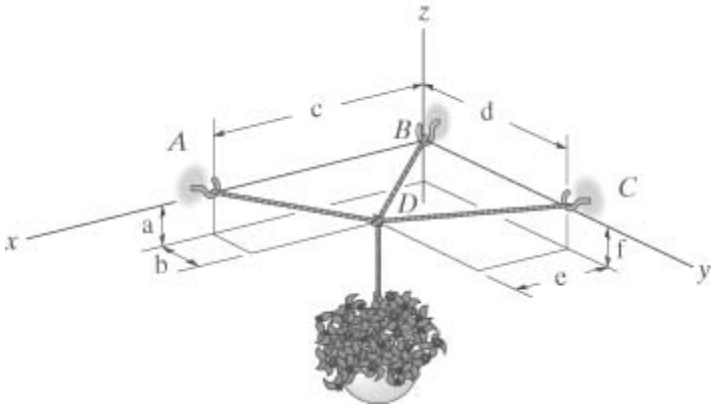
$b = 2$ ft

$c = 8$ ft

$d = 7$ ft

$e = 3$ ft

$f = a$



Solution:

Initial Guesses: $F_{DA} = 20$ lb $F_{DB} = 10$ lb $F_{DC} = 15$ lb

Given

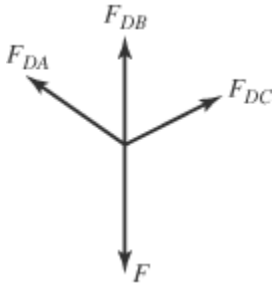
$$\Sigma F_x = 0; \quad \frac{c-e}{\sqrt{(c-e)^2+b^2+a^2}}F_{DA} + \frac{-e}{\sqrt{e^2+(d-b)^2+f^2}}F_{DC} + \frac{-e}{\sqrt{e^2+b^2+f^2}}F_{DB} = 0$$

$$\Sigma F_x = 0; \quad \frac{-b}{\sqrt{(c-e)^2+b^2+a^2}}F_{DA} + \frac{d-b}{\sqrt{e^2+(d-b)^2+f^2}}F_{DC} + \frac{-b}{\sqrt{e^2+b^2+f^2}}F_{DB} = 0$$

$$\Sigma F_y = 0; \quad \frac{a}{\sqrt{(c-e)^2+b^2+a^2}}F_{DA} + \frac{f}{\sqrt{e^2+(d-b)^2+f^2}}F_{DC} + \frac{f}{\sqrt{e^2+b^2+f^2}}F_{DB} - W = 0$$

$$\begin{pmatrix} F_{DA} \\ F_{DB} \\ F_{DC} \end{pmatrix} = \text{Find}\left(F_{DA}, F_{DB}, F_{DC}\right)$$

$$\begin{pmatrix} F_{DA} \\ F_{DB} \\ F_{DC} \end{pmatrix} = \begin{pmatrix} 21.54 \\ 13.99 \\ 17.61 \end{pmatrix} \text{lb}$$



Problem 3-54

The mast *OA* is supported by three cables. If cable *AB* is subjected to tension *T*, determine the tension in cables *AC* and *AD* and the vertical force **F** which the mast exerts along its axis on the collar at *A*.

Given:

$T = 500$ N

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$a = 6$ m

$b = 3$ m

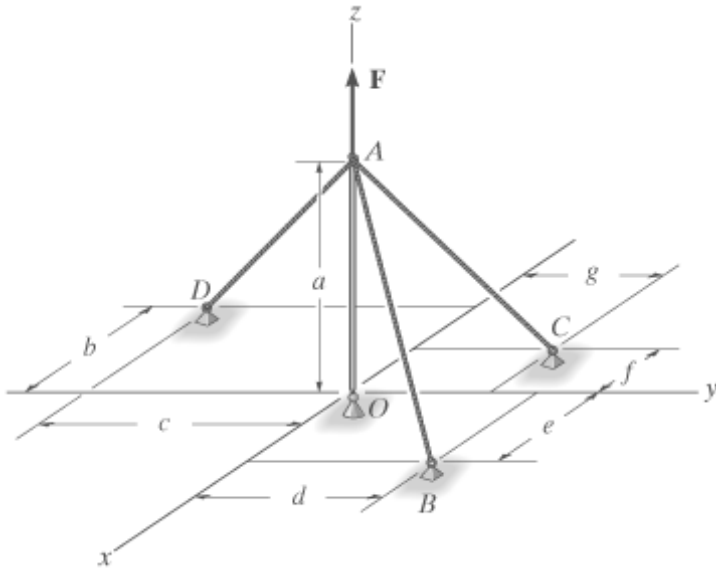
$c = 6$ m

$d = 3$ m

$e = 2$ m

$f = 1.5$ m

$g = 2$ m



Solution:

Initial Guesses: $F_{AC} = 90$ N $F_{AD} = 350$ N $F = 750$ N

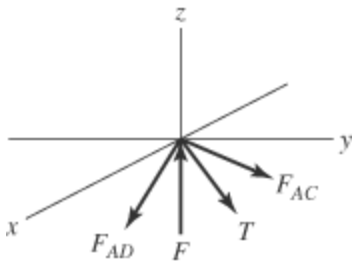
Given

$$\Sigma F_x = 0; \quad \frac{e}{\sqrt{e^2 + d^2 + a^2}} T - \frac{f}{\sqrt{f^2 + g^2 + a^2}} F_{AC} - \frac{b}{\sqrt{b^2 + c^2 + a^2}} F_{AD} = 0$$

$$\Sigma F_y = 0; \quad \frac{d}{\sqrt{e^2 + d^2 + a^2}} T + \frac{g}{\sqrt{f^2 + g^2 + a^2}} F_{AC} - \frac{c}{\sqrt{b^2 + c^2 + a^2}} F_{AD} = 0$$

$$\Sigma F_z = 0; \quad \frac{-a}{\sqrt{e^2 + d^2 + a^2}} T - \frac{a}{\sqrt{f^2 + g^2 + a^2}} F_{AC} - \frac{a}{\sqrt{b^2 + c^2 + a^2}} F_{AD} + F = 0$$

$$\begin{pmatrix} F_{AC} \\ F_{AD} \\ F \end{pmatrix} = \text{Find}(F_{AC}, F_{AD}, F) \qquad \begin{pmatrix} F_{AC} \\ F_{AD} \\ F \end{pmatrix} = \begin{pmatrix} 92.9 \\ 364.3 \\ 757.1 \end{pmatrix} \text{N}$$



Problem 3-55

The ends of the three cables are attached to a ring at A and to the edge of the uniform plate of mass M. Determine the tension in each of the cables for equilibrium.

Given:

$M = 150$ kg $e = 4$ m

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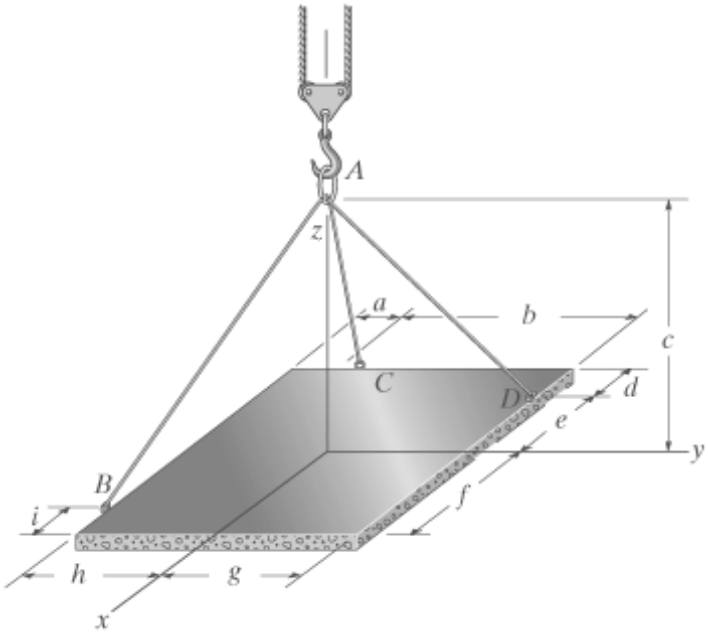
$a = 2$ m $\quad f = 6$ m

$b = 10$ m $\quad g = 6$ m

$c = 12$ m $\quad h = 6$ m

$d = 2$ m $\quad i = 2$ m

$gravity = 9.81 \dfrac{\text{m}}{\text{s}^2}$



Solution:

The initial guesses**:**

$F_B = 15$ N

$F_C = 16$ N

$F_D = 16$ N

Given

$$\Sigma F_x = 0; \quad \frac{F_B(f-i)}{\sqrt{(f-i)^2+h^2+c^2}} + \frac{F_C(-d-e)}{\sqrt{(d+e)^2+(h-a)^2+c^2}} + \frac{F_D(-e)}{\sqrt{e^2+g^2+c^2}} = 0$$

$$\Sigma F_y = 0; \quad \frac{F_B(-h)}{\sqrt{(f-i)^2+h^2+c^2}} + \frac{F_C[-(h-a)]}{\sqrt{(d+e)^2+(h-a)^2+c^2}} + \frac{F_D(g)}{\sqrt{e^2+g^2+c^2}} = 0$$

$$\Sigma F_z = 0; \quad \frac{F_B(-c)}{\sqrt{(f-i)^2+h^2+c^2}} + \frac{F_C(-c)}{\sqrt{(d+e)^2+(h-a)^2+c^2}} + \frac{F_D(-c)}{\sqrt{e^2+g^2+c^2}} + M\,gravity = 0$$

$$\begin{pmatrix} F_B \\ F_C \\ F_D \end{pmatrix} = \text{Find}(F_B, F_C, F_D) \qquad \begin{pmatrix} F_B \\ F_C \\ F_D \end{pmatrix} = \begin{pmatrix} 858 \\ 0 \\ 858 \end{pmatrix} \text{N}$$

Problem 3-56

The ends of the three cables are attached to a ring at A and to the edge of the uniform plate. Determine the largest mass the plate can have if each cable can support a maximum tension of T.

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kN $= 10^3$ N

Given:

$T = 15$ kN $\quad e = 4$ m

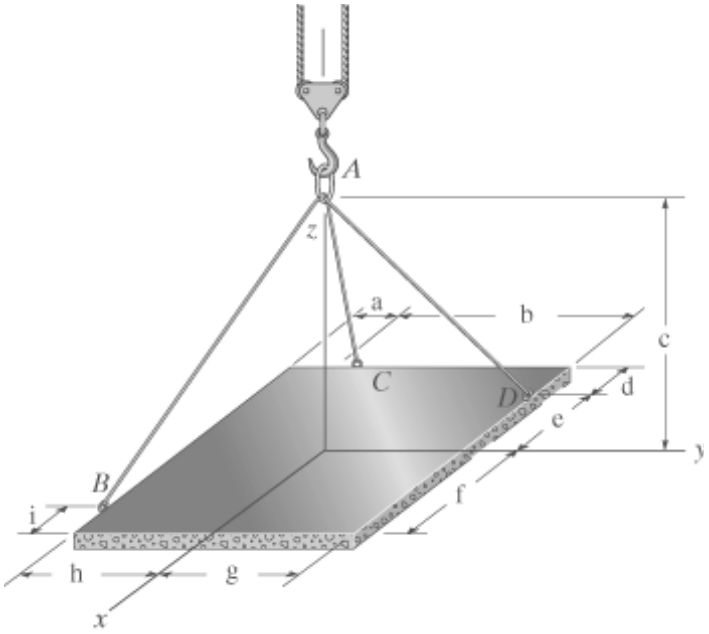
$a = 2$ m $\quad f = 6$ m

$b = 10$ m $\quad g = 6$ m

$c = 12$ m $\quad h = 6$ m

$d = 2$ m $\quad i = 2$ m

$gravity = 9.81 \dfrac{\text{m}}{\text{s}^2}$



Solution:

The initial guesses**:** $F_B = T \quad F_C = T \quad F_D = T \quad M = 1$ kg

Case 1: Assume that cable B reaches maximum tension

Given

$$\Sigma F_x = 0; \quad \frac{F_B(f-i)}{\sqrt{(f-i)^2+h^2+c^2}} + \frac{F_C(-d-e)}{\sqrt{(d+e)^2+(h-a)^2+c^2}} + \frac{F_D(-e)}{\sqrt{e^2+g^2+c^2}} = 0$$

$$\Sigma F_y = 0; \quad \frac{F_B(-h)}{\sqrt{(f-i)^2+h^2+c^2}} + \frac{F_C[-(h-a)]}{\sqrt{(d+e)^2+(h-a)^2+c^2}} + \frac{F_D(g)}{\sqrt{e^2+g^2+c^2}} = 0$$

$$\Sigma F_z = 0; \quad \frac{F_B(-c)}{\sqrt{(f-i)^2+h^2+c^2}} + \frac{F_C(-c)}{\sqrt{(d+e)^2+(h-a)^2+c^2}} + \frac{F_D(-c)}{\sqrt{e^2+g^2+c^2}} + M\,gravity = 0$$

$$\begin{pmatrix} M_1 \\ F_{C1} \\ F_{D1} \end{pmatrix} = \text{Find}(M, F_C, F_D) \qquad \begin{pmatrix} F_{C1} \\ F_{D1} \end{pmatrix} = \begin{pmatrix} -0.00 \\ 15.00 \end{pmatrix} \text{kN} \qquad M_1 = 2621.23 \text{ kg}$$

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Case 2: Assume that cable D reaches maximum tension

Given

$$\Sigma F_x = 0; \qquad \frac{F_B(f-i)}{\sqrt{(f-i)^2+h^2+c^2}} + \frac{F_C(-d-e)}{\sqrt{(d+e)^2+(h-a)^2+c^2}} + \frac{F_D(-e)}{\sqrt{e^2+g^2+c^2}} = 0$$

$$\Sigma F_y = 0; \qquad \frac{F_B(-h)}{\sqrt{(f-i)^2+h^2+c^2}} + \frac{F_C[-(h-a)]}{\sqrt{(d+e)^2+(h-a)^2+c^2}} + \frac{F_D(g)}{\sqrt{e^2+g^2+c^2}} = 0$$

$$\Sigma F_z = 0; \qquad \frac{F_B(-c)}{\sqrt{(f-i)^2+h^2+c^2}} + \frac{F_C(-c)}{\sqrt{(d+e)^2+(h-a)^2+c^2}} + \frac{F_D(-c)}{\sqrt{e^2+g^2+c^2}} + M\,gravity = 0$$

$$\begin{pmatrix} M_2 \\ F_{B2} \\ F_{C2} \end{pmatrix} = \text{Find}(M, F_B, F_C) \qquad \begin{pmatrix} F_{B2} \\ F_{C2} \end{pmatrix} = \begin{pmatrix} 15.00 \\ 0.00 \end{pmatrix} \text{kN} \qquad M_2 = 2621.23 \text{ kg}$$

For this set of number $F_C = 0$ for any mass that is applied. For a different set of numbers it would be necessary to also check case 3: Assume that the cable C reaches a maximum.

$$M = \min(M_1, M_2) \qquad M = 2621.23\,\text{kg}$$

Problem 3-57

The crate of weight W is suspended from the cable system shown. Determine the force in each segment of the cable, i.e., *AB*, *AC*, *CD*, *CE*, and *CF*. *Hint:* First analyze the equilibrium of point *A*, then using the result for *AC*, analyze the equilibrium of point *C*.

Units Used:

kip = 1000 lb

Given:

$W = 500$ lb

$a = 10$ ft

$b = 24$ ft

$c = 24$ ft

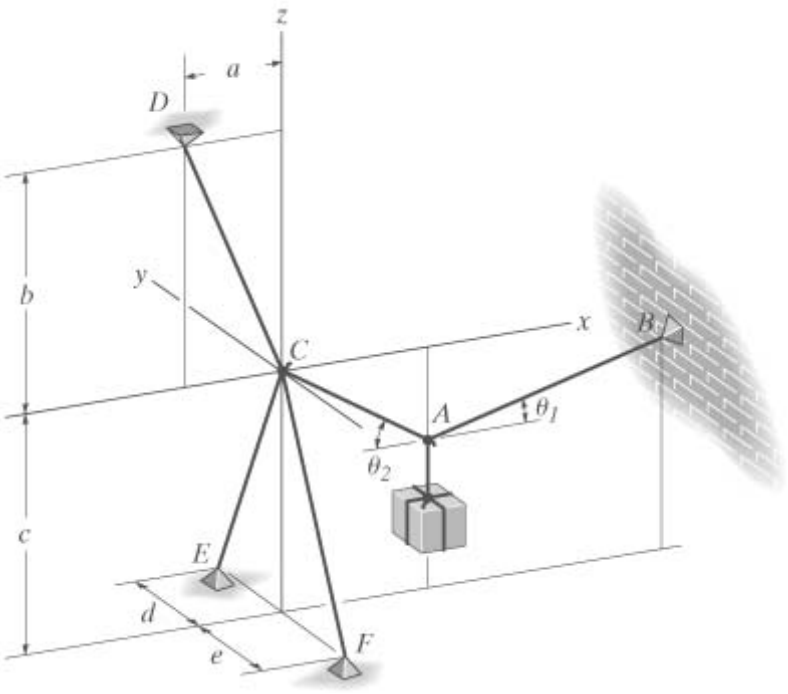
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$d = 7$ ft

$e = 7$ ft

$\theta_1 = 20$ deg

$\theta_2 = 35$ deg



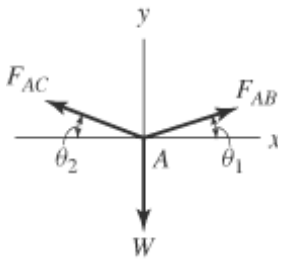
Solution: At *A:*

Initial guesses: $F_{AC} = 570$ lb $F_{AB} = 500$ lb

Given

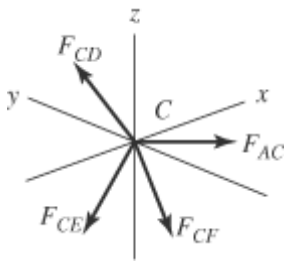
$$\xrightarrow{+} \Sigma F_x = 0; \quad F_{AB}\cos(\theta_1) - F_{AC}\cos(\theta_2) = 0$$

$$+\uparrow \Sigma F_y = 0; \quad F_{AB}\sin(\theta_1) + F_{AC}\sin(\theta_2) - W = 0$$



$$\begin{pmatrix} F_{AC} \\ F_{AB} \end{pmatrix} = \text{Find}(F_{AC}, F_{AB}) \qquad \begin{pmatrix} F_{AC} \\ F_{AB} \end{pmatrix} = \begin{pmatrix} 574 \\ 500 \end{pmatrix} \text{lb}$$

At *C:* Initial Guesses



$F_{CD} = 1$ lb $F_{CE} = 1$ lb $F_{CF} = 1$ lb

Given

$$F_{AC}\begin{pmatrix} \cos(\theta_2) \\ 0 \\ -\sin(\theta_2) \end{pmatrix} + \frac{F_{CD}}{\sqrt{a^2 + b^2}}\begin{pmatrix} -a \\ 0 \\ b \end{pmatrix} + \frac{F_{CE}}{\sqrt{c^2 + d^2}}\begin{pmatrix} 0 \\ d \\ -c \end{pmatrix} + \frac{F_{CF}}{\sqrt{c^2 + e^2}}\begin{pmatrix} 0 \\ -e \\ -c \end{pmatrix} = 0$$

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$$\begin{pmatrix} F_{CD} \\ F_{CE} \\ F_{CF} \end{pmatrix} = \text{Find}\left(F_{CD}, F_{CE}, F_{CF}\right) \qquad F_{CD} = 1.22\,\text{kip} \qquad \begin{pmatrix} F_{CE} \\ F_{CF} \end{pmatrix} = \begin{pmatrix} 416 \\ 416 \end{pmatrix} \text{lb}$$

Problem 3-58

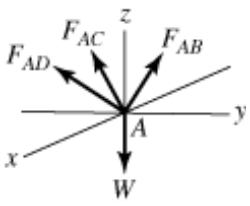
The chandelier of weight W is supported by three wires as shown. Determine the force in each wire for equilibrium.

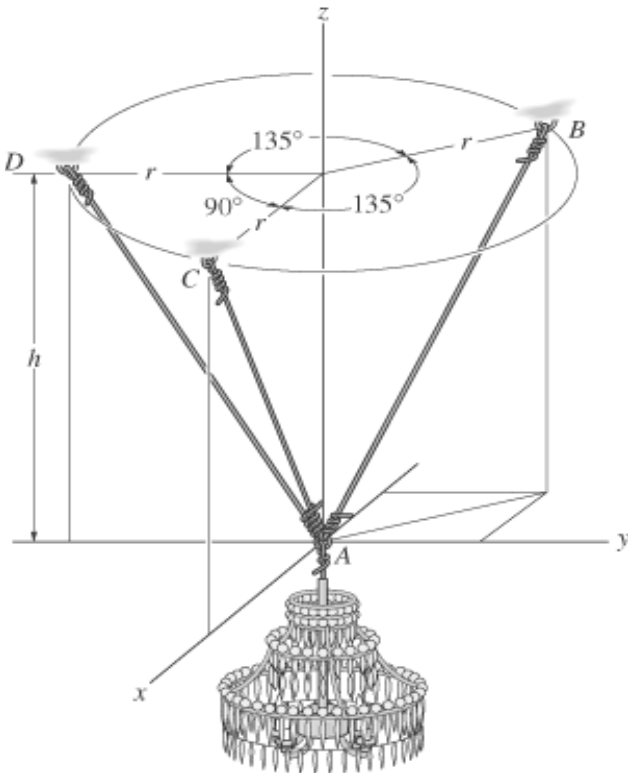
Given:

$W = 80$ lb

$r = 1$ ft

$h = 2.4$ ft





Solution:

The initial guesses:

$F_{AB} = 40$ lb

$F_{AC} = 30$ lb

$F_{AD} = 30$ lb

Given

$$\Sigma F_x = 0; \qquad \frac{r}{\sqrt{r^2 + h^2}} F_{AC} - \frac{r\cos(45\ \text{deg})}{\sqrt{r^2 + h^2}} F_{AB} = 0$$

$$\Sigma F_y = 0; \qquad \frac{-r}{\sqrt{r^2 + h^2}} F_{AD} + \frac{r\cos(45\ \text{deg})}{\sqrt{r^2 + h^2}} F_{AB} = 0$$

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$$\Sigma F_z = 0; \qquad \frac{h}{\sqrt{r^2+h^2}}F_{AC} + \frac{h}{\sqrt{r^2+h^2}}F_{AD} + \frac{h}{\sqrt{r^2+h^2}}F_{AB} - W = 0$$

$$\begin{pmatrix} F_{AB} \\ F_{AC} \\ F_{AD} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{AC}, F_{AD}\right) \qquad \begin{pmatrix} F_{AB} \\ F_{AC} \\ F_{AD} \end{pmatrix} = \begin{pmatrix} 35.9 \\ 25.4 \\ 25.4 \end{pmatrix} \text{lb}$$

Problem 3-59

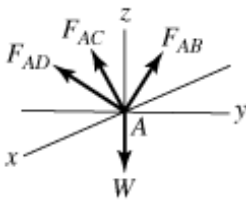
If each wire can sustain a maximum tension T_{max} before it fails, determine the greatest weight of the chandelier the wires will support in the position shown.

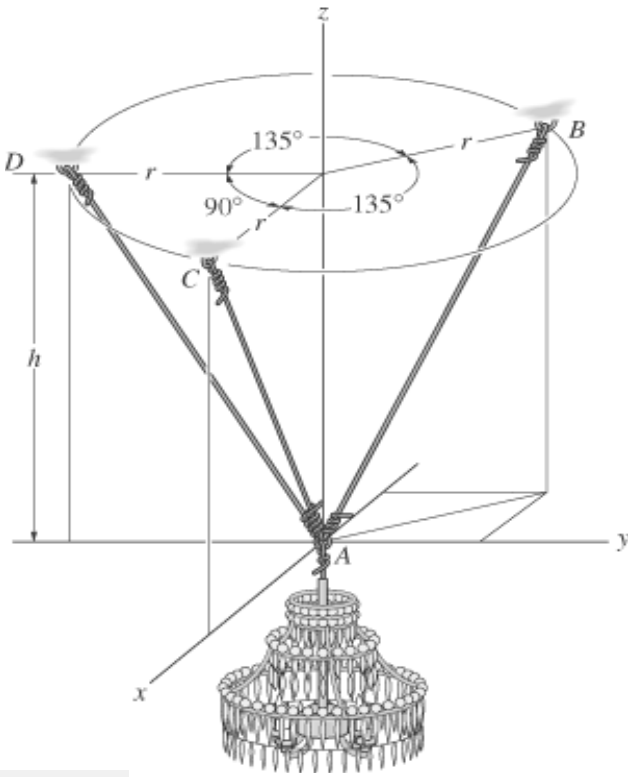
Given:

$T_{max} = 120$ lb

$r = 1$ ft

$h = 2.4$ ft





Solution:

The initial guesses:

$F_{AB} = T_{max}$

$F_{AC} = T_{max}$

$F_{AD} = T_{max}$

$W = T_{max}$

Case 1 Assume that cable AB has maximum tension

Given

$$\Sigma F_x = 0; \qquad \frac{r}{\sqrt{r^2+h^2}}F_{AC} - \frac{r\cos(45\text{deg})}{\sqrt{r^2+h^2}}F_{AB} = 0$$

$$\Sigma F_y = 0; \qquad \frac{-r}{\sqrt{r^2+h^2}}F_{AD} + \frac{r\cos(45\text{deg})}{\sqrt{r^2+h^2}}F_{AB} = 0$$

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$$\Sigma F_z = 0; \qquad \frac{h}{\sqrt{r^2+h^2}}F_{AC} + \frac{h}{\sqrt{r^2+h^2}}F_{AD} + \frac{h}{\sqrt{r^2+h^2}}F_{AB} - W = 0$$

$$\begin{pmatrix} W_1 \\ F_{AC1} \\ F_{AD1} \end{pmatrix} = \text{Find}\left(W, F_{AC}, F_{AD}\right) \qquad \begin{pmatrix} W_1 \\ F_{AC1} \\ F_{AD1} \end{pmatrix} = \begin{pmatrix} 267.4 \\ 84.9 \\ 84.9 \end{pmatrix} \text{lb}$$

Case 2 Assume that cable *AC* has maximum tension

Given

$$\Sigma F_x = 0; \qquad \frac{r}{\sqrt{r^2+h^2}}F_{AC} - \frac{r\cos(45\ \text{deg})}{\sqrt{r^2+h^2}}F_{AB} = 0$$

$$\Sigma F_y = 0; \qquad \frac{-r}{\sqrt{r^2+h^2}}F_{AD} + \frac{r\cos(45\ \text{deg})}{\sqrt{r^2+h^2}}F_{AB} = 0$$

$$\Sigma F_z = 0; \qquad \frac{h}{\sqrt{r^2+h^2}}F_{AC} + \frac{h}{\sqrt{r^2+h^2}}F_{AD} + \frac{h}{\sqrt{r^2+h^2}}F_{AB} - W = 0$$

$$\begin{pmatrix} W_2 \\ F_{AB2} \\ F_{AD2} \end{pmatrix} = \text{Find}\left(W, F_{AB}, F_{AD}\right) \qquad \begin{pmatrix} W_2 \\ F_{AB2} \\ F_{AD2} \end{pmatrix} = \begin{pmatrix} 378.2 \\ 169.7 \\ 120 \end{pmatrix} \text{lb}$$

Case 3 Assume that cable *AD* has maximum tension

Given

$$\Sigma F_x = 0; \qquad \frac{r}{\sqrt{r^2+h^2}}F_{AC} - \frac{r\cos(45\text{deg})}{\sqrt{r^2+h^2}}F_{AB} = 0$$

$$\Sigma F_y = 0; \qquad \frac{-r}{\sqrt{r^2+h^2}}F_{AD} + \frac{r\cos(45\text{deg})}{\sqrt{r^2+h^2}}F_{AB} = 0$$

$$\Sigma F_z = 0; \qquad \frac{h}{\sqrt{r^2+h^2}}F_{AC} + \frac{h}{\sqrt{r^2+h^2}}F_{AD} + \frac{h}{\sqrt{r^2+h^2}}F_{AB} - W = 0$$

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$$\begin{pmatrix} W_3 \\ F_{AB3} \\ F_{AC3} \end{pmatrix} = \text{Find}\left(W, F_{AB}, F_{AC}\right) \qquad \begin{pmatrix} W_3 \\ F_{AB3} \\ F_{AC3} \end{pmatrix} = \begin{pmatrix} 378.2 \\ 169.7 \\ 120 \end{pmatrix} \text{lb}$$

$$W = \min\left(W_1, W_2, W_3\right) \qquad W = 267.42\,\text{lb}$$

Problem 3-60

Determine the force in each cable used to lift the surge arrester of mass *M* at constant velocity.

Units Used:

$\text{kN} = 10^3\ \text{N}$

$\text{Mg} = 10^3\ \text{kg}$

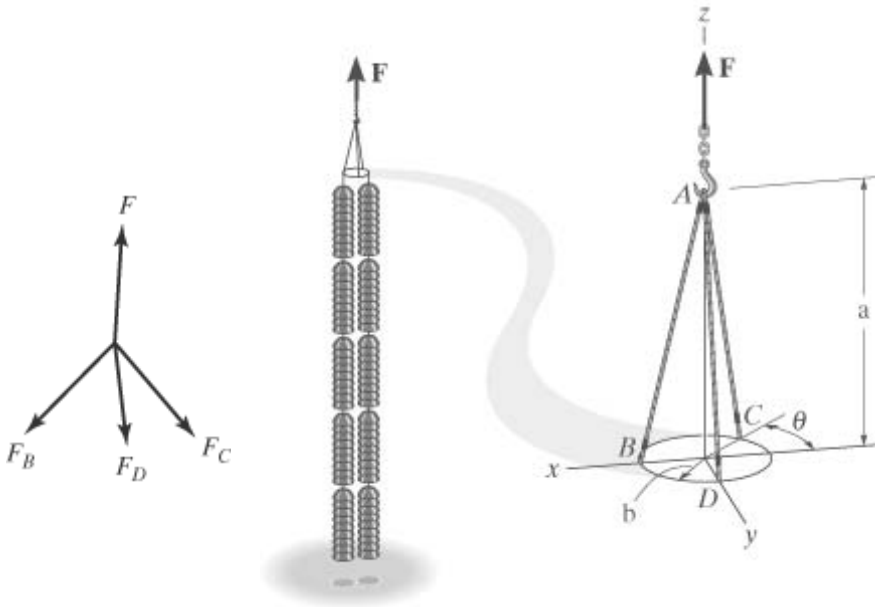
Given:

$M = 9.50\ \text{Mg}$

$a = 2\ \text{m}$

$b = 0.5\ \text{m}$

$\theta = 45\ \text{deg}$



Solution:

Initial guesses: $F_B = 50\ \text{kN}$ $F_C = 30\ \text{kN}$ $F_D = 10\ \text{kN}$

Given

$$\Sigma F_x = 0; \qquad F_B \frac{b}{\sqrt{b^2 + a^2}} - F_C \frac{b\cos(\theta)}{\sqrt{b^2 + a^2}} = 0$$

$$\Sigma F_y = 0; \qquad F_C \frac{-b\sin(\theta)}{\sqrt{b^2 + a^2}} + F_D \frac{b}{\sqrt{b^2 + a^2}} = 0$$

$$\Sigma F_z = 0; \qquad Mg - F_B \frac{a}{\sqrt{b^2 + a^2}} - F_C \frac{a}{\sqrt{b^2 + a^2}} - F_D \frac{a}{\sqrt{b^2 + a^2}} = 0$$

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$$\begin{pmatrix} F_B \\ F_C \\ F_D \end{pmatrix} = \text{Find}(F_B, F_C, F_D) \qquad \begin{pmatrix} F_B \\ F_C \\ F_D \end{pmatrix} = \begin{pmatrix} 28.13 \\ 39.78 \\ 28.13 \end{pmatrix} \text{kN}$$

Problem 3-61

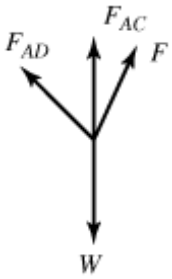
The cylinder of weight W is supported by three chains as shown. Determine the force in each chain for equilibrium.

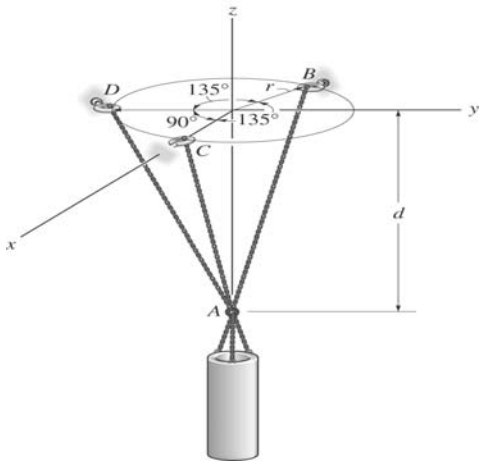
Given:

$W = 800$ lb

$r = 1$ ft

$d = 1$ ft





Solution:

The initial guesses**:**

$F_{AB} = 1$ lb

$F_{AC} = 1$ lb

$F_{AD} = 1$ lb

Given

$$\Sigma F_x = 0; \qquad \frac{r}{\sqrt{r^2 + d^2}} F_{AC} - \frac{r\cos(45 \text{ deg})}{\sqrt{r^2 + d^2}} F_{AB} = 0$$

$$\Sigma F_y = 0; \qquad \frac{-r}{\sqrt{r^2 + d^2}} F_{AD} + \frac{r\sin(45 \text{ deg})}{\sqrt{r^2 + d^2}} F_{AB} = 0$$

$$\Sigma F_z = 0; \qquad \frac{d}{\sqrt{r^2 + d^2}} F_{AD} + \frac{d}{\sqrt{r^2 + d^2}} F_{AC} + \frac{d}{\sqrt{r^2 + d^2}} F_{AB} - W = 0$$

$$\begin{pmatrix} F_{AB} \\ F_{AC} \\ F_{AD} \end{pmatrix} = \text{Find}(F_{AB}, F_{AC}, F_{AD}) \qquad \begin{pmatrix} F_{AB} \\ F_{AC} \\ F_{AD} \end{pmatrix} = \begin{pmatrix} 469 \\ 331 \\ 331 \end{pmatrix} \text{lb}$$

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Problem 3-62

The triangular frame *ABC* can be adjusted vertically between the three equal-length cords. If it remains in a horizontal plane, determine the required distance *s* so that the tension in each of the cords, *OA*, *OB*, and *OC*, equals **F**. The lamp has a mass *M*.

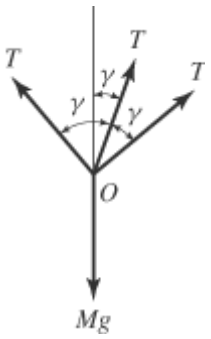
Given:

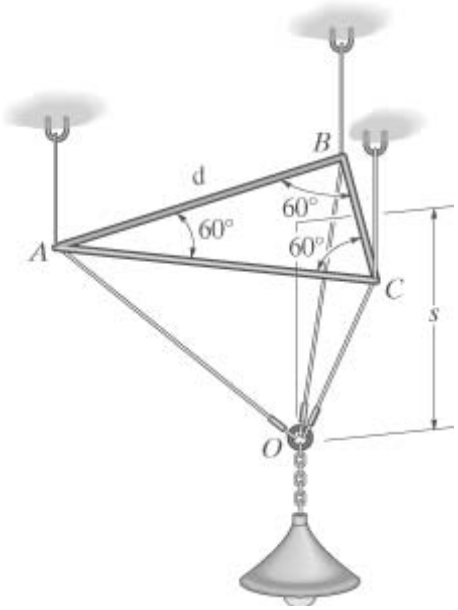
$F = 20\ \text{N}$

$M = 5\ \text{kg}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$

$d = 0.5\ \text{m}$





Solution:

$\Sigma F_z = 0; \qquad 3F\cos(\gamma) = Mg$

$\gamma = \text{acos}\left(\frac{Mg}{3F}\right) \qquad \gamma = 35.16\,\text{deg}$

Geometry $\frac{2d\cos(30\ \text{deg})}{3} = s\tan(\gamma)$

$s = \frac{2d\cos(30\ \text{deg})}{3\tan(\gamma)} \qquad s = 410\,\text{mm}$

Problem 3-63

Determine the force in each cable needed to support the platform of weight *W*.

Units Used:

$\text{kip} = 10^3\ \text{lb}$

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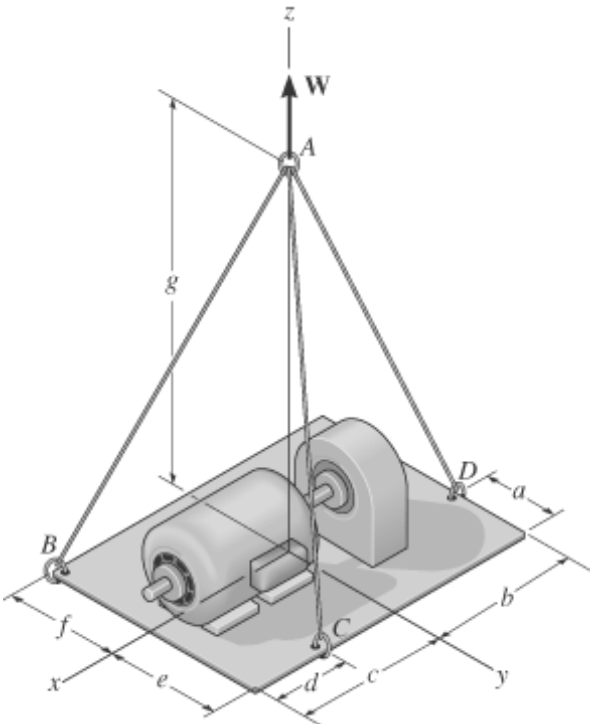
Given:

$W = 3500$ lb $\quad d = 4$ ft

$a = 2$ ft $\quad e = 3$ ft

$b = 4$ ft $\quad f = 3$ ft

$c = 4$ ft $\quad g = 10$ ft



Solution:



The initial guesses**:**

$F_{AB} = 1$ lb $\quad F_{AC} = 1$ lb $\quad F_{AD} = 1$ lb

Given

$$\frac{-b}{\sqrt{g^2+(e-a)^2+b^2}}F_{AD}+\frac{c-d}{\sqrt{(c-d)^2+e^2+g^2}}F_{AC}+\frac{c}{\sqrt{c^2+f^2+g^2}}F_{AB}=0$$

$$\frac{e-a}{\sqrt{g^2+(e-a)^2+b^2}}F_{AD}+\frac{e}{\sqrt{e^2+(c-d)^2+g^2}}F_{AC}-\frac{f}{\sqrt{c^2+f^2+g^2}}F_{AB}=0$$

$$\frac{-g}{\sqrt{g^2+(e-a)^2+b^2}}F_{AD}-\frac{g}{\sqrt{e^2+(c-d)^2+g^2}}F_{AC}-\frac{g}{\sqrt{g^2+f^2+c^2}}F_{AB}+W=0$$

$$\begin{pmatrix}F_{AB}\\F_{AC}\\F_{AD}\end{pmatrix}=\text{Find}\left(F_{AB},F_{AC},F_{AD}\right) \qquad \begin{pmatrix}F_{AB}\\F_{AC}\\F_{AD}\end{pmatrix}=\begin{pmatrix}1.467\\0.914\\1.42\end{pmatrix}\text{kip}$$

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Problem 3-64

A flowerpot of mass *M* is supported at *A* by the three cords. Determine the force acting in each cord for equilibrium.

Given:

$M = 25\ \text{kg}$

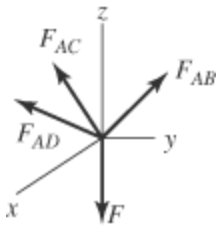
$g = 9.81\ \frac{\text{m}}{\text{s}^2}$

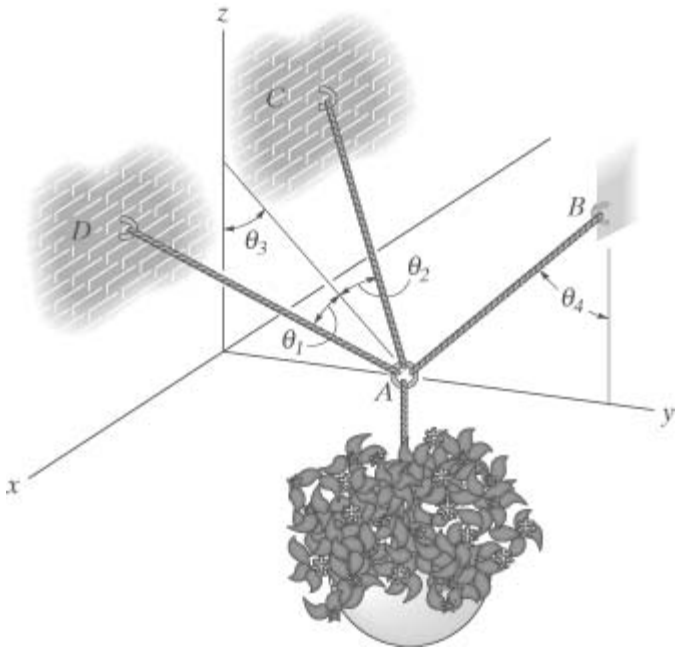
$\theta_1 = 30\ \text{deg}$

$\theta_2 = 30\ \text{deg}$

$\theta_3 = 60\ \text{deg}$

$\theta_4 = 45\ \text{deg}$





Solution:

Initial guesses:

$F_{AB} = 1\ \text{N}$

$F_{AD} = 1\ \text{N}$

$F_{AC} = 1\ \text{N}$

Given

$\Sigma F_x = 0; \quad F_{AD}\sin(\theta_1) - F_{AC}\sin(\theta_2) = 0$

$\Sigma F_y = 0; \quad -F_{AD}\cos(\theta_1)\sin(\theta_3) - F_{AC}\cos(\theta_2)\sin(\theta_3) + F_{AB}\sin(\theta_4) = 0$

$\Sigma F_z = 0; \quad F_{AD}\cos(\theta_1)\cos(\theta_3) + F_{AC}\cos(\theta_2)\cos(\theta_3) + F_{AB}\cos(\theta_4) - M\,g = 0$

$$\begin{pmatrix} F_{AB} \\ F_{AC} \\ F_{AD} \end{pmatrix} = \text{Find}(F_{AB}, F_{AC}, F_{AD}) \qquad \begin{pmatrix} F_{AB} \\ F_{AC} \\ F_{AD} \end{pmatrix} = \begin{pmatrix} 219.89 \\ 103.65 \\ 103.65 \end{pmatrix}\ \text{N}$$

Problem 3-65

If each cord can sustain a maximum tension of **T** before it fails, determine the greatest weight of the flowerpot the cords can support.

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Given:

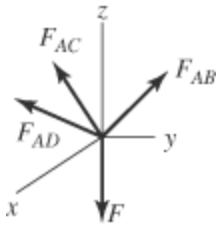
$T = 50$ N

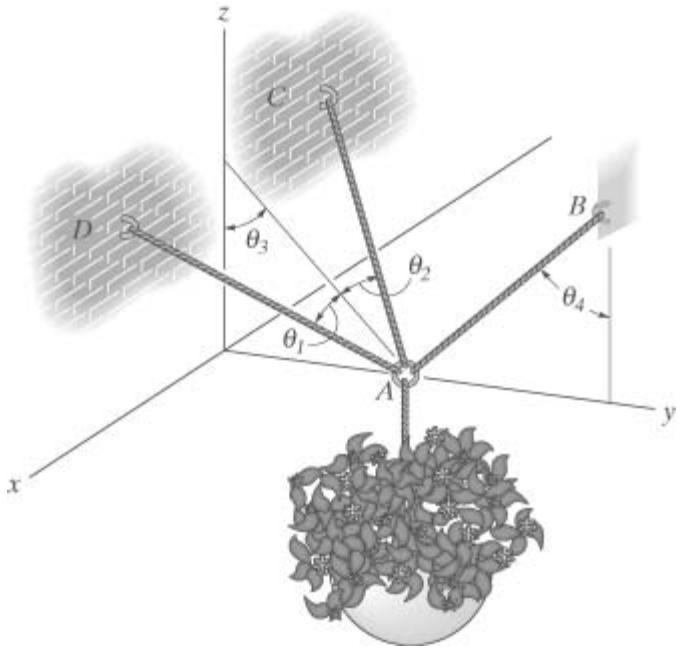
$\theta_1 = 30$ deg

$\theta_2 = 30$ deg

$\theta_3 = 60$ deg

$\theta_4 = 45$ deg





Solution:

Initial guesses:

$F_{AB} = T$

$F_{AD} = T$

$F_{AC} = T$

$W = T$

Case 1 Assume that *AB* reaches maximum tension

Given

$\Sigma F_x = 0; F_{AD}\sin(\theta_1) - F_{AC}\sin(\theta_2) = 0$

$\Sigma F_y = 0; -F_{AD}\cos(\theta_1)\sin(\theta_3) - F_{AC}\cos(\theta_2)\sin(\theta_3) + F_{AB}\sin(\theta_4) = 0$

$\Sigma F_z = 0; F_{AD}\cos(\theta_1)\cos(\theta_3) + F_{AC}\cos(\theta_2)\cos(\theta_3) + F_{AB}\cos(\theta_4) - W = 0$

$$\begin{pmatrix} W_1 \\ F_{AC1} \\ F_{AD1} \end{pmatrix} = \text{Find}(W, F_{AC}, F_{AD}) \qquad \begin{pmatrix} W_1 \\ F_{AC1} \\ F_{AD1} \end{pmatrix} = \begin{pmatrix} 55.77 \\ 23.57 \\ 23.57 \end{pmatrix} \text{N}$$

Case 2 Assume that *AC* reaches maximum tension

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Given

$$\Sigma F_x = 0;\ F_{AD}\sin(\theta_1) - F_{AC}\sin(\theta_2) = 0$$

$$\Sigma F_y = 0;\ -F_{AD}\cos(\theta_1)\sin(\theta_3) - F_{AC}\cos(\theta_2)\sin(\theta_3) + F_{AB}\sin(\theta_4) = 0$$

$$\Sigma F_z = 0;\ F_{AD}\cos(\theta_1)\cos(\theta_3) + F_{AC}\cos(\theta_2)\cos(\theta_3) + F_{AB}\cos(\theta_4) - W = 0$$

$$\begin{pmatrix} W_2 \\ F_{AB2} \\ F_{AD2} \end{pmatrix} = \text{Find}(W, F_{AB}, F_{AD}) \qquad \begin{pmatrix} W_2 \\ F_{AB2} \\ F_{AD2} \end{pmatrix} = \begin{pmatrix} 118.30 \\ 106.07 \\ 50.00 \end{pmatrix} \text{N}$$

Case 3 Assume that *AD* reaches maximum tension

Given

$$\Sigma F_x = 0;\ F_{AD}\sin(\theta_1) - F_{AC}\sin(\theta_2) = 0$$

$$\Sigma F_y = 0;\ -F_{AD}\cos(\theta_1)\sin(\theta_3) - F_{AC}\cos(\theta_2)\sin(\theta_3) + F_{AB}\sin(\theta_4) = 0$$

$$\Sigma F_z = 0;\ F_{AD}\cos(\theta_1)\cos(\theta_3) + F_{AC}\cos(\theta_2)\cos(\theta_3) + F_{AB}\cos(\theta_4) - W = 0$$

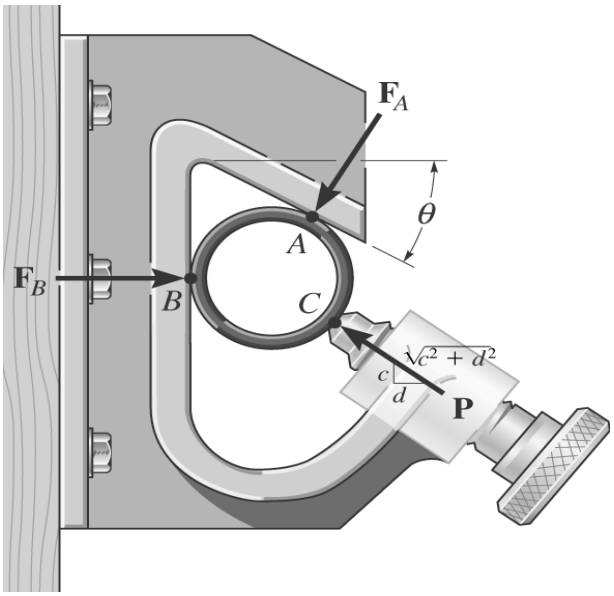
$$\begin{pmatrix} W_3 \\ F_{AB3} \\ F_{AC3} \end{pmatrix} = \text{Find}(W, F_{AB}, F_{AC}) \qquad \begin{pmatrix} W_3 \\ F_{AB3} \\ F_{AC3} \end{pmatrix} = \begin{pmatrix} 118.30 \\ 106.07 \\ 50.00 \end{pmatrix} \text{N}$$

$$W = \min(W_1, W_2, W_3) \qquad W = 55.77\ \text{N}$$

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Problem 3-66

The pipe is held in place by the vice. If the bolt exerts force P on the pipe in the direction shown, determine the forces F_A and F_B that the smooth contacts at A and B exerton the pipe.



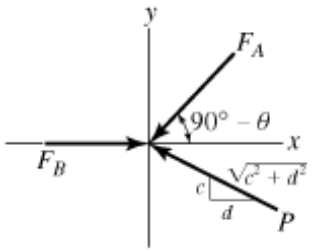
Given:

$P = 50$ lb

$\theta = 30$ deg

$c = 3$

$d = 4$



Solution:

Initial Guesses $F_A = 1$ lb $F_B = 1$ lb

Given

$$\xrightarrow{+} \Sigma F_x = 0; \qquad F_B - F_A \sin(\theta) - P\left(\frac{d}{\sqrt{c^2 + d^2}}\right) = 0$$

$$+\uparrow \Sigma F_y = 0; \qquad -F_A \cos(\theta) + P\left(\frac{c}{\sqrt{c^2 + d^2}}\right) = 0$$

$$\begin{pmatrix} F_A \\ F_B \end{pmatrix} = \text{Find}(F_A, F_B) \qquad \begin{pmatrix} F_A \\ F_B \end{pmatrix} = \begin{pmatrix} 34.6 \\ 57.3 \end{pmatrix} \text{lb}$$

Problem 3-67

When y is zero, the springs sustain force F_0. Determine the magnitude of the applied vertical forces **F** and **-F** required to pull point A away from point B a distance y_1. The ends of cords CAD and CBD are attached to rings at C and D.

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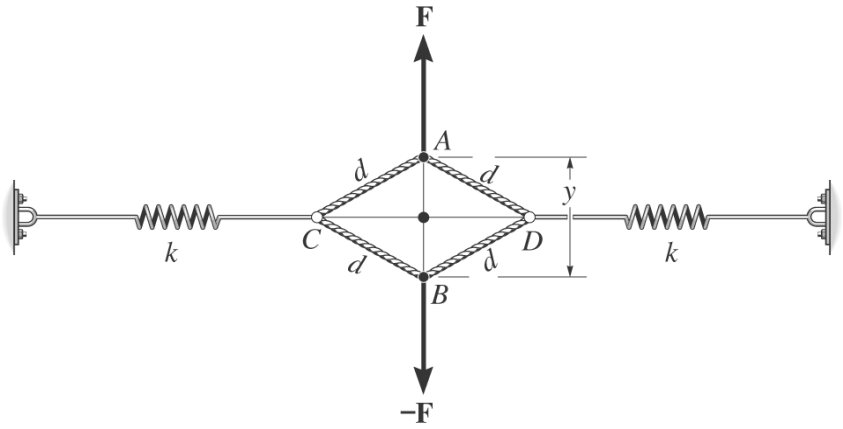
Given:

$F_0 = 60$ lb

$k = 40 \frac{\text{lb}}{\text{ft}}$

$d = 2$ ft

$y_1 = 2$ ft



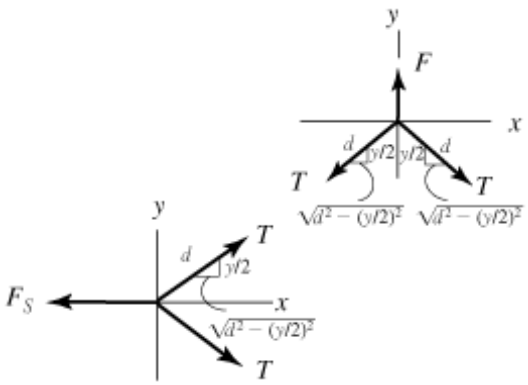
Solution:

Initial spring stretch:

$$s_1 = \frac{F_0}{k} \qquad s_1 = 1.50\,\text{ft}$$

Initial guesses:

$F_s = 1$ lb $\quad T = 1$ lb $\quad F = 1$ lb



Given

$$F - \frac{y_1}{2d}T = 0 \qquad \frac{\sqrt{d^2 - \left(\frac{y_1}{2}\right)^2}}{d}T - F_s = 0 \qquad F_s = k\left[d - \sqrt{d^2 - \left(\frac{y_1}{2}\right)^2} + s_1\right]$$

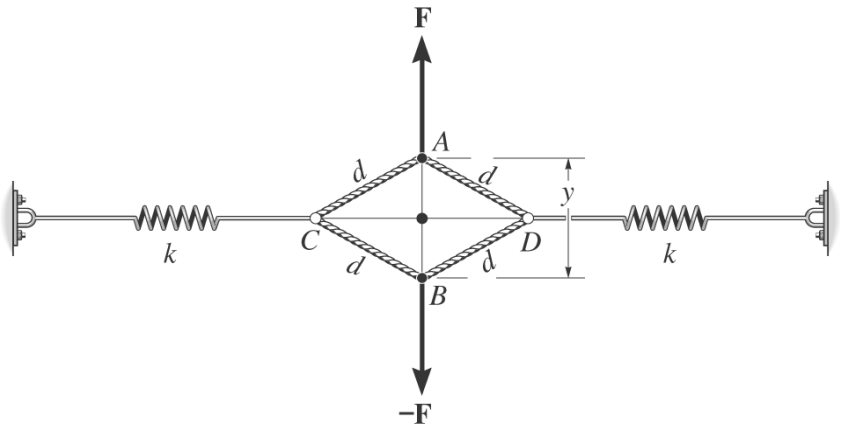
$$\begin{pmatrix} F_s \\ T \\ F \end{pmatrix} = \text{Find}(F_s, T, F) \qquad \begin{pmatrix} F_s \\ T \end{pmatrix} = \begin{pmatrix} 70.72 \\ 81.66 \end{pmatrix}\text{lb} \qquad F = 40.83\,\text{lb}$$

Problem 3-68

When y is zero, the springs are each stretched a distance δ. Determine the distance y if a force F is applied to points A and B as shown. The ends of cords CAD and CBD are attached to

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rings at C and D.



Given:

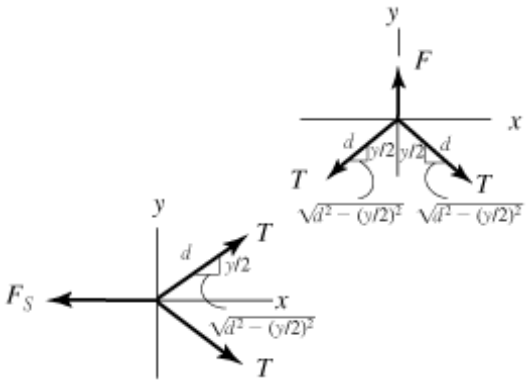
$\delta = 1.5$ ft

$k = 40 \frac{\text{lb}}{\text{ft}}$

$d = 2$ ft

$F = 60$ lb

Solution:



Initial guesses:

$F_S = 1$ lb $\quad T = 1$ lb $\quad y = 1$ ft

Given

$$F - \frac{y}{2d}T = 0$$

$$\frac{\sqrt{d^2 - \left(\frac{y}{2}\right)^2}}{d}T - F_S = 0 \qquad F_S = k\left[d - \sqrt{d^2 - \left(\frac{y}{2}\right)^2} + \delta\right]$$

$$\begin{pmatrix} F_S \\ T \\ y \end{pmatrix} = \text{Find}(F_S, T, y) \qquad \begin{pmatrix} F_S \\ T \end{pmatrix} = \begin{pmatrix} 76.92 \\ 97.55 \end{pmatrix} \text{lb} \qquad y = 2.46\,\text{ft}$$

Problem 3-69

Cord AB of length a is attached to the end B of a spring having an unstretched length b. The other end of the spring is attached to a roller C so that the spring remains horizontal as it stretches. If a weight W is suspended from B, determine the angle θ of cord AB for equilibrium.

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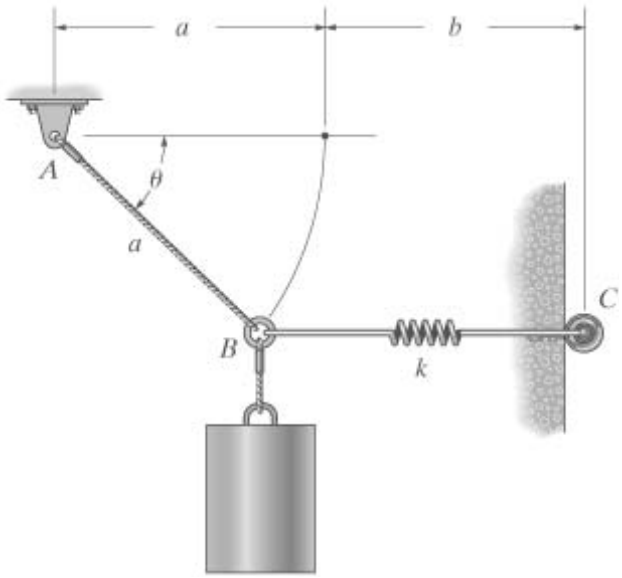
Given:

$a = 5$ ft

$b = 5$ ft

$k = 10 \frac{\text{lb}}{\text{ft}}$

$W = 10$ lb



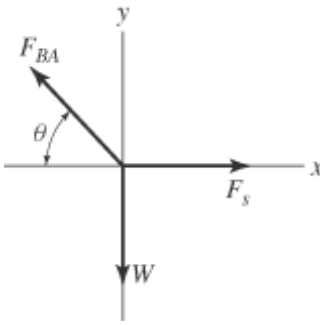
Solution:

Initial Guesses

$F_{BA} = 1$ lb

$F_{sp} = 1$ lb

$\theta = 30$ deg



Given

$F_{sp} - F_{BA}\cos(\theta) = 0$

$F_{BA}\sin(\theta) - W = 0$

$F_{sp} = k(a - a\cos(\theta))$

$$\begin{pmatrix} F_{sp} \\ F_{BA} \\ \theta \end{pmatrix} = \text{Find}(F_{sp}, F_{BA}, \theta) \qquad \begin{pmatrix} F_{sp} \\ F_{BA} \end{pmatrix} = \begin{pmatrix} 11.82 \\ 15.49 \end{pmatrix} \text{lb} \qquad \theta = 40.22 \text{ deg}$$

Problem 3-70

The uniform crate of mass M is suspended by using a cord of length l that is attached to the sides of the crate and passes over the small pulley at O. If the cord can be attached at either points A and B, or C and D, determine which attachment produces the least amount of tension in the cord and specify the cord tension in this case.

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Given:

$M = 50\ \text{kg}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$

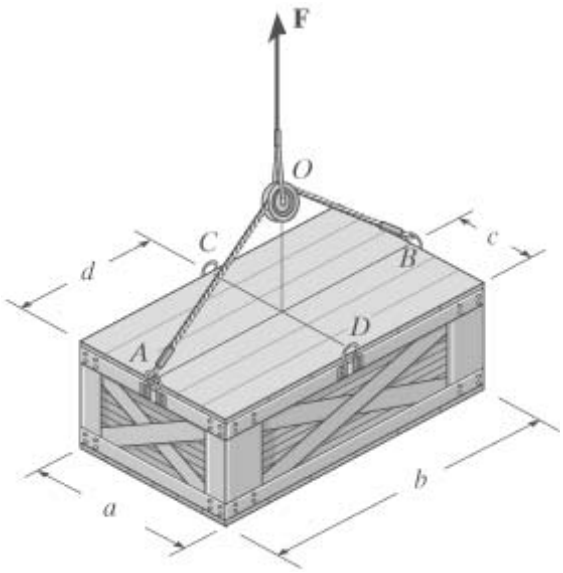
$a = 0.6\ \text{m}$

$b = 1.5\ \text{m}$

$l = 2\ \text{m}$

$c = \frac{a}{2}$

$d = \frac{b}{2}$



Solution:

Case 1 Attached at A and B Guess $T = 1\ \text{N}$

Given $Mg - \left[\frac{\sqrt{\left(\frac{l}{2}\right)^2 - d^2}}{\frac{l}{2}}\right] 2T = 0$ $T_1 = \text{Find}(T)$ $T_1 = 370.78\ \text{N}$

Case 2 Attached at C and D Guess $T = 1\ \text{N}$

Given $Mg - \left[\frac{\sqrt{\left(\frac{l}{2}\right)^2 - c^2}}{\frac{l}{2}}\right] 2T = 0$ $T_2 = \text{Find}(T)$ $T_2 = 257.09\ \text{N}$

Choose the arrangement that gives the smallest tension. $T = \min(T_1, T_2)$ $T = 257.09\ \text{N}$

Problem 3-71

The man attempts to pull the log at C by using the three ropes. Determines the direction θ in which he should pull on his rope with a force P, so that he exerts a maximum

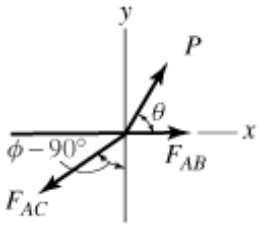
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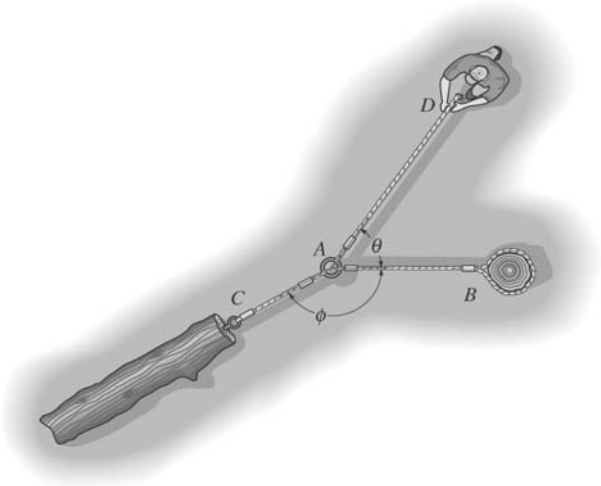
force on the log. What is the force on the log for this case ? Also, determine the direction in which he should pull in order to maximize the force in the rope attached to B.What is this maximum force?

Given:

$P = 80$ lb

$\phi = 150$ deg





Solution:

$\xrightarrow{+} \Sigma F_x = 0; \qquad F_{AB} + P\cos(\theta) - F_{AC}\sin(\phi - 90\ \text{deg}) = 0$

$+\uparrow \Sigma F_y = 0; \qquad P\sin(\theta) - F_{AC}\cos(\phi - 90\ \text{deg}) = 0$

$F_{AC} = \dfrac{P\sin(\theta)}{\cos(\phi - 90\text{deg})}$ In order to maximize *FAC* we choose $\sin(\theta) = 1.$

Thus $\theta = 90$ deg $F_{AC} = \dfrac{P\sin(\theta)}{\cos(\phi - 90\text{deg})}$ $F_{AC} = 160.00$ lb

Now let's find the force in the rope *AB.*

$F_{AB} = -P\cos(\theta) + F_{AC}\sin(\phi - 90\ \text{deg})$

$F_{AB} = -P\cos(\theta) + \dfrac{P\sin(\theta)\sin(\phi - 90\ \text{deg})}{\cos(\phi - 90\ \text{deg})}$

$F_{AB} = P\dfrac{\sin(\theta)\sin(\phi - 90\ \text{deg}) - \cos(\theta)\cos(\phi - 90\ \text{deg})}{\cos(\phi - 90\ \text{deg})} = -P\ \dfrac{\cos(\theta + \phi - 90\text{deg})}{\cos(\phi - 90\ \text{deg})}$

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In order to maximize the force we set $\cos(\theta + \phi - 90\ \text{deg}) = -1$

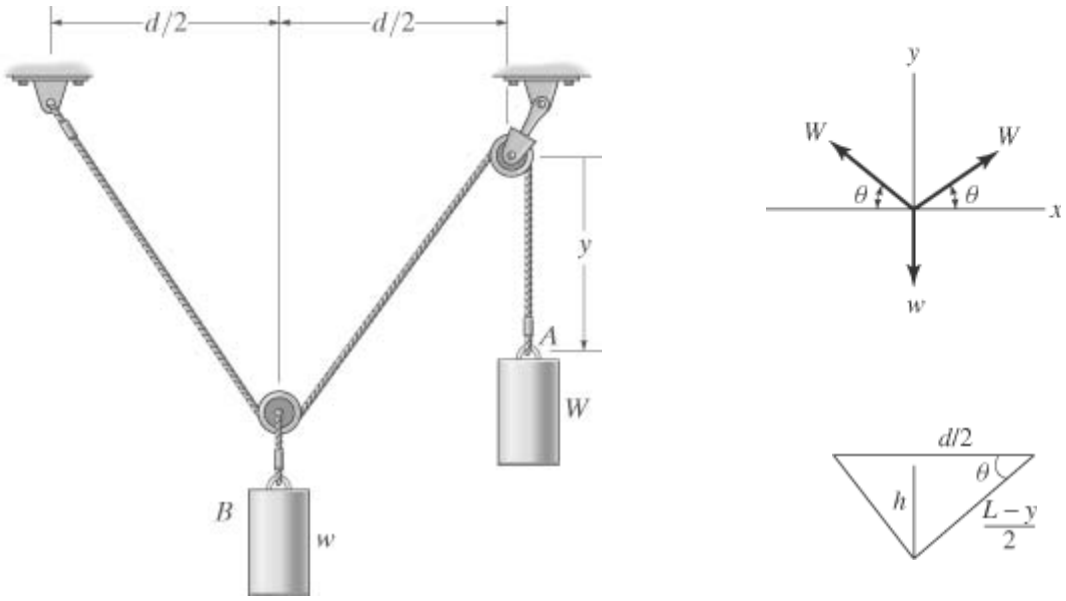
$\theta + \phi - 90\ \text{deg} = 180\ \text{deg}$

$\theta = 270\ \text{deg} - \phi$ $\theta = 120.00\,\text{deg}$

$F_{AB} = -P\dfrac{\cos(\theta + \phi - 90\ \text{deg})}{\cos(\phi - 90\ \text{deg})}$ $F_{AB} = 160.00\,\text{lb}$

Problem 3-72

The "scale" consists of a known weight W which is suspended at A from a cord of total length L. Determine the weight w at B if A is at a distance y for equilibrium. Neglect the sizes and weights of the pulleys.



Solution:

$+\uparrow\ \Sigma F_y = 0; \quad 2W\sin(\theta) - w = 0$

Geometry $h = \sqrt{\left(\dfrac{L-y}{2}\right)^2 - \left(\dfrac{d}{2}\right)^2} = \dfrac{1}{2}\sqrt{(L-y)^2 - d^2}$

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$$w = 2W\left[\frac{\frac{1}{2}\sqrt{(L-y)^2 + d^2}}{\frac{L-y}{2}}\right] \qquad w = \frac{2W}{L-y}\sqrt{(L-y)^2 - d^2}$$

Problem 3-73

Determine the maximum weight W that can be supported in the position shown if each cable AC and AB can support a maximum tension of **F** before it fails.



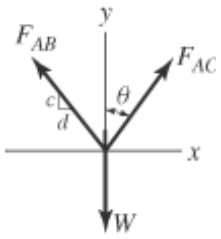
Given:

$\theta = 30\ \text{deg}$

$F = 600\ \text{lb}$

$c = 12$

$d = 5$



Solution:

Initial Guesses $F_{AB} = F \qquad F_{AC} = F \qquad W = F$

Case 1 Assume that cable AC reaches maximum tension

Given $$F_{AC}\sin(\theta) - \frac{d}{\sqrt{c^2 + d^2}}F_{AB} = 0$$

$$F_{AC}\cos(\theta) + \frac{c}{\sqrt{c^2 + d^2}}F_{AB} - W = 0$$

$$\begin{pmatrix} W_1 \\ F_{AB1} \end{pmatrix} = \text{Find}(W, F_{AB}) \qquad \begin{pmatrix} W_1 \\ F_{AB1} \end{pmatrix} = \begin{pmatrix} 1239.62 \\ 780.00 \end{pmatrix}\ \text{lb}$$

Case 2 Assume that cable AB reaches maximum tension

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Given $F_{AC}\sin(\theta) - \frac{d}{\sqrt{c^2 + d^2}}F_{AB} = 0$

$F_{AC}\cos(\theta) + \frac{c}{\sqrt{c^2 + d^2}}F_{AB} - W = 0$

$\begin{pmatrix} W_2 \\ F_{AC2} \end{pmatrix} = \text{Find}(W, F_{AC})$ $\begin{pmatrix} W_2 \\ F_{AC2} \end{pmatrix} = \begin{pmatrix} 953.55 \\ 461.54 \end{pmatrix}$ lb

$W = \min(W_1, W_2)$ $W = 953.6$ lb

Problem 3-74

If the spring on rope *OB* has been stretched a distance δ and fixed in place as shown, determine the tension developed in each of the other three ropes in order to hold the weight W in equilibrium. Rope *OD* lies in the *x-y* plane.

Given:

$a = 2$ ft

$b = 4$ ft

$c = 3$ ft

$d = 4$ ft

$e = 4$ ft

$f = 4$ ft

$x_B = -2$ ft

$y_B = -3$ ft

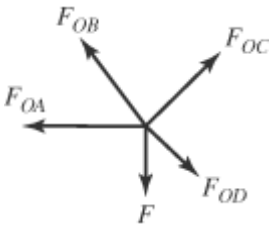
$z_B = 3$ ft

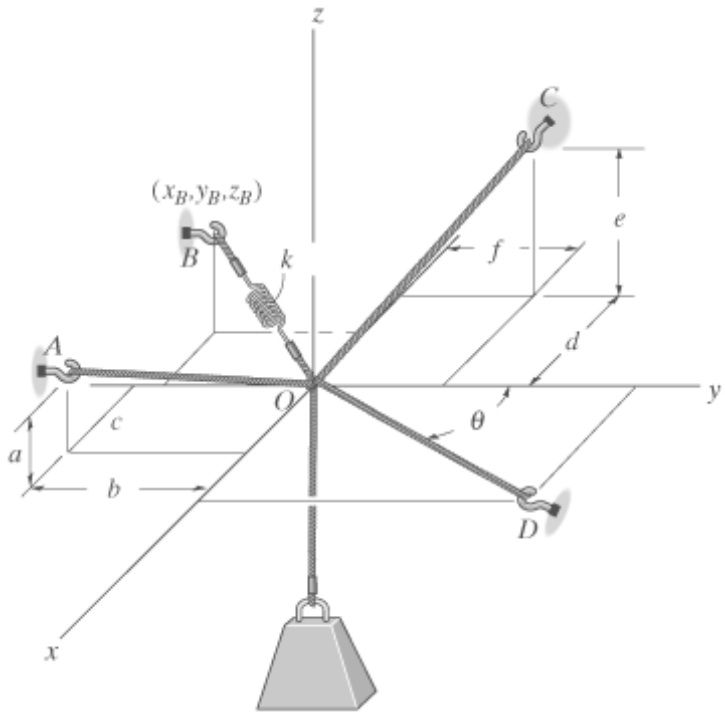
$\theta = 30$ deg

$k = 20 \frac{\text{lb}}{\text{in}}$

$\delta = 2$ in

$W = 225$ lb





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Solution:

Initial Guesses $F_{OA} = 10$ lb $F_{OC} = 10$ lb $F_{OD} = 10$ lb

Given

$$\frac{c}{\sqrt{a^2+b^2+c^2}}F_{OA} + k\delta\frac{x_B}{\sqrt{x_B{}^2+y_B{}^2+z_B{}^2}} + \frac{-d}{\sqrt{d^2+e^2+f^2}}F_{OC} + F_{OD}\sin(\theta) = 0$$

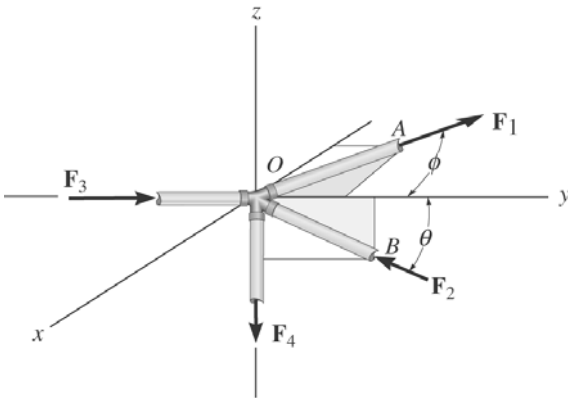
$$\frac{-b}{\sqrt{a^2+b^2+c^2}}F_{OA} + k\delta\frac{y_B}{\sqrt{x_B{}^2+y_B{}^2+z_B{}^2}} + \frac{f}{\sqrt{d^2+e^2+f^2}}F_{OC} + F_{OD}\cos(\theta) = 0$$

$$\frac{a}{\sqrt{a^2+b^2+c^2}}F_{OA} + k\delta\frac{z_B}{\sqrt{x_B{}^2+y_B{}^2+z_B{}^2}} + \frac{e}{\sqrt{d^2+e^2+f^2}}F_{OC} - W = 0$$

$$\begin{pmatrix} F_{OA} \\ F_{OC} \\ F_{OD} \end{pmatrix} = \text{Find}(F_{OA}, F_{OC}, F_{OD}) \qquad \begin{pmatrix} F_{OA} \\ F_{OC} \\ F_{OD} \end{pmatrix} = \begin{pmatrix} 201.6 \\ 215.7 \\ 58.6 \end{pmatrix} \text{lb}$$

Problem 3-75

The joint of a space frame is subjected to four member forces. Member *OA* lies in the *x* - *y* plane and member *OB* lies in the *y* - *z* plane. Determine the forces acting in each of the members required for equilibrium of the joint.



Given:

$F_4 = 200$ lb

$\theta = 40$ deg

$\phi = 45$ deg

Solution:

The initial guesses: $F_1 = 200$ lb $F_2 = 200$ lb $F_3 = 200$ lb

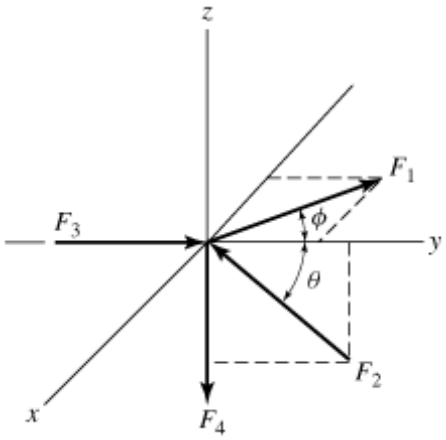
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Given

$$\Sigma F_y = 0; \quad F_3 + F_1 \cos(\phi) - F_2 \cos(\theta) = 0$$

$$\Sigma F_x = 0; \quad -F_1 \sin(\phi) = 0$$

$$\Sigma F_z = 0; \quad F_2 \sin(\theta) - F_4 = 0$$



$$\begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = \text{Find}(F_1, F_2, F_3)$$

$$\begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 311.1 \\ 238.4 \end{pmatrix} \text{lb}$$

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